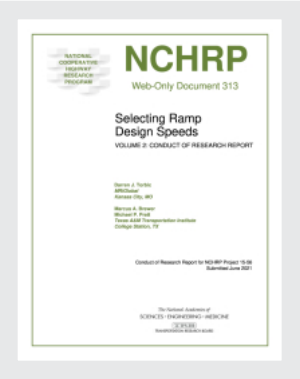
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NCHRP

Web-Only Document 313

Selecting Ramp Design Speeds

VOL. 2: CONDUCT OF RESEARCH REPORT

Darren J. Torbic
MRIGlobal
Kansas City, MO

Marcus A. Brewer
Michael P. Pratt
Eun Sug Park
Texas A&M Transportation Institute
College Station, TX

Conduct of Research Report for NCHRP Project 15-56
Submitted June 2021

NATIONAL COOPERATIVE HIGHWAY RESEARCH PROGRAM
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Tyler Smith, *Senior Program Assistant*
Natalie Barnes, *Director of Publications*
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Mark A. Doctor, *FHWA Liaison*
Stephen F. Maher, *TRB Liaison*

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Summary

The design speed concept has been the centerpiece of American Association of State Highway and Transportation Officials (AASHTO's) geometric design policy since the 1930s. Designing extended sections of highway based on the design speed process is relatively straightforward. However, when applied to interchange ramps where high-speed facilities meet low-speed facilities and drivers are expected to accelerate or decelerate over short distances, application of the design speed process is more complex.

The objectives of this research were to (a) conduct a review of the state of practice for selecting ramp design speeds, (b) develop new guidelines for selecting appropriate ramp design speeds, and (c) recommend revisions to the AASHTO *Green Book* based upon the results of this research. This final report presents the overall research findings including the literature review, survey, and interviews with selected designers; details about the speed data collected during the study; statistical analyses to develop speed prediction models for ramps; project summary and future research needs; and recommended changes to the AASHTO *Green Book*. A companion document to this final report presents enhanced guidelines for selecting an appropriate ramp design speed, details about the ramp speed profile model, and case studies that illustrate how to use the guidelines and ramp speed file model to help make design decisions that are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads.

Based upon the literature review, the survey results, and interviews with selected professionals, it was assessed that the design guidelines presented in the AASHTO *Green Book* related to application of the design speed process for ramps and the appropriate selection of a ramp design speed have several overarching challenges that lead to confusion and inconsistent interpretation of the guidelines. In particular, the *Green Book* does not specifically define ramp design speed. *Green Book* Table 10-1, which is the centerpiece of AASHTO's guidance on selecting an appropriate ramp design speed for the design of all ramp types, does not provide sufficient detail for selecting ramp design speeds for specific ramp types and configurations. Little guidance is provided on how the selected ramp design speed is to be used to develop a balanced design among all components of a ramp, nor is there consensus on the portion of the ramp to which the ramp design speed applies or whether the ramp design speed should be a single value or vary along the ramp. Therefore, development of enhanced design guidelines is necessary for selecting appropriate ramp design speeds to enable practitioners to design ramps in a consistent manner based on the selected ramp design speed, accounting for sequential speed transitions from one component or section to the next, consistent with performance capabilities of vehicles and driver expectations.

An observational field study was conducted to collect vehicle speeds at interchange ramps. The purpose of the study was to gather supplemental speed data along interchange ramps that could be used in combination with speed data collected along interchange ramps during previous research to further develop and refine speed prediction models for use in a spreadsheet-based ramp speed profile model (RSPM) tool that designers and traffic engineers can use to estimate vehicle speeds along interchange ramps. A database was assembled to develop ramp speed prediction models that could be incorporated into the spreadsheet-based RSPM tool. Separate

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statistical analyses were conducted to develop ramp speed prediction models for curves and tangents.

While developing the RSPM tool, the research team also developed an initial set of enhanced guidelines for selecting an appropriate ramp design speed. The research team also identified a few ramps to serve as case studies to apply and evaluate the enhanced guidelines for selecting an appropriate ramp design speed and refined the guidelines accordingly. The enhanced guidelines are summarized in this final report but are presented in more detail in the companion document.

The enhanced guidelines define ramp design speed and related terminology and discuss in detail the most relevant elements that affect selecting an appropriate ramp design speed. A new table developed as part of the guidelines provides a range of guide values for ramp design speeds for system and service interchanges, and the guide values are presented as a range of values given the context, ramp configuration, and type of interchange. The guidelines also address how to apply design controls and criteria so that individual components and features of ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next, and several case studies are presented to demonstrate how to evaluate the consistency of a ramp design based on the selected ramp design speed. These case studies demonstrate use of the RSPM tool and how it can be used by designers and traffic engineers to conduct operational analyses and to help make design decisions.

The enhanced guidelines are intended to facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information and are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads. The enhanced guidelines (1) are not expected to introduce unforeseen safety or operational issues, create the need for more design exceptions, nor increase or decrease construction costs; (2) may help to improve safety and operational issues related to ramp design by reducing the number of ramps designed and built in an inconsistent manner due to confusion or inconsistent interpretation of existing AASHTO policy; (3) may benefit an agency’s overall program budget by reducing the number of ramps that need to be reconstructed at a later date because they were originally designed and built in an inconsistent manner; and (4) are not expected to create any specific tort liability concerns related to ramp design.

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Section 1. Introduction

1.1 Background

In the AASHTO *A Policy on Geometric Design of Highways and Streets*, commonly referred to as the *Green Book*, design speed is defined as a selected speed used to determine the various geometric design features of the roadway (AASHTO, 2018). In principle, design speed is used to determine all pertinent highway features to obtain a balanced design. The design speed concept has been the centerpiece of the geometric design process since introduced in 1936 (Barnett, 1936).

Over an extended length of highway, it is desirable to select a uniform design speed. Selection of a design speed should be based upon the anticipated operating speed, topography, adjacent land use, modal mix, and functional classification of the highway (AASHTO, 2018). Where high-speed facilities meet low-speed facilities, drivers are expected to reduce their speeds suitable to the roadway environment they are entering. Ideally, this change in speed is achieved over an extended distance, but in the case of an interchange ramp, this is not possible. Conceptually, designing extended sections of highway based on the design speed process is relatively simple; however, when applied to interchange ramps, application of the design speed process and selection of an appropriate ramp design speed is a complex process.

According to AASHTO policy, the ramp design speed should approximate the low-volume running speed on the intersecting highways; however, this is not always practical. *Green Book* Table 10-1 (Table 1) provides guidance for ramp design speeds based upon a range of percentages (i.e., upper, middle, and lower) of the higher-speed highway design speed. Further guidance is provided for selecting a ramp design speed from the table based on various conditions and ramp types. For example, design speeds between the middle and upper ranges of the highway speeds should be used for direct and semidirect connections, and upper-range values of design speeds generally are not attainable on loop ramps. Table 2 summarizes preferred and practical minimum ramp design speeds as recommended in the Institute of Transportation Engineers (ITE) *Freeway and Interchange Geometric Design Handbook (Leisch, 2005)*.

Table 1. Guide Values for Ramp Design Speed as Related to Highway Design Speed (AASHTO, 2018)

U.S. Customary											
Highway design speed (mph)	30	35	40	45	50	55	60	65	70	75	80
Ramp design speed (mph)											
Upper range (85%)	25	30	35	40	45	50	50	55	60	65	70
Middle range (70%)	20	25	30	30	35	40	45	45	50	55	60
Lower range (50%)	15	20	20	25	25	30	30	30	35	40	45
Corresponding minimum radius (ft)	See (*Green Book*) Table 3-7										

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Table 2. Design Speed Guidelines for Ramps (Leisch, 2005)

Ramp Type	Preferred	Practical Minimum
Right turning	Upper range	Middle range
Loops	30-35 mph	25 mph
Semidirect	Middle-upper range	30 mph
Direct	Middle-upper range	40 mph

The design values in *Green Book* Table 10-1 are applicable to the sharpest, or controlling, curve on the ramp, usually on the ramp proper, and do not pertain to the ramp terminals. The selection of a ramp design speed affects many aspects and elements of a ramp, including right-of-way requirements, horizontal and vertical alignments, speed-change lane lengths, ramp type, superelevation, and stopping sight distance, in addition to project costs and potential environmental and social impacts. Directly or indirectly, physical and economic constraints and anticipated operating speeds on the ramp should be considered when selecting the ramp design speed. The primary components of a ramp include the freeway mainline ramp terminal (i.e., an acceleration or deceleration lane), the ramp proper (i.e., the turning roadway), and the crossroad ramp terminal, and they must be considered together during the design of a ramp (Figure 1). The actual boundaries or transitions of where one component ends (e.g., freeway mainline ramp terminal) and the adjacent component begins (e.g., ramp proper) are not clearly defined in all cases.

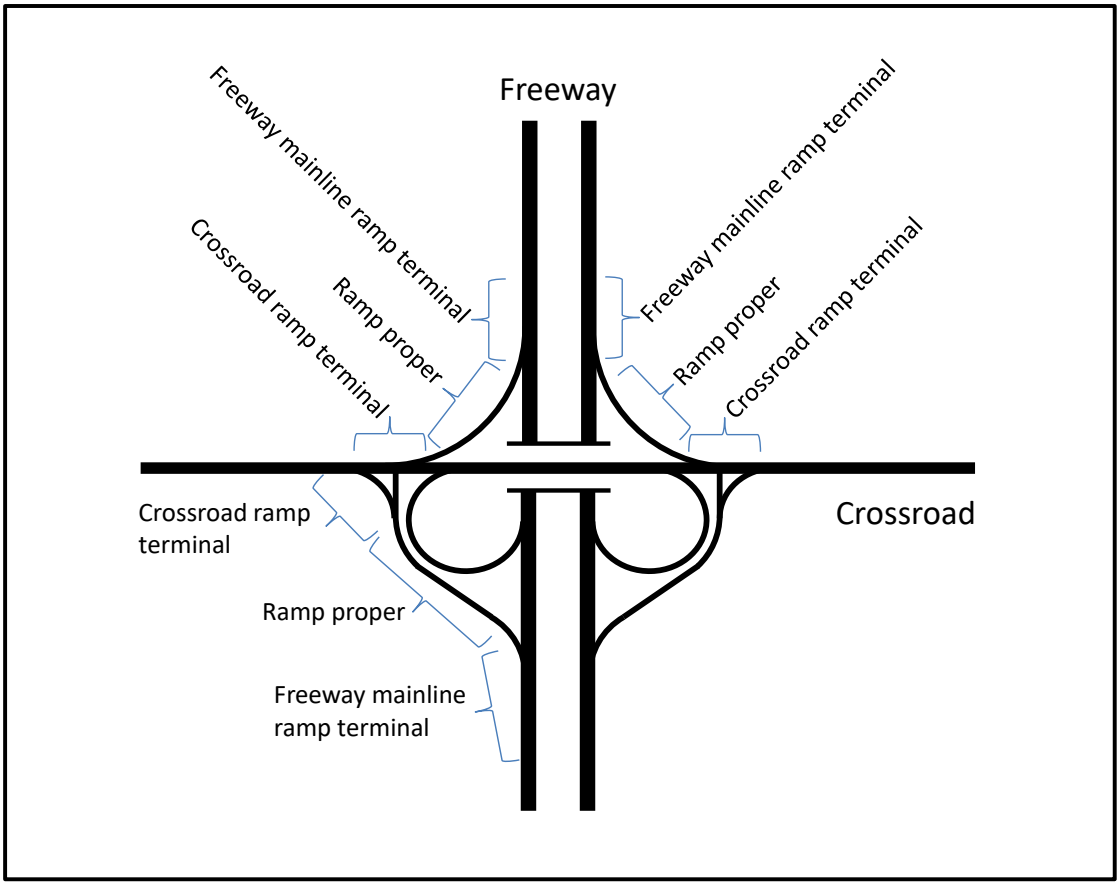


Figure 1. Primary Components of an Interchange Ramp

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The rationale of AASHTO's current policy on ramp design speed is therefore in need of review. Guidance in the current *Green Book* is complex and may lead to insufficient or poor ramp designs. At the very least, the current guidance for selecting ramp design speed needs to be reviewed to determine if it can be improved and/or is consistent with the overall design speed process and concept.

1.2 Research Objective and Scope

The objectives of this research were to (a) conduct a review of the state of practice for selecting ramp design speeds, (b) develop new guidelines for selecting appropriate ramp design speeds, and (c) recommend revisions to the AASHTO *Green Book* based upon the results of this research. The state of practice review was based on information gathered from published and unpublished sources, a survey of engineers, designers, safety professionals, and others; and interviews with selected engineers with practical experience designing interchange ramps. While developing new guidelines for selecting appropriate ramp design speeds, the research team considered the following:

- The guidelines should facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information.
- The guidelines should clearly define ramp design speed and all terms and elements related to its application, including the portion(s) of the ramp to which the design speed should be applicable.
- The guidelines should be based upon known safety and operational relationships of ramps over a full range of interchange forms, ramp types, and area environments (i.e., rural and urban).
- The guidelines should result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads.
- Development of the guidelines should include the examination of field data and site observations supplemented with safety modeling considerations and traffic operational simulations, and speed transitions within ramps should be reviewed for performance and design policy implications.
- The ramp design speed guidelines should address, at a minimum:
  - System and service interchanges.
  - Entrance and exit ramps.
  - Different types of ramp interfaces at crossroads.
  - Single- and multilane ramps.
  - Degree of physical site constraints, environmental and social impacts, and right-of-way costs.
  - Effects of ramp grade.
  - Use of transition curves into the controlling alignment feature.

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- Presence of ramp meters.
- Ramp terminal configuration (e.g., tapered and parallel, presence of auxiliary lane).
- Applicability of ramp design controls as opposed to mainline design controls for certain connections.

Furthermore, the guidelines should be presented in a straightforward, easy-to-comprehend format so they are readily understood by potential users and for incorporation into the *Green Book*.

1.3 General Research Approach

The first step towards accomplishing the research objective was to summarize the state of practice and other information related to ramp design speed selection. The research team reviewed published research studies, guidance documents, and design manuals; conducted an online survey of engineers, designers, safety professionals, planners, construction managers, operations managers, and others; and interviewed several survey respondents by telephone to explore issues covered by the survey in greater depth. Based upon the literature review, the survey results, and interviews with selected professionals, the research team identified key issues and limitations of AASHTO's current policy on selection of an appropriate ramp design speed and developed a work plan to achieve the research objectives.

The research plan consisted of developing a speed profile model to predict vehicle speeds on ramps. To do so, the research team reviewed several speed prediction models that estimate speeds of vehicles at certain locations along entrance and exit ramps. In addition, the research team collected speed data at several ramps to further refine and calibrate acceleration and deceleration/speed/distance models and curve speed prediction models so that models could be combined to predict speeds along an entire ramp. Priority was given to collecting speed data on ramps of system interchanges and ramps along freeways with posted speeds limits of 75 mph or higher.

While developing the RSPM, the research team also developed an initial set of enhanced guidelines for selecting an appropriate ramp design speed. The research team also identified a few ramps to serve as case studies to apply and evaluate the enhanced guidelines for selecting an appropriate ramp design speed and refined the guidelines accordingly.

The research results are presented in two companion documents. This final report presents the overall research including the literature review, survey, and interviews with selected designers; details about the speed data collected during the study; the statistical analyses to develop speed prediction models for ramps; the general conclusions from the research and future research needs; and recommended changes to the AASHTO *Green Book*. The companion document to this final report presents the enhanced guidelines for selecting an appropriate ramp design speed, details about the RSPM, and case studies that illustrate how to use the guidelines and RSPM to help make design decisions that are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads (Torbic et al., 2021).

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1.4 Outline of Report

This report presents an overview of the work conducted in National Cooperative Highway Research Program (NCHRP) Project 15-56. The remainder of this report is organized as follows:

Section 2—Literature Review and Survey of Practice
Section 3—Observational Field Study
Section 4—Development of RSPM
Section 5—Summary of Design Guidelines
Section 6—Project Summary and Future Research Needs
Section 7—References
Section 8—Abbreviations, Acronyms, Initialisms, and Symbols

Appendix A—Suggested Text for Future Edition of AASHTO *Green Book*

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Section 2.
Literature Review and Survey of Practice

This section summarizes the state of practice and other information related to ramp design speed selection. It is based on a review of published research studies, guidance documents, and design manuals; a survey of engineers, designers, safety professionals, planners, construction managers, operations managers, and others; and interviews with selected professionals.

2.1 Literature Review

The published literature is summarized according to the following topics:

- Current AASHTO policy on ramp design speed.
- Summary of state departments of transportation (DOT) design manuals on ramp design speed.
- Speed prediction models for ramps.
- Interchange ramp design and safety performance.
- Summary of other key resources.

2.1.1 Current AASHTO Policy on Ramp Design Speed

To begin the review of AASHTO's current policy on selecting appropriate ramp design speeds, it is necessary to first review AASHTO's definition of design speed and the basic concepts of design speed most pertinent to ramp design speed. At the start of this project, the 2011 edition of the *Green Book* was the most recent version, so this section refers to the 2011 version of the *Green Book*.

2.1.1.1 Design Speed

Section 2.3.6 of the 2011 *Green Book* states that,

> *"Design speed is a selected speed used to determine the various geometric design features of the roadway. The selected design speed should be a logical one with respect to the anticipated operating speed, topography, the adjacent land use, and the functional classification of the highway. In selection of design speed, every effort should be made to attain a desired combination of safety, mobility, and efficiency within the constraints of environmental quality, economics, aesthetics, and social or political impacts. Once the design speed is selected, all of the pertinent highway features should be related to it to obtain a balanced design. Above-minimum design criteria for specific design elements should be used, where practical, particularly on high-speed facilities. On lower-speed facilities, use of above-minimum design criteria may encourage travel at speeds higher than the design speed. Some design features, such as curvature, superelevation, and sight distance, are directly related to, and vary appreciably with, design speed. Other features, such as widths of lanes and shoulders and clearances to walls and rails, are not directly related to design speed, but they do affect*

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> *vehicle speeds. Thus, when a change is made in design speed, many elements of the highway design will change accordingly."*

AASHTO policy continues to explain that the selected design speed should be consistent with the speeds that drivers are likely to expect on a given highway facility and should fit the travel desires and habits of all drivers expected to use the particular facility. It is also desirable that the running speed of a large proportion of drivers be lower than the design speed.

2.1.1.2 Ramp Design Speed

Referring specifically to guidance on selecting a design speed for ramps, Section 10.9.6 of the 2011 AASHTO *Green Book* states that it is desirable for ramp design speeds to approximate the low-volume running speed on the intersecting highways, but this is not always practical. Thus, lower design speeds may be selected but should not be less than the lower range of speeds shown in *Green Book* Table 10-1. AASHTO policy provides further guidance on selecting appropriate design speed values from *Green Book* Table 10-1 based on various conditions and ramp types.

The *Green Book* also states that the guide values for ramp design speed in *Green Book* Table 10-1 only apply to the sharpest or controlling ramp curve, which is usually on the ramp proper, and that the speed values in *Green Book* Table 10-1 do not pertain to the ramp terminals. The ramp terminals are to be properly transitioned and provided with speed-change facilities adequate for the speed of the highway being considered.

On ramps for right turns, the *Green Book* states that an upper-range value of design speed is often attainable, while a value in the middle range is usually practical. A diagonal ramp of a diamond interchange may also be used for right turns. For diagonal ramps, a middle-range value for the design speed is suggested as practical.

Upper-range values of design speed are typically not attainable on loop ramps, because higher design speeds (above 30 mph) require greater travel distance and larger radii, which require more land often unavailable in urban areas. The *Green Book* advises that minimum design speed values usually control the design of loop ramps, but the design speed should be no less than 25 mph for highway design speeds above 50 mph. Under less restrictive conditions, the design speed and radius of a loop ramp may be increased.

Design speeds between the middle and upper ranges in *Green Book* Table 10-1 are recommended for semidirect connections and are typically in the 30 to 40 mph range. A design speed less than 30 mph should not be used for semidirect connections. For short single-lane semidirect connections, design speeds greater than 50 mph are not practical.

For direct connections, design speeds between the middle and upper ranges in *Green Book* Table 10-1 are recommended, and the preferred minimum design speed is 40 mph.

The *Green Book* states that for intersecting highways with different design speeds, the highway with the greater design speed should be the control for selecting the design speed for the ramp as a whole, with the provision that, as stated above, design speeds apply to certain portions of the ramp more than others. As a result, the portion of the ramp nearer the highway with the lower design speed could also be designed for the lower speed, particularly for ramps on an upgrade from a higher-speed roadway to a lower-speed roadway.

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The *Green Book* also states that *Green Book* Table 10-1 is not applicable to the portion of ramps near at-grade intersections, because stop signs or signals typically control those locations.

The sight distance along a ramp should be at least as great as the design stopping sight distance. Sight distance for passing on a ramp is not needed.

Ramp grades are not directly related to the ramp design speed. However, the ramp design speed is a general indication of the quality of the ramp design. Generally, ramps with higher design speeds should have gradients flatter than ramps with lower design speeds. Table 3 specifies the desirable limiting upgrades for ramps based on the ramp design speeds. Steeper grades may be used where appropriate based on topographic conditions. For downgrades on one-way ramps, the limiting values in Table 3 still apply, but in special cases they may be 2 percent greater.

Table 3. Desirable Limiting Upgrades for Ramps Based on Ramp Design Speeds (based on AASHTO, 2011)

Ramp Design Speed	Desirable Upgrade Limits
45 to 50 mph	3 to 5 percent
40 mph	4 to 6 percent
25 to 30 mph	5 to 7 percent
15 to 25 mph	6 to 8 percent

The three segments of a ramp (exit terminal, ramp proper, and entrance terminal) should be evaluated in combination to determine appropriate design speeds and superelevation rates for the given ramp configuration. For a diamond ramp, the superelevation rate and curve radii should reflect a decreasing sequence of design speeds for the exit terminal, the ramp proper, and the entrance terminal. Loop ramps consist of a moderate-speed exit terminal connecting to a slow-speed ramp proper which connects to a moderate-speed acceleration lane. Direct and semidirect ramps usually consist of a high-speed exit, a moderate- or high-speed ramp proper, and a high-speed entrance.

2.1.1.3 Freeway Mainline Ramp Terminals

AASHTO defines a speed-change lane (SCL) as an auxiliary lane, including tapered areas, used for acceleration or deceleration of vehicles entering or leaving the through traffic lanes. The three terms—SCL, acceleration lane, and deceleration lane—apply broadly to the added pavement joining the traveled way of the highway or street with that of the turning roadway. A SCL should have sufficient length to enable a driver to perform an appropriate change in speed between the highway and the turning roadway in a safe and comfortable manner.

In some situations, AASHTO policy uses a speed other than design speed to determine the various geometric design features. In the case of freeway mainline ramp terminals, minimum acceleration and deceleration lengths are based upon operating speeds along the freeway and the entrance/exit curve of the ramp.

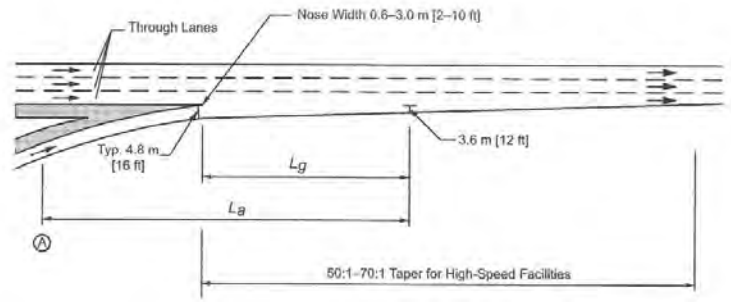
Two general types of SCLs are tapered and parallel designs. The tapered design provides direct entry or exit at a flat angle, while the parallel design has an added lane for changing speed.

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2.1.1.3.1 Entrance Ramp Terminals

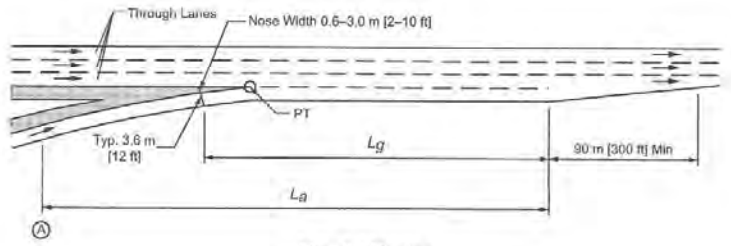
A taper-type entrance, as shown in Figure 2A, merges into the freeway with a long, uniform taper. The geometrics of a taper-type entrance should be designed in a way to allow drivers to attain a speed within 5 mph of the operating speed of the freeway by the time they reach the point where the left edge of the SCL joins the traveled way of the freeway. AASHTO defines this point of convergence as the location where the right edge of the SCL and traveled way are 12-ft apart.

The length required for the vehicle to accelerate to this point of convergence is governed by the speed differential between the operating speed on the entrance curve of the ramp proper and the operating speed of the highway. *Green Book* Table 10-3 (Table 4) shows the minimum lengths of acceleration distances for entrance terminals. *Green Book* Figure 10-69 (Figure 2) shows the minimum lengths for gap acceptance. Adjustment factors in *Green Book* Table 10-4 (Table 5) are used to increase the recommended acceleration lengths for ramps with positive grades and decrease acceleration lengths for ramps with negative grades.



Tapered Design

– A –



Parallel Design

– B –

Notes:

1. L_a is the required acceleration length as shown in Table 10-3 or as adjusted by Table 10-4.
2. Point A controls speed on the ramp. L_a should not start back on the curvature of the ramp unless the radius equals 300 m [100 ft] or more.
3. L_g is the required gap acceptance length. L_g should be a minimum of 150 m [300 ft to 500 ft] depending on the nose width.
4. The value of L_a or L_g, whichever produces the greater distance downstream from where the nose equals 0.6 m [2 ft], is suggested for use in the design of the ramp distance.

Figure 2. Tapered and Parallel Designs for Entrance Ramps (AASHTO, 2011).

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Table 4. Minimum Acceleration Lengths for Entrance Terminals with Flat Grades of Two Percent or Less (AASHTO, 2011)

Acceleration Length, *L* (ft) for Entrance Curve Design Speed (mph)										
Highway		Stop Condition	15	20	25	30	35	40	45	50
Design Speed, *V* (mph)	Speed Reached, V_a (mph)	and Initial speed, V'_a (mph)								
		0	14	18	22	26	30	36	40	44
30	23	180	140	–	–	–	–	–	–	–
35	27	280	220	160	–	–	–	–	–	–
40	31	360	300	270	210	120	–	–	–	–
45	35	560	490	440	380	280	160	–	–	–
50	39	720	660	610	550	450	350	130	–	–
55	43	960	900	810	780	670	550	320	150	–
60	47	1,200	1,140	1,100	1,020	910	800	550	420	180
65	50	1,410	1,350	1,310	1,220	1,120	1000	770	600	370
70	53	1,620	1,560	1,520	1,420	1,350	1230	1,000	820	580
75	55	1,790	1,730	1,630	1,580	1,510	1,420	1,160	1,040	780

NOTE: Uniform 50:1 to 70:1 tapers are recommended where lengths of acceleration lanes exceed 1,300 ft.

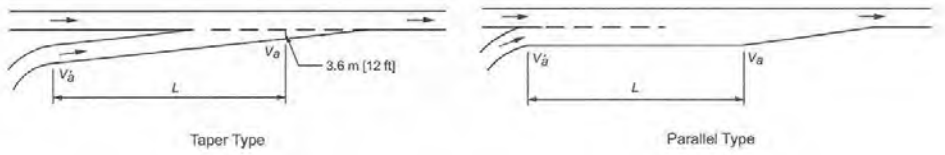


Table 5. SCL Adjustment Factors as a Function of Grade (AASHTO, 2011)

Design Speed of Highway (mph)	Deceleration Lanes				
	Ratio of Length on Grade to Length on Level for Design Speed of Turning Curve (mph)[a]				
All Speeds	3 to 4% upgrade 0.9			3 to 4% downgrade 1.2	
All speeds	5 to 6% upgrade 0.8			5 to 6% downgrade 1.35	
Acceleration Lanes					
Design Speed of Highway (mph)	Ratio of Length on Grade to Length on Level for Design Speed of Turning Curve (mph)[a]				
	20	30	40	50	All Speeds
	3 to 4% Upgrade				3 to 4% Downgrade
40	1.3	1.3	–	–	0.7
45	1.3	1.35	–	–	0.675
50	1.3	1.4	1.4	–	0.65
55	1.35	1.45	1.45	–	0.625
60	1.4	1.5	1.5	1.6	0.6
65	1.45	1.55	1.6	1.7	0.6
70	1.5	1.6	1.7	1.8	0.6
	5 to 6% Upgrade				5 to 6% Downgrade
40	1.5	1.5	–	–	0.6
45	1.5	1.6	–	–	0.575
50	1.5	1.7	1.9	–	0.55
55	1.6	1.8	2.05	–	0.525
60	1.7	1.9	2.2	2.5	0.5
65	1.85	2.05	2.4	2.75	0.5
70	2.0	2.2	2.6	3.0	0.5

[a] Ratio from this table multiplied by length of Table 10-3 or Table 10-5 gives length of SCL on grade.

A parallel-type entrance, as shown in Figure 2B, provides an added lane of sufficient length to enable a vehicle to accelerate to near-freeway speed prior to merging. A taper is provided at the end of the added lane. The length required for the vehicle to accelerate in advance of the merge point is governed by the speed differential between the operating speed on the entrance curve of

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the ramp proper and the operating speed of the highway. Similar to a taper-type entrance, *Green Book* Table 10-3 (Table 4) shows the minimum lengths of acceleration distances for a parallel-type entrance; *Green Book* Figure 10-69 (Figure 2) shows the minimum lengths for gap acceptance; and *Green Book* Table 10-4 (Table 5) provides adjustment factors to increase the recommended acceleration lengths for ramps with positive grades and decrease acceleration lane lengths for ramps with negative grades.

For acceleration lanes, AASHTO policy assumes part of the ramp proper may be considered in the acceleration lane, provided the curve approaching the acceleration lane has a radius of 1,000 ft or more and the motorist on the ramp has an unobstructed view of traffic on the freeway to the left.

2.1.1.3.2 Exit Ramp Terminals

A taper-type exit, as shown in Figure 3A and 3B, provides a direct path to the ramp proper. The deceleration length is measured from the point where the lane is a 12-ft width on the ramp to the first horizontal curve on the exit ramp. *Green Book* Table 10-5 (Table 6) provides recommended minimum deceleration lengths for various combinations of design conditions for both taper- and parallel-type exit terminals. The minimum recommended deceleration lengths are to be adjusted in accordance with *Green Book* Table 10-4 (Table 5) where grades are present.

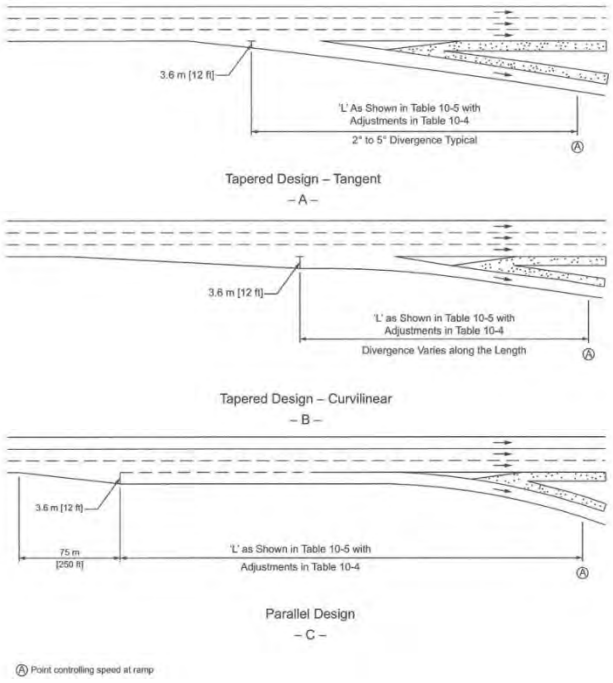
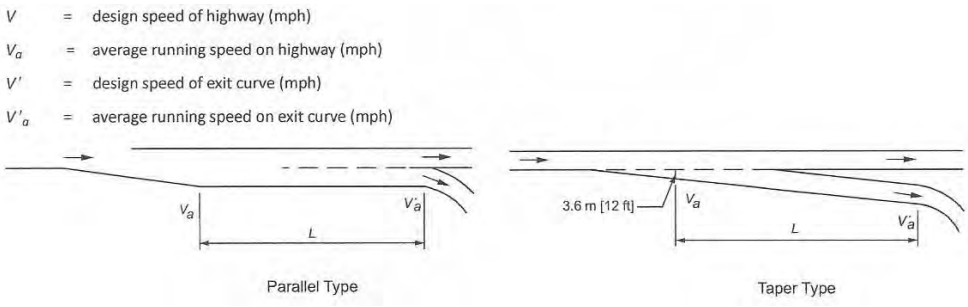


Figure 3. Tapered and Parallel Designs for Exit Ramps (AASHTO, 2011)

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Table 6. Minimum Deceleration Lengths for Exit Terminals with Flat Grades of Two Percent or Less (AASHTO, 2011)

Deceleration Length, *L* (ft) for Design Speed of Exit Curve *V'* (mph)										
Highway Design Speed, *V* (mph)	Speed Reached, V_a (mph)	Stop Condition	15	20	25	30	35	40	45	50
		For Average Running Speed on Exit Curve, V'_a (mph)								
		0	14	18	22	26	30	36	40	44
30	28	235	200	170	140	–	–	–	–	–
35	32	280	250	210	185	150	–	–	–	–
40	36	320	295	265	235	185	155	–	–	–
45	40	385	350	325	295	250	220	–	–	–
50	44	435	405	385	355	315	285	225	175	–
55	48	480	455	440	410	380	350	285	235	–
60	52	530	500	480	460	430	405	350	300	240
65	55	570	540	520	500	470	440	390	340	280
70	58	615	590	570	550	520	490	440	390	340
75	61	660	635	620	600	575	535	490	440	390



A parallel-type exit terminal, as shown in Figure 3C, generally begins with a taper, followed by an added lane parallel to the traveled way. The length of a parallel-type deceleration lane is measured from where the added lane attains a 12-ft width to the point where the alignment of the ramp roadway departs from the alignment of the freeway. Again, *Green Book* Table 10-5 (Table 6) provides recommended minimum deceleration lengths for various combinations of design conditions for both taper- and parallel-type exit terminals, and *Green Book* Table 10-4 (Table 5) provides adjustments to the deceleration lane lengths where grades are present.

2.1.2 Summary of State DOT Design Manuals on Ramp Design Speed

Individual state DOTs sometimes have guidelines that differ from, or add to, the material found in the *Green Book*. To gain an appreciation for the differences that might exist, the research team conducted an online search of state DOT design manuals. Rather than reviewing design manuals for all 50 states, the research team reviewed a sample of design manuals from 20 states, including:

- Arizona
- California
- Colorado
- Florida
- Georgia
- Illinois
- Kansas
- Kentucky
- Maryland
- Massachusetts
- Michigan
- Minnesota
- Missouri
- New Jersey
- North Carolina
- Ohio
- Pennsylvania
- Texas
- Virginia
- Washington

Of these 20 states, 16 had design manuals that contained sections or chapters corresponding to the relevant material in the *Green Book*. Of these 16 states, 11 provided guidance that was

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essentially the same as the *Green Book* and/or specifically referred the reader to the *Green Book*. For the remaining five states, much of their guidance was also very similar to the *Green Book* but contained some unique features. For example, the ramp design speed values presented in the Illinois and Washington design manuals, shown in tables analogous to *Green Book* Table 10-1, are in multiples of 5 mph, while in the *Green Book* the values can be any integer (e.g., the mainline design speed of 55 mph has an upper-range ramp design speed of 50 mph in the state manuals but 48 mph in the *Green Book*). The ramp design speed table from the Washington manual is shown in Table 7; values different from *Green Book* Table 10-1 are shown in red. The Florida manual advises that minimum acceleration/deceleration lengths are provided with a minimum length of taper, but those values correspond to *Green Book* values. The California manual bases the minimum deceleration lengths on the radius of the controlling curve, and the Georgia manual states that ramp design speed should be no less than 10 mph below the design speed of the mainline.

Table 7. Ramp Design Speed from Washington DOT Design Manual Exhibit 1360-4 (WSDOT, 2015)

Main Line Design Speed (mph)		50	55	60	65	70	80
Ramp Design Speed (mph)	Upper Range	45	50	50	55	60	70
	Midrange	30	40	45	45	50	60
	Lower Range	25	25	25	25	25	25

NOTE: Values in red differ from design values in *Green Book* Table 10-1.

2.1.3 Speed Prediction Models for Ramps

Speed prediction models can be used to develop an estimated speed profile of vehicles along the entire ramp or at certain locations along a ramp. Some speed prediction models estimate speeds for passenger vehicles, while others estimate speeds for trucks.

2.1.3.1 Ramp Speed Trends—HSM Methodology

For use in the *Highway Safety Manual* (HSM) (AASHTO, 2010), Bonneson et al. (2012) developed crash prediction methodologies for freeways and interchanges. The methodologies were recently incorporated into the HSM as a supplement (i.e., Chapter 18—Predictive Method for Freeways, and Chapter 19—Predictive Method for Ramps) (AASHTO, 2014) to the original three-volume edition published in 2010 (AASHTO, 2010). The general form of the safety performance function (SPF) for estimating the crash frequency for a ramp is as follows:

$$N = L_r \times exp\,[a + b \times ln(c\,AADT_r) + d\,(c \times AADT_r)] \quad (1)$$

where: N = Crash frequency per year on the ramp.
L_r = Ramp length (mi).
$AADT_r$ = Average annual daily traffic volume on the ramp (veh/day).
a, b, c, d = Regression coefficients.

The SPF uses different regression coefficients for one-lane and two-lane ramps, for fatal-and-injury (FI) and property-damage-only (PDO) crashes, and for multiple-and single-vehicle

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crashes. The crash modification factors (CMFs) developed for use with the SPFs account for the following factors on ramp segments:

- Horizontal curvature
- Lane width
- Right shoulder width
- Right-side barrier
- Left-side barrier
- Lane addition or drop
- Ramp SCL
- Left shoulder width

For horizontal curvature, the base condition is a tangent ramp proper, and the CMF value is a function of the radius of curvature, the average entry speed for the curve, and the proportion of the ramp proper with a curvilinear alignment. The CMF value predicts an increase in crashes as the radius of curvature decreases, the average entry speed increases, and the proportion of the ramp proper with a curvilinear alignment increases.

A curve speed prediction model, used with the horizontal curvature CMF, was based on data from five interchange loop ramp curves and 20 rural two-lane highway curves. The speed profile models included in the HSM methodology are applied in the direction of travel and account for the variables listed in Table 8. The speed profile models are implemented in the spreadsheet-based Enhanced Interchange Safety Analysis Tool (ISATe). Bonneson et al. note that these speed models were not developed for the purpose of predicting vehicle speeds in the context of operational or design analyses.

Table 8. Input Data for Ramp Curve Speed Prediction Procedures in ISATe (Bonneson et al., 2012)

Variable	Description	Default Value	Applicable Procedure
X_i	Milepost of the point of change from tangent to curve (PC) for curve i[1], mi	None	All
R_i	Radius of curve i[2], ft	None	All
$L_{C,i}$	Length of horizontal curve i, mi	None	All
V_{Frwy}	Average traffic speed on freeway during off-peak periods of the typical day, mph	Estimate is equal to the speed limit	All
V_{xroad}	Average speed at point where ramp connects to crossroad, mph	15—ramps with stop-, yield-, or signal-controlled crossroad ramp terminals 30—all other ramps at service interchanges	Entrance ramp, exit ramp, connector ramp at service interchange

NOTES:

[1] If the curve is preceded by a spiral transition, then X_i *is* the average of the TS and SC mileposts, where TS is the milepost of the point of change from tangent to spiral and SC is the milepost of the point of change from spiral to curve.

[2] If the curve has spiral transitions, then R_i is equal to the radius of the central circular portion of the curve.

When applied, the speed profile models included in the HSM methodology yield average entry and exit speeds for each curve on a ramp. Separate seven-step procedures were developed for entrance and exit ramps and are described in the following subsections.

Entrance Ramp Procedure

Step 1—Gather Input Data: The input data for this procedure are identified in Table 8.

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Step 2—Compute Limiting Curve Speed: The limiting curve speed is computed for each curve on the ramp using Equation 2.

$$v_{max,i} = 3.24\ (32.2\ R_i)^{0.30} \quad (2)$$

where, $v_{max,i}$ equals the limiting speed for curve *i*, ft/s.

The analysis proceeds in the direction of travel with the first curve encountered on the ramp designated as Curve 1 (*i*=1). The value of v_{max} is computed for all curves prior to, and including, the curve of interest. The value obtained from Equation 2 is an upper limit on the curve speed. Vehicles may reach this speed if the distance between curves is long enough or the crossroad speed is high.

Step 3—Calculate Curve 1 Entry Speed: The average entry speed at Curve 1 is computed using Equation 3.

$$v_{ent,1} = ([1.47\ V_{XRoad}]^3 + 495 \times 5280\ X_1)^{1/3}\ ;\ \leq 1.47\ V_{Frwy} \quad (3)$$

where, $v_{ent,1}$ equals the average entry speed for Curve 1, ft/s.

The boundary condition of Equation 3 indicates the value computed ($v_{ent,1}$) cannot exceed the average freeway speed (V_{Frwy}).

Step 4—Calculate Curve 1 Exit Speed: The average exit speed at Curve 1 is computed using Equation 4.

$$v_{ext,1} = (V^3_{ent,1} + 495 \times 5280\ L_{C,1})^{1/3}\ ;\ \leq v_{max,1} \text{ and} \leq 1.47\ V_{Frwy} \quad (4)$$

where, $v_{ext,1}$ equals the average exit speed for Curve 1, ft/s.

The boundary conditions of Equation 4 indicate the value computed ($v_{ext,1}$) should not exceed the limiting curve speed ($v_{max,i}$) or the average freeway speed (V_{Frwy}).

Step 5—Calculate Curve *i* Entry Speed: The average entry speed at Curve 2 (and all subsequent curves) is computed using Equation 5.

$$v_{ent,i} = (\ V^3_{ext,i-1} + 495 \times 5280\ [X_i - X_{i-1} - L_{C,i-1}]\ )^{1/3}\ ;\ \leq 1.47\ V_{Frwy} \quad (5)$$

where, $v_{ent,i}$ equals the average entry speed for curve *i* (*i* = 2, 3, ...), ft/s; and $v_{ext,i}$ equals the average exit speed for curve *i*, ft/s.

Step 6—Calculate Curve *i* Exit Speed: The average exit speed at Curve 2 (and all subsequent curves) is computed using Equation 6.

$$v_{ext,i} = (V^3_{ent,i} + 495 \times 5280\ L_{C,i})^{1/3}\ ;\ \leq v_{max,i} \text{ and} \leq 1.47\ V_{Frwy} \quad (6)$$

Step 7—Calculate Speed on Successive Curves: The entry and exit speeds for Curve 3 and successive curves are computed by applying Steps 5 and 6 for each curve.

Exit Ramp Procedure

Step 1—Gather Input Data: The input data for this procedure are identified in Table 8.

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Step 2—Compute Limiting Curve Speed: This step is the same as Step 2 for the entrance ramp procedure. A lower curve speed than that obtained from Equation 2 is possible as deceleration may occur along the ramp as the driver transitions from the freeway speed to the crossroad speed.

Step 3—Calculate Curve 1 Entry Speed: The average entry speed at Curve 1 is computed using Equation 7.

$$v_{ent,1} = 1.47\ V_{Frwy} - 0.034 \times 5280\ X_1 \quad ;\ \geq 1.47\ V_{XRoad} \tag{7}$$

The boundary condition of Equation 7 indicates the value computed ($v_{ent,1}$) cannot be less than the average speed at the point where the ramp connects to the crossroad (V_{XRoad}).

Step 4—Calculate Curve 1 Exit Speed: The average exit speed at Curve 1 is computed using Equation 8.

$$v_{ext,1} = v_{ent,1} - 0.034 \times 5280\ L_{C,1} \quad ;\ \leq v_{max,1} \text{ and } \geq 1.47\ V_{XRoad} \tag{8}$$

The boundary conditions of Equation 8 indicate the value computed ($v_{ent,1}$) should not exceed the limiting curve speed ($v_{max,i}$) and should not be less than the average speed at the point where the ramp connects to the crossroad (V_{XRoad}).

Step 5—Calculate Curve i Entry Speed: The average entry speed at Curve 2 (and all subsequent curves) is computed using Equation 9.

$$v_{ent,i} = v_{ext,i-1} - 0.034 \times 5280\ (X_1 - X_{i-1} - L_{C,i-1}) \quad ;\ \geq 1.47\ V_{XRoad} \tag{9}$$

Step 6—Calculate Curve *i* Exit Speed: The average exit speed at Curve 2 (and all subsequent curves) is computed using Equation 10.

$$v_{ext,i} = v_{ent,i} - 0.034 \times 5280\ L_{C,i} \quad ;\ \leq v_{max,i} \text{ and } \geq 1.47\ V_{XRoad} \tag{10}$$

Step 7—Calculate Speed on Successive Curves: The entry and exit speeds for Curve 3 and successive curves are computed by applying Steps 5 and 6 for each curve.

2.1.3.2 Ramp Speed Trends—Other Sources

A focused analysis of vehicle speeds on loop ramps was conducted by Torbic et al. (2016). Field data showed that the HSM methodology models tended to overestimate vehicle speeds on the controlling curves (i.e., sharpest curves) of loop ramps by the following magnitudes:

- Entrance ramp: 2.6 mph at the midpoint, 1.8 mph at the PT (PT is the point of tangency—the ending point of the horizontal curve).
- Exit ramp: 10.6 mph at the PC (PC is the point of curve—the beginning point of the horizontal curve), 2.2 mph at the midpoint.

The aforementioned analysis was based on 15 entrance ramp sites and 13 exit ramp sites. The following models were developed to provide more accurate estimates of ramp speeds:

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$$v_{ent,c,MC} = 8.359 + 1.978\,I_{l2} + 0.040\,R + 0.313\,W_l + 0.912\,W_{os} + 0.682\,W_{is} - 4.333\,I_{tk} \quad (11)$$

$$v_{ent,c,PT} = 16.276 + 1.444\,I_{l2} + 0.054\,R + 1.079\,W_{os} - 4.051\,I_{tk} \quad (12)$$

$$v_{ext,c,PC} = 17.515 + 0.090\,R - 5.967\,I_{tk} \quad (13)$$

$$v_{ext,c,MC} = 9.512 + 1.241 I_{l2-3} + 0.053R + 1.008W_{os} - 4.873I_{tk} + 3.551I_{rs} + 2.911I_d + 3.975I_p + 4.334I_w \quad (14)$$

where:

- $v_{ent,c,MC}$ = Average passenger car speed at the midpoint of the entrance ramp controlling curve, mph.
- $v_{ent,c,PT}$ = Average passenger car speed at the PT of the entrance ramp controlling curve, mph.
- $v_{ext,c,PC}$ = Average passenger car speed at the PC of the exit ramp controlling curve, mph.
- $v_{ext,c,MC}$ = Average passenger car speed at the midpoint of the exit ramp controlling curve, mph.
- I_{l2} = Indicator variable for lane 2 (= 1 if predicting speed in the outside lane, 0 otherwise).
- $I_{l2\text{-}3}$ = Indicator variable for lanes 2 and 3 (= 1 if predicting speed in the middle or outside lanes, 0 otherwise).
- R = Radius (measured to the inside of the traveled way), ft.
- W_l = Lane width, ft.
- W_{os} = Outside (left) shoulder width, ft.
- W_{is} = Inside (right) shoulder width, ft.
- I_{tk} = Indicator variable for trucks (= 1 if predicting truck speed, 0 otherwise).
- I_{rs} = Indicator variable for curve radius type (= 1 if simple, 0 if compound).
- I_d = Indicator variable for drop SCL (= 1 if present, 0 otherwise).
- I_p = Indicator variable for parallel SCL (= 1 if present, 0 otherwise).
- I_w = Indicator variable for weaving SCL (= 1 if present, 0 otherwise).

A RSPM was developed by Venglar et al. (2009) for the purpose of setting exit ramp advisory speeds. Their model is described as follows:

$$v_c = -20.872 - 0.758DC + 9.864\ln(Z) \quad (15)$$

where:

- v_c = Average passenger car speed, mph.
- DC = Degree of horizontal curvature.
- Z = Distance to the first downstream at-grade signalized or stop-controlled intersection, ft.

Venglar et al. (2009) suggested applying a multiplier of 0.95 to Equation 15 to estimate the average truck speed, and then using the average truck speed to set the advisory speed on an exit ramp.

Studies have shown that speed consistency is more important than speed alone in influencing road safety. Guo et al. (2011) compared speed differentials to evaluate the level of safety at

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freeway exits. In the process of this evaluation, Guo et al. developed speed prediction models and speed differential models along different segments of a freeway exit, and suggested using the computed speeds and differentials as a method of assessing design and safety in a manner similar to the design consistency framework suggested by Lamm et al. (1988). The freeway exit was divided into five segments: freeway mainline, transition zone, deceleration lane, upstream ramp curve, and downstream ramp curve. The following models were proposed to predict speeds at different areas of freeway exit ramps:

$$v_{TZ} = \frac{v_{FW}}{(1+9.6\%)} \tag{16}$$

$$v_{85,DL} = 38.901 + 2.021\, v_{85,FW} - 0.041\, L_{DL} + 0.073\, SL_{RP} \tag{17}$$

$$v_{85,UR} = 35.232 - \frac{2674.210}{R_1} \tag{18}$$

$$v_{85,LR} = 33.279 - \frac{2394.180}{R_2} \tag{19}$$

where:

- v_{TZ} = Point speeds in the transition zone, mph.
- v_{FW} = Point speeds on the freeway mainline, mph.
- $v_{85,DL}$ = 85th-percentile operating speeds on the deceleration lane, mph.
- $v_{85,FW}$ = 85th-percentile operating speeds on the freeway mainline, mph.
- $v_{85,UR}$ = 85th-percentile operating speeds on the upstream ramp curve, mph.
- $v_{85,LR}$ = 85th-percentile operating speeds on the downstream ramp curve, mph.
- L_{DL} = Length of deceleration lane, ft.
- SL_{RP} = Speed limit on the ramp, mph.
- R_1 = Radius of the upstream ramp curve, ft.
- R_2 = Radius of the downstream ramp curve, ft.

Guo et al. also developed models for estimating the operating speed differences between the segments on the freeway exit; these are expressed as follows:

$$85\Delta V_{FW-DL} = 38.252 - 1.032\, v_{85,FW} + 0.046\, L_{DL} \tag{20}$$

$$85\Delta V_{DL-UR} = 14.957 - 0.002\, R_1 - 1.193\, D_1 \tag{21}$$

$$85\Delta V_{UR-LR} = 3.162 + 3217.475 \left| \frac{1}{R_1} - \frac{1}{R_2} \right| \tag{22}$$

where:

- $85\Delta V_{FW\text{-}DL}$ = 85th-percentile speed difference between freeway mainline and the deceleration lane, mph.
- $85\Delta V_{DL\text{-}UR}$ = 85th-percentile speed difference between deceleration lane and the upstream ramp curve, mph.
- $85\Delta V_{UR\text{-}LR}$ = 85th-percentile individual speed difference between upstream ramp curve and the downstream ramp curve, mph.
- D_1 = Deflection angle of the upstream ramp portion, degrees.

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Xia et al. (2012) conducted a partial correlation analysis of vehicle operating speeds and off-ramp geometric parameters under free-flow conditions and concluded that the circular curve radius, superelevation, and longitudinal gradient are the most influential variables. Speed prediction models were developed based on data collected at 12 exit ramp sites in China. The following models were proposed to predict operating speeds for passenger cars and trucks at the PC and the midpoint of the exit ramp controlling curve:

$$v_{c,PC} = 9.342 \ln R + 2.805 \quad (23)$$

$$v_{t,PC} = 7.872 \ln R + 0.976 \quad (24)$$

$$v_{c,MC} = 7.673 \ln R - 105.631 \mu + 5.836 \quad (25)$$

$$v_{t,MC} = 7.119 \ln R - 42.609 \mu + 2.419 \quad (26)$$

where:

- $v_{c,PC}$ = Average passenger car speed at the PC of the exit ramp controlling curve, mph.
- $v_{t,PC}$ = Average truck speed at the PC of the exit ramp controlling curve, mph.
- $v_{c,MC}$ = Average passenger car speed at the midpoint of the exit ramp controlling curve, mph.
- $v_{t,MC}$ = Average truck speed at the midpoint of the exit ramp controlling curve, mph;
- R = Radius, ft.
- μ = Longitudinal gradient, %.

Gattis et al. (2010) conducted an analysis of truck speeds and acceleration rates as trucks departed weigh stations to re-enter the highway mainline. The dataset included hundreds of trucks across five sites; each site was a ramp connecting a weigh station to a highway mainline. The sites represented a range of grades and ramp lengths. Gattis et al. categorized sites as downhill, level, or uphill and each truck as impeded or unimpeded. A truck was considered impeded if its headway with the preceding truck on the ramp was less than 7 seconds or if its headway with the leading or trailing vehicle on the highway mainline was less than 3 seconds. They developed six models to estimate the average truck speed including all trucks (impeded and unimpeded) for each grade category (downhill, level, or uphill) and unimpeded trucks for each grade category. They developed an additional set of six models using the same combinations of grade and trucks to estimate the 10th-percentile truck speed. Their model for the average speed of unimpeded trucks at a level site is as follows:

$$v_{a,u,l} = -22.272 + 0.0169d - 1.8x10^{-6}d^2 \quad (27)$$

where:

- $v_{a,u,l}$ = Average speed of unimpeded trucks at a level site, mph.
- d = Distance from end of scale, ft.

Based on their model trends, Gattis et al. observed that accommodating truck acceleration will generally result in substantially longer entrance ramps than what are typically provided.

In a detailed analysis of vehicle speeds at freeway entrances during off-peak periods, Ahammed et al. (2008) developed several models to estimate entering vehicle speeds at the merge point.

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Their dataset included speed and acceleration profiles of hundreds of vehicles at 16 freeway entrances in Ottawa, Canada, where the SCL ended before reaching the next downstream exit ramp. They developed the following models to predict the 85th-percentile speeds of merging (entering) vehicles and vehicles in the right lane of the freeway mainline:

$$v_{85Merge} = 42.348 + \frac{34.809}{\theta} + 0.013L - 0.0005\,AADT_{SCL} \quad (28)$$

$$v_{85Merge} = 42.523 + \frac{34.447}{\theta} + 0.013L - 0.007\,Q_{SCL} \quad (29)$$

$$v_{85RL} = 56.512 - 0.009Q_{RL} + 0.324v_{85Merge} \quad (30)$$

where:

$v_{85Merge}$ = 85th-percentile speed of merging vehicles, mph.
v_{85RL} = 85th-percentile speed of vehicles in the rightmost lane of the freeway mainline, mph.
θ = Angle of convergence between entrance ramp and mainline, degrees.
L = Entrance ramp length, ft.
$AADT_{SCL}$ = Annual average daily traffic volume on entrance ramp SCL, veh/day.
Q_{SCL} = Equivalent hourly traffic volume on entrance ramp SCL, pc/h.
Q_{RL} = Equivalent hourly traffic volume in rightmost freeway mainline lane, pc/h.

Based on these models, Ahammed et al. observed that entering vehicles will merge onto the freeway mainline at a higher speed if a longer SCL or smaller angle of convergence is provided. Vehicle speeds will decrease as ramp volume increases.

Additionally, they calibrated the following model to predict the 85th-percentile passenger car merging distance:

$$D_{85Merge} = 261.28 + 0.030L^2 \quad (31)$$

where:

$D_{85Merge}$ = 85th-percentile passenger car merging distance, ft.

Based on this model, Ahammed et al. observed that increasing the SCL length will not necessarily result in improved merging operations, as drivers do not necessarily use all of the added length. They suggested that it is more effective to improve merging operations by selecting ramp geometry that will result in higher merging speeds. Based on their findings and those of others, the key geometric variables affecting merging speed are controlling curve radius and distance between the controlling curve and the entrance SCL.

2.1.4 Interchange Ramp Design and Safety Performance

Several researchers investigated the relationship of geometric design elements and safety performance of ramps. Several relevant findings are as follows:

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- Exit ramps have higher crash rates than entrance ramps (Twomey et al., 1993; Khaoshadi, 1998).
- Ramps show increasing crash rates with increasing degrees of curvature (Twomey et al., 1993).
- When comparing four types of exit ramps (Type 1: parallel from a tangent single-lane exit ramp; Type 2: single-lane exit ramp without a taper; Type 3: two-lane exit ramp with an optional lane; and Type 4: two-lane exit ramp without an optional lane), the Type 1 exit ramp had the best safety performance in terms of lowest crash frequency and crash rate (Lu et al., 2010).

Key geometric variables and site characteristics that influence the safety performance of interchange ramps and SCLs include (Bauer and Harwood, 1998; Khaoshadi, 1998; Yi and Mulinazzi, 2007):

- Freeway volume.
- Ramp volume.
- Speed of right freeway lane.
- Area type (urban or rural).
- Ramp type (entrance or exit ramp).
- Ramp configuration.
- Length of ramp.
- Length of SCL.

Several recent studies developed safety prediction models related to freeways and ramps. Bonneson and Pratt (2009) examined freeway safety trends using Texas data and developed safety prediction models for freeway segments, including CMFs that account for the presence of ramp entrances and weaving sections. The ramp entrance CMF is described by the following equation:

$$CMF_{enr|agg} = (1 - P_{enr}) + P_{enr} e^{1529/I_{enr}} \quad (32)$$

where:

$CMF_{enr|agg}$ = Aggregated ramp entrance CMF.
P_{enr} = Proportion of freeway segment length adjacent to a ramp entrance.
I_{enr} = Average ramp entrance length (ft).

Equation 32 is applied to a freeway segment that may have one or more ramp entrances along its length, and it yields an aggregated CMF value based on all ramp entrances on the segment. The CMF is illustrated in Figure 4. As shown, the presence of a ramp entrance increases the crash frequency along a freeway segment, and the magnitude of the increase is greater if the ramp entrance length is shorter. A similar CMF was developed for weaving sections.

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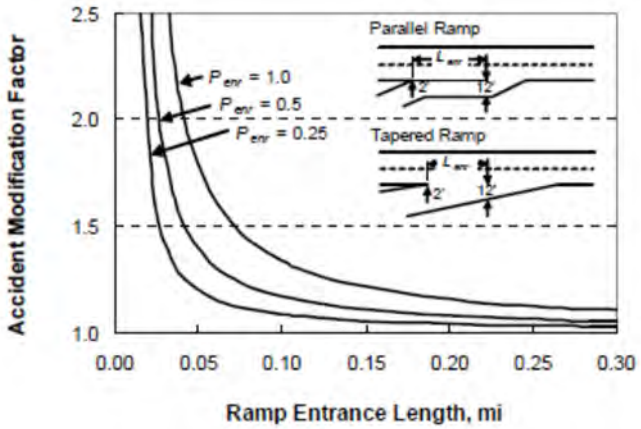


Figure 4. Aggregated Ramp Entrance CMF (Bonneson and Pratt, 2009)

Bonneson et al. (2012) developed safety prediction models for both freeway mainline segments and ramp segments and developed chapters for inclusion in the HSM. The supplemental material for the HSM contains several CMFs for freeway mainline segments relevant to the issue of selecting ramp design speed (AASHTO, 2014). These CMFs include:

- Lane-change maneuvers, accounting for the following ramp entrance and exit characteristics:
  - Presence of, and distance to, upstream ramp entrances and downstream ramp exits.
  - Ramp entrance or exit length.
  - Ramp entrance or exit side (connects to right or left side of mainline).
- Type B weaving sections (which allow one weaving movement to be completed without a lane change but require at least one lane change for the other weaving movement).

Even more relevant to the issue of selecting ramp design speed are the safety prediction models for ramp segments. These models include CMFs for the following ramp characteristics:

- Horizontal curvature.
- Lane width.
- Shoulder width.
- Longitudinal barrier presence.
- Weaving section presence (e.g., on collector-distributor roads within cloverleaf interchanges).
- Speed-change lane presence (at locations where ramps join freeway mainlines or crossroads).
- Lane add or drop presence.

The crash prediction methodology involves describing the characteristics of homogeneous ramp segments and estimating the crash frequency on those segments. This approach is more detailed than approaches in earlier sources, which often involved simply applying base crash rates for entire ramps based on their configuration (e.g., diamond, free-flow loop, non-free-flow loop, connector). In particular, the horizontal curvature CMF for ramp segments is influenced by vehicle speeds as depicted in Figure 5. The crash prediction methodology includes basic procedures for estimating vehicle speeds at different locations along ramps.

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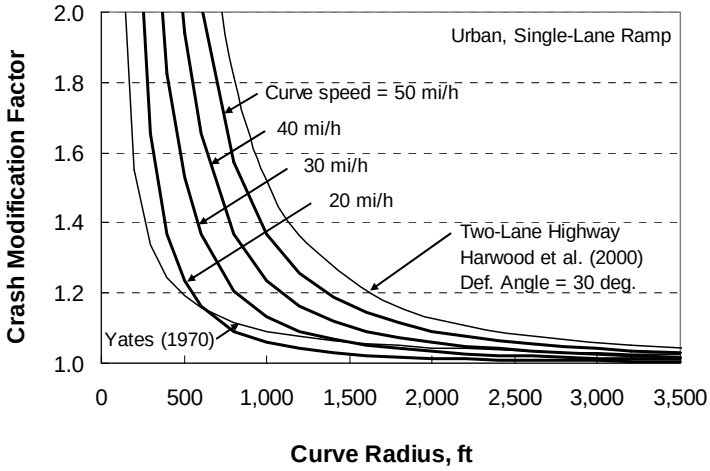


Figure 5. Ramp Horizontal Curve CMF for FI Crashes (Bonneson et al., 2012)

In other research, Torbic et al. (2016) assessed how well the crash prediction models for ramps represent the safety performance of two ramp types with distinctly different geometrics: loop ramps and diagonal ramps. The HSM crash prediction procedures were applied to 235 loop ramps and 243 diagonal ramps in two-states – California and Washington – and the results were compared to five years of actual crash data for the same ramps. The results indicate that the HSM crash prediction can be applied to both loop and diagonal ramps, but to properly compare the safety performance of these two ramp types, separate calibration of loop and diagonal ramps is recommended.

2.1.5 Summary of Other Key Resources

This section summarizes other key resources not discussed in the previous sections.

2.1.5.1 FHWA Ramp Design Speed Research

In the late 1990s the Federal Highway Administration (FWHA) sponsored research on ramp design speed which resulted in multiple published reports and papers by Hunter and Machemehl (1997), Hunter et al. (2000), and Hunter et al. (2001). The objectives of the research were to (1) evaluate through literature review and analysis current ramp design speed criteria, (2) evaluate current ramp design speed criteria through a carefully designed sample of ramp operational data, and (3) provide evidence to validate current ramp design speed policy or to modify current procedures. The authors note that evaluation of entry ramp design speed criteria requires examination of assumptions regarding vehicle acceleration and deceleration rates and gap seeking and acceptance behavior, and should include examination of freeway driver activity and ramp driver actions. Traffic counts were collected; individual vehicles were tracked along merging areas; and times and locations where ramp vehicles merge into the freeway were recorded. The relationships between entry ramp geometric design features, speed, and other operational characteristics were investigated by comparing characteristics of ramps having "good" versus "poor" geometrics. The primary findings and recommendations from the research are as follows:

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1. Average ramp speeds on all observed entry ramps are consistently greater than 50 percent of the freeway design speed, even where the freeway design speed is 70 mph. Designing facilities for speeds lower than the speeds at which drivers typically operate would be inappropriate. Thus, the design criterion allowing a ramp design speed to be 50 percent of the freeway design speed should be deleted from AASHTO and TxDOT policy.
2. Speed-distance data on ramps with adequate sight distance and SCL lengths exhibit smooth profiles, having no abrupt speed changes. Speed-distance data for ramp-freeway facilities with vertical profiles limiting sight distance on ramps and marginal SCL lengths exhibit undulating waveforms, indicating significant speed changes on ramps.
3. Acceleration/deceleration rates for ramps with "good" and "poor" geometrics have patterns similar to speed histories. Ramps having adequate sight distance and SCL lengths produce small positive acceleration rates (0 to 2 mphps). Ramps with inadequate sight distance and/or inadequate acceleration lane lengths produce larger values of positive and negative acceleration (−4 to 4 mphps). Thus, the AASHTO acceleration rate model used to estimate acceleration lane lengths should not be changed.
4. Freeway right-lane speeds are not largely affected by ramp vehicles if the ramp has adequate sight distance and SCL lengths. Inadequate sight distance and/or acceleration lane lengths tend to cause significant reductions in freeway right-lane speeds, particularly under high freeway and ramp traffic volumes.
5. Freeway right-lane time headways tend to be influenced not by complex ramp design but by traffic volume.
6. Freeway right-lane volume is the factor that primarily seems to influence the size of the time gap accepted by merging ramp drivers.
7. During high traffic volume conditions, most ramp drivers travel to the end of a ramp with adequate sight distance and SCL length before merging. If a ramp has inadequate sight distance and/or acceleration lane length, this trend is much less pronounced, with some drivers aggressively merging at different locations to avoid being trapped at the end of the SCL.
8. Drivers tend to begin the merge acceleration process only after gaining a clear view of freeway right-lane traffic. If the ramp driver's view is obstructed, acceleration does not begin until near the ramp gore where the view becomes unobstructed. Thus, acceleration lengths for taper-type entry ramps should include only the portions from which ramp drivers can clearly view the freeway right-lane traffic. AASHTO policy should clearly include this specification.

The authors also noted that driver behavior upstream and downstream of the ramp gore may exhibit different characteristics. A policy that allows flexibility of choosing separate design speeds upstream and downstream of the gore may provide for overall superior designs. For example, the current 50 percent criterion produces a very desirable long acceleration lane but permits speed limiting horizontal and vertical alignment elements upstream of the ramp gore. Provision of a high design speed for upstream features and a low design speed for downstream features may provide an optimal design standard. Such a change in design philosophy should be considered.

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2.1.5.2 ITE Freeway and Interchange Geometric Design Handbook

This handbook presents fundamental concepts and practices related to freeway and interchange geometric design, often referencing the AASHTO *Green Book* (Leisch, 2005). This section summarizes information from the handbook relevant to the selection of ramp design speeds.

2.1.5.2.1 Ramp Design Speed

The design speed of ramps is related to the design speed of the intersecting highways to which the ramps connect. Drivers anticipate some speed reduction when exiting a highway. From a practical perspective, there is a need to limit travel distance and speed on ramps and to maintain a degree of compactness of the interchange, especially in urban areas. At the upper limit, ramp design speeds equivalent to 80 to 85 percent of design speeds on intersecting highways are considered appropriate. At the lower limit, ramp design speed equivalent to about 50 percent of the design speeds on intersection roadways are considered appropriate.

AASHTO's guide values for ramp design speeds as related to highway design speed are provided in Table 1 of this report. The higher ramp design speed allows the driver to exit the highway with little effort and close to the average running speed on that highway, while the lower ramp design speed requires deceleration by the exiting driver, accomplished through appropriate speed transition. For the entering driver, a proper ramp acceleration transition allows the merge to occur at the approximate speed of traffic on the highway.

Ramp design speed depends largely upon the intersecting highways, the area types (urban or rural), and the form of interchange. Site controls and topography also have an effect.

Direct connections that involve right-hand ramps for all system interchanges and some service interchanges, as Parclo-Bs and cloverleafs, call for design speeds in the upper part of the range. For system interchanges, ramp design speeds are usually 40 to 60 mph, and for service interchanges, ramp design speeds are usually 30 to 50 mph.

Semidirect connections for major left-turn movements are designed close to the upper range of design speeds. Because of reverse-curve constraints, design speeds are generally limited to about 50 mph and may be as low as 40 mph in urban areas with constrained conditions.

Loop ramps serve as the primary ramp configuration to accommodate left-turn movements for both service and system interchanges. Considering their spatial influence upon other ramps within the interchange, loop ramp configurations limit curve radii and ramp design speeds. Operational experience, including the aspect of extra travel distance, indicates that the practical radii for loop ramps generally range between 150 and 250 ft; approximately equivalent to design speeds of 25 to 30 mph. Table 9 presents several operational measures related to the effect of differences in loop size for curve radii ranging from 100 to 250 ft.

Table 9. Loop Ramp Characteristics (Leisch, 2005)

Radius (ft)	100	150	250
Design speed (mph)	20	25	30
Travel distance (ft)	630	950	1,600
Travel time (sec)	22	26	37

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The ramp design speeds given in Table 1 as a function of the mainline highway design speed are applicable to ramps on arterials and collectors, as well as freeway ramps. For arterials and collectors, the design speed of the arterial or collector is used as the highway design speed.

Recall the 2001 *Green Book* provided two sets of design criteria for horizontal curves: high-speed and low-speed design criteria. The low-speed design criteria for horizontal curves are based on higher net lateral acceleration values than the high-speed criteria, not to be confused with the "upper, middle, and lower ranges" of ramp design speeds. Table 10 illustrates that the choice between horizontal curve design criteria based on AASHTO's earlier high-speed and low-speed classifications for a specific ramp should be based on the highway functional classification and design speed.

Table 10. Horizontal Curve Design Criteria for Ramps as a Function of Design Speed and Functional Classification of the Mainline Highway (adapted from Leisch, 2005)

Functional Classification	Highway Design Speed (mph)								
	30	35	40	45	50	55	60	65	70
Freeway or expressway	Not applicable				HSD				
Arterial	LSD	HSD	HSD	HSD					
Collector	LSD	LSD	HSD	HSD					

HSD (high-speed design criteria for horizontal curves).
LSD (low-speed design criteria for horizontal curves).

2.1.5.2.2 Speed-Curve Relationship

Curvature of ramps is predicated on design speed, superelevation, and an acceptable side friction factor. On ramps and turns at intersections, drivers tend to operate at higher speeds in relation to the curve radius than on open highway curves largely due to the acceptance by drivers of a higher side friction (i.e., net lateral acceleration) on turning roadways than on highways for through travel.

It has been the practice to establish a single minimum radius for each design speed for ramps. This is logical since there is a range of design speeds that may be designated on ramps for a given highway approach. It would be an undue refinement to vary the controlling curvature by a relatively small amount for a selected ramp design speed due to possible use of differing superelevation rates. In deriving the controlling curvatures, rates of superelevation have been assumed that can be achieved in design, ranging from zero at crown slopes and at-grade ramp terminals to 0.10 for ramps at higher design speeds. The superelevation rates also reflect the condition that the sharper the curve, the shorter the length available for developing superelevation. The side friction factor varies from 0.38 for sharp turns to 0.15 at relatively high-speed ramps.

The minimum radii for turning roadways for various design speeds are provided in Table 11. These are normally applied to the inner edge of the traveled way.

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Table 11. Minimum Radii for Interchange Ramp Curves (Leisch, 2005)

Design (turning) speed, V (mph)	10	15	20	25	30	35	40	45
Side friction factor, f	0.38	0.32	0.27	0.23	0.20	0.18	0.16	0.15
Assumed minimum superelevation, e/100	0.00	0.00	0.02	0.04	0.06	0.08	0.09	0.10
Total e/100 + f	0.38	0.32	0.29	0.27	0.26	0.26	0.25	0.25
Calculated minimum radius, R (ft)	18	47	92	154	231	314	426	540
Suggested minimum radius curve for design (ft)	25	50	90	150	230	310	430	540
Average running speed (mph)	10	14	18	22	26	30	34	36

NOTE: For design speeds greater than 45 mph, use values for open highway conditions.

On the controlling curve at each design speed, the ramp should be fully superelevated if feasible. Ramps should be superelevated at no more than 0.08, particularly on long curves such as jughandles and loops.

Ramp design speeds are often dictated by physical design constraints. The most critical design conditions for horizontal curves occur at lower design speeds, particularly at design speeds of 20 to 30 mph. Under worst-case conditions, a truck on a curve with a ramp design speed of 20 mph can roll over when traveling at 25 mph and may skid off the road under wet pavement conditions at 27 mph. Similarly, a truck on a curve with a ramp design speed of 30 mph can rollover at 38 mph and skid off the roadway at 30 mph. Passenger vehicle operations are less critical than for trucks and higher design speeds are less critical than lower design speeds.

The following guidelines should be considered when selecting the design speed for an exit ramp.

1. Consider physical and economic constraints when selecting a tentative design speed for the ramp. Use the upper or middle range of ramp design speeds if possible. Avoid the lower range of ramp design speeds on ramps that will carry substantial truck traffic.
2. Identify the most critical curve on the ramp (usually, but not necessarily, the first curve downstream of the gore).
3. Develop a forecast of operating speeds at the most critical curve on the ramp based on actual speeds on existing ramps with similar mainline design speeds, mainline operating speeds, and similar geometrics for the SCL and portion of the ramp prior to the most critical curve. This forecast should be based on mainline design and operating speeds, but not on ramp design speed.
4. If the forecast ramp operating speed exceeds the design speed, raise the design speed. If the ramp design speed cannot be raised due to physical, environmental, or economic constraints, consider implementing speed-control measures.

For exit ramps where the anticipated operating speeds exceed the maximum feasible ramp design speed, the following speed-control measures should be considered:

1. Provide signing with an appropriate advisory speed for the ramp.
2. Place the advisory speed signing so that drivers have sufficient time to slow down before the most critical curve.
3. Increase the length of the deceleration lane.
4. Realign the ramp to increase the distance from the gore to the most critical curve.

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5. Supplement standard advisory speed signs to make the signing more conspicuous, increase the distance from the signing to the most critical curve, and draw attention of truck drivers to the signing. This can be accomplished by:
   a. Using more than one ramp speed advisory sign.
   b. Placing ramp speed advisory signing on the mainline highway in advance of the ramp.
   c. Incorporating an exit speed panel on the guide sign for the exit ramp.
   d. Using overhead signing.
   e. Using a TRUCK SPEED advisory sign.
   f. Using flashing beacons to call attention to the advisory speed.
6. Avoid designs where the critical curve on a ramp is not obvious (e.g., where a "tight" horizontal curve follows a larger-radius curve).
7. Consider the use of collector-distributor roads in the interchange as they may reduce ramp speeds.

2.1.5.2.3 Transitions and Compound Curves

Interchanges involve curvilinear alignment with relatively rapid changes in direction. While traversing ramps, drivers naturally follow transitional paths. Spirals in combination with circular curves are well suited for this purpose. Spirals on ramps may be shorter in length than on open highways as drivers tend to drive more aggressively on turning roadways. Spiral lengths for design of ramps are shown in Table 12. The suggested lengths are generally applicable to the ramp proper where speed operation is relatively uniform. Where significant speed changes take place, considerably longer spirals are required.

Table 12. Suggested Lengths of Spirals for Ramps (adapted from Leisch, 2005)

Design speed (mph)	20	25	30	35	40	45	50
Minimum radius (ft)	90	150	230	310	430	550	700
Assumed C (ft/s^2)	4.0	3.75	3.5	3.25	3.0	2.75	2.50
Suggested length of spiral (ft)	150	150	200	250	250	300	350
Begin using spiral when radius is (ft)	700	900	1000	1100	1300	1500	1800
Suggested length of spiral (ft)	150	150	200	200	200	200	250

2.1.5.2.4 Grades and Profile

A general guideline for maximum gradients on ramps is presented in Table 13. There is no direct relationship between maximum gradients and ramp design speeds, but lower gradients are more applicable to the higher ramp design speeds.

Table 13. Guidelines for Maximum Grades for Design of Ramps (Leisch, 2005)

Normal Conditions	Heavy Truck Traffic	Rugged Terrain (Special Cases)
4%-6%	3%-4%	6%-8%

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Table 14 provides suggested design speeds for exit and entrance vertical curves based on highway design speeds. These values are 80 to 90 percent of the highway design speed. This provides the driver with a sufficient view of the ramp and smooth transition between the highway grade and ramp grade.

Table 14. Suggested Ramp Exit/Entrance Vertical Curve Design Speed (Leisch, 2005)

Roadway Design Speed (mph)	Suggested Design Speed (mph) Ramp Exit/Entrance Vertical Curve
70-80	60-65
60-65	55
50-55	45-50
40-50	40

2.1.5.2.5 Cross Slope and Superelevation

Superelevation should be predicated on the design speed of each segment of the ramp. The development of superelevation is the transition from a crown slope to the full superelevation of a curve. The rate of cross-slope change for the design of ramps is a function of the ramp design speed. Another control in developing superelevation is the pavement/shoulder cross-slope break. To maintain comfort and safety as drivers encounter the pavement/shoulder cross-slope break, the algebraic difference in pavement slopes should not exceed the values in Table 15.

Table 15. Maximum Algebraic Difference in Pavement Cross Slope for Design of Ramp Terminals at Crossover Crownline (Leisch, 2005)

Design Speed of Highway (mph)	Freeway/Highway Ramp Terminal	Ramp/Ramp Terminal
15-25	N/A	0.07
30-40	0.05	0.06
50-60	0.04	0.05
70-80	0.03	N/A

2.1.5.2.6 Sight Distance

Sufficient ramp sight distance should be provided throughout based on the ramp design speed. Where feasible, longer sight distances should be provided.

2.1.5.2.7 Speed-Change Lanes

Lengths of SCLs are predicated on free-flow operating speeds for the designated design speed of the highway and ramp. Lengths of SCLs are based on the performance capabilities of passenger vehicles. If SCL lengths were based on trucks, longer lengths would be required.

For tapered exit terminals, the angle of divergence is typically 3 degrees (i.e., a departure rate of just under 20:1) for higher design speeds. Angles of 4 to 5 degrees may be used, but these generally pertain to lower design speeds (highways and ramps). For tapered entrance terminals, the

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angle of convergence is 1 degree, or a departure rate between 50:1 and 70:1. For highway design speeds of 50 mph or less and ramps, a rate of convergence of 40:1 may be appropriate.

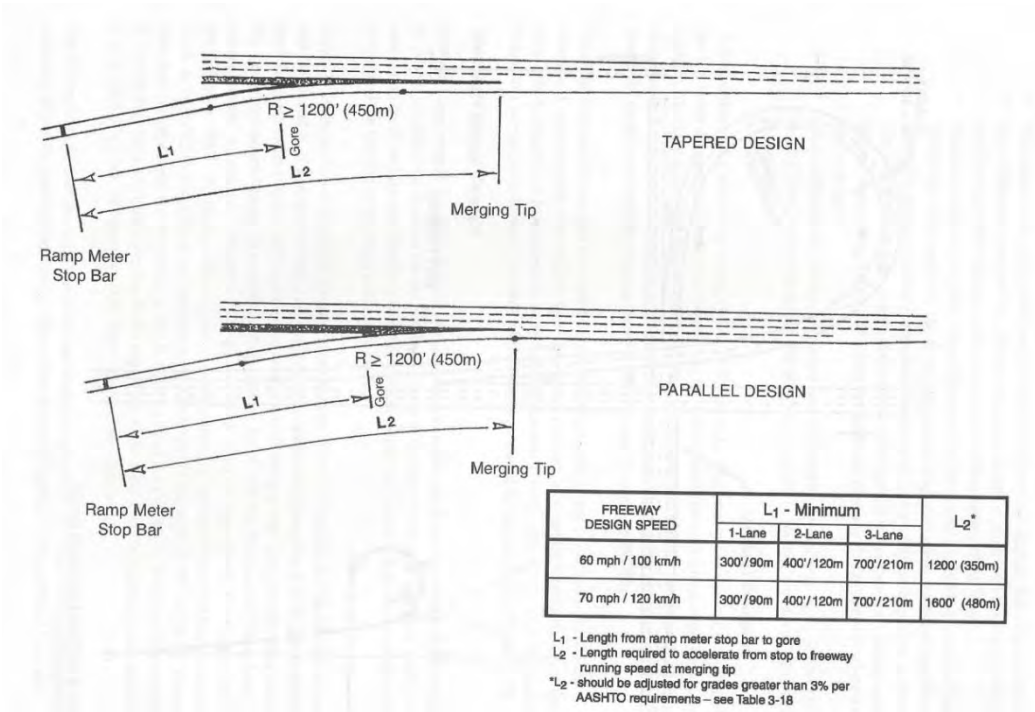
2.1.5.2.8 Design of Two-Lane Loop Ramps

The loop radius of the inner edge of the traveled way normally should be not less than 200 ft. A radius of 180 ft may be employed under restricted conditions or where the design speed of the approach roadway is 50 mph or less. A radius of 150 ft has been observed to satisfactorily operate with moderate volumes and sufficient width, and where the approach roadway design speed is 45 mph or less.

2.1.5.2.9 Metered Ramp Design

Metered ramps require the application of design criteria to reflect the stop condition on the ramp proper. Also taken into consideration are the number of lanes at the ramp meter stop bar, ramp traffic volume, ramp grade, and freeway design speed. All of these dictate the length and geometry of the ramp proper and the location of the ramp meter stop bar.

Figure 6 shows two examples of metered ramp designs. The dimension (L2) from the ramp meter stop bar to the merging tip at the beginning of the merging area is for acceleration. The dimension (L1) is for vehicles to accelerate from a stop to the running speed of the freeways, assuming the controlling ramp curve radius approaching the ramp terminal is 1,200 ft or greater. If the radius is less, then the dimension should be increased to reflect the design speed of the controlling curve. Also, if the ramp grade is 3 percent or greater, the L2 dimension should be adjusted.



FREEWAY DESIGN SPEED	L_1 - Minimum			L_2*
	1-Lane	2-Lane	3-Lane	
60 mph / 100 km/h	300'/90m	400'/120m	700'/210m	1200' (350m)
70 mph / 120 km/h	300'/90m	400'/120m	700'/210m	1600' (480m)

L_1 - Length from ramp meter stop bar to gore
L_2 - Length required to accelerate from stop to freeway running speed at merging tip
*L_2 - should be adjusted for grades greater than 3% per AASHTO requirements – see Table 3-18

Figure 6. Example Designs—Metered Ramp (Leisch, 2005)

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2.1.5.3 NCHRP Report 730 (Design Guidance for Freeway Mainline Ramp Terminals)

The objective of this research was to develop improved design guidance for freeway mainline ramp terminals based on modern driver behavior and vehicle performance capabilities (Torbic et al., 2012). The primary steps of the research included a crash analysis and field studies. Several general conclusions from this research most relevant to the selection of ramp design speeds are as follows:

- Passenger cars should remain the principal design vehicle for freeway mainline ramp terminals, which is consistent with current AASHTO policy. The exception to this rule is at freeway mainline ramp terminals where the truck volumes on the ramps are substantial, in which case further consideration should be given to more fully accommodating trucks within the design.
- Freeway mainline ramp terminals should be designed based upon free-flow conditions, consistent with current AASHTO policy. By designing for free-flow conditions, sufficient length is provided to accommodate merge and diverge maneuvers during more congested operating conditions.
- Consistent with current AASHTO policy, freeway mainline ramp terminal design should be based upon average operating speeds of vehicles, rather than design speeds. Designs based upon design speeds would provide terminals that are over designed.

Several conclusions specific to entrance ramps from this research most relevant to the selection of ramp design speeds are as follows:

- Merging vehicles do not accelerate at a constant rate along the length of the ramp and SCL.
- A clear view of the freeway and mainline traffic is important to accommodate merging vehicles. Drivers begin glancing at the freeway and mainline traffic prior to reaching the painted nose of the ramp, and while drivers are taking a glance at the freeway traffic, they continue to accelerate along the ramp and SCL.
- Vehicles exit curves on ramps at speeds much higher than the values given for “initial speed” in *Green Book* Table 10-5, suggesting that vehicle performance and driver preferences have changed since the design values were determined.
- In free-merge conditions, many drivers choose to enter the freeway at speeds much lower than the speed of freeway traffic.
- Constrained-merge conditions appear to be the most difficult for drivers, as reaching freeway speeds becomes more critical since gaps are smaller and do not provide as much opportunity for accelerating in the freeway lane. Additionally, drivers have a more difficult task identifying appropriate gaps. Some drivers take the first available gap they find even if they have not reached an ideal merge speed, while other drivers use the full length of the SCL, and in some cases the taper, to reach a speed near the speed of the freeway traffic before merging.
- Heavy vehicles accelerate at lower rates than passenger vehicles, and they merge onto the freeway at lower speeds. However, their merging behavior does not appear to negatively impact the overall operation of the ramp terminals.

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- Vehicles are more likely to use the full length of a tapered SCL to accelerate to near-freeway speeds before merging in contrast to a parallel SCL, where vehicles may merge earlier along the ramp and at lower speeds.
- Because most situations do not require that vehicles accelerate to the speeds assumed with the design, many drivers choose to accelerate at lower rates than assumed within AASHTO policy, and many vehicles are capable of accelerating at higher rates than the assumed acceleration rates used to determine minimum acceleration lane lengths for entrance terminals in the *Green Book*.
- Upgrades as steep as 3 to 4 percent do not impact the acceleration capabilities of passenger cars, at least over lengths necessary for entrance terminals. As grades increase to 5 or 6 percent, the acceleration rates of passenger cars tend to decrease.
- Assuming constant acceleration is a reasonable approach for determining minimum acceleration lane lengths for design. The current values provided in *Green Book* Table 10-3 are conservative estimates for minimum acceleration lane lengths, given the current vehicle fleet and driver population. They provide sufficient length for vehicles to merge onto the freeway under a range of freeway operating conditions. In situations where free-flow conditions are expected for the foreseeable future and constraints make it difficult to provide the recommended minimum acceleration lengths, the minimum acceleration lane lengths can be reduced by 15 percent without causing expected operational problems.

Several conclusions specific to exit ramps from this research most relevant to the selection of ramp design speeds are as follows:

- Most diverge maneuvers begin before or within the taper or within the first or middle third of the SCL, while few diverge maneuvers take place in the final third of the SCL or beyond the painted nose.
- Vehicles that diverge earlier along the deceleration lane diverge closer to freeway speeds than vehicles that diverge later along the deceleration lane, closer to the painted nose.
- Deceleration rates of exiting vehicles are greater for vehicles that diverge closer to the painted nose than for vehicles that diverge further upstream from the painted nose.
- Deceleration rates of diverging vehicles along the speed-change lane and ramp are not constant.
- Where the deceleration lane length is longer than the *Green Book* recommended minimum length, most vehicles decelerate at rates lower than those assumed by the *Green Book*.
- Free-flow vehicles decelerate at greater rates than platooned vehicles.
- When exiting the freeway, trucks decelerate at rates very comparable to those of passenger cars and typically diverge from the freeway at lower speeds than passenger cars.
- Crash rates for trucks are higher on parclo, free-flow and “other” ramp configurations than at diamond; outer connection; direct or semidirect connection; and button hook, scissor, and slip ramp configurations.

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- Drivers exiting on loop ramps tend to reduce their speed in the freeway lane more, and decelerate along the speed-change lane at a greater rate, than drivers exiting on straight ramps. This may be due the visual perceptions of drivers as they approach the loop ramp.
- Parallel deceleration lanes generally lead to substantially higher deceleration rates than on tapered deceleration lanes. The disparity between deceleration rates is most apparent on straight ramps.
- AASHTO policy assumes a two-step process for establishing design criteria for minimum deceleration lane lengths. Deceleration is accomplished first while coasting in gear without the use of brakes and then during the application of the brake. AASHTO assumes 3.0 sec for the coasting period. The 3.0 sec of coasting time is a valid assumption for describing the diverge maneuver, but 2 sec of the coasting time typically occurs prior to the diverge maneuver while in the freeway, and 1 sec of the coasting time occurs within the speed-change lane following the diverge maneuver.
- The two-step process of deceleration is a reasonable approach for determining minimum deceleration lane lengths for design and provides conservative estimates for minimum deceleration lane lengths, given the current vehicle fleet and driver population. Vehicles decelerated at rates well within the capabilities of the vehicle fleet and driver preferences. This was in part due to some deceleration by diverging vehicles in the freeway mainline prior to the diverge maneuver. It is beneficial to have a conservative design process that does not assume vehicles begin decelerating in the freeway mainline.

2.1.5.4 NCHRP Report 687 (Guidelines for Ramp and Interchange Spacing)

These guidelines assist planners and designers considering ramp and interchange spacing and the feasibility of new or rebuilt interchanges and ramps (Ray et al., 2011). This report does not provide much guidance relevant to the selection of ramp design speed beyond indicating that the design speed on system interchange ramps vary from 25 to 70 mph, and that interchange spacing and ramp density influence a freeway's estimated free-flow speed. As the ramp density increases, there is a corresponding decrease in free-flow speed on the freeway. The guidelines for ramp and interchange spacing address various issues, including geometry, operations, safety, and signing, and are based on balancing the competing needs for mobility and access.

2.1.5.5 Acceleration Characteristics at Metered On-Ramps

Yang et al. (2015) state that ramp metering has significant impacts on the design and operation of freeway on-ramps; however, the majority of the metered ramps in the United States are retrofitted to existing ramps. A ramp meter should be strategically placed to prevent on-ramp queue spillback as well as to provide enough acceleration length for a vehicle to leave the ramp meter and reach desired merge speeds.

Chaudhary and Messer (2000) explained that the design of a metered entrance ramp consists of three length components:

- Sufficient distance should be provided for vehicles upstream of the entrance ramp to decelerate and stop as they encounter the back of a queue.
- Sufficient queue storage length should be provided for vehicles waiting to enter the freeway.

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- Sufficient length should be provided for a vehicle stopped at the meter to accelerate to freeway merging speed.

When metering is implemented on an existing ramp, it is necessary to verify that sufficient acceleration length can still be provided after a portion of the ramp is allocated for deceleration and queue storage needs.

Additionally, Zheng and McDonald (2007) observed that the merging operation may become more difficult for the entering driver as well as drivers on the mainline when ramp metering is implemented. Their analysis was based on observations of speed, lane changes, and eye movements of human subjects who drove through metered ramps (with the metering on or off) in an instrumented vehicle. They observed that when metering is on, entering drivers exhibit increased eye movements, merge onto the mainline at a lower speed, and more frequently induce lane changes by drivers already on the mainline. Zheng and McDonald suggested that it may be necessary to locate meters farther upstream along entrance ramps to mitigate these complications. This suggestion is feasible if the ramp is long enough to provide the needed queue storage space after the meter is moved upstream.

2.2 Survey of Practice

A web-based survey of state DOT designers and design consultants was conducted to determine the current practice pertaining to the choice of ramp design speed. Interviews with selected design professionals were also conducted. The results of the web-based survey and interviews are reported in Sections 2.2.1 and 2.2.2. When the survey was distributed, the most recent version of the *Green Book* was 2011, so the survey questions reference the 2011 *Green Book*, not the 2018 *Green Book*.

2.2.1 Web-Based Survey

A web-based survey of state DOT and consultants was conducted regarding their policies and practices for the determination of ramp design speeds. One survey was sent to each of the 50 state DOT agencies. This survey focused not only on practice, but also on agency policy. A second survey targeted planners, engineers, designers, safety professionals, construction managers, operations managers, and others in the private sector and academia or at local highway agencies. This second survey was shorter and contained some of the same (or similar) questions as the survey for the state DOT agencies, but it focused more on state-of-the-practice. Participants invited to complete the surveys were identified from the following AASHTO and TRB committees and subcommittees:

- AASHTO Subcommittee on Design—Technical Committee on Geometric Design.
- AASHTO Highway Traffic Safety Subcommittee on Safety Management.
- AASHTO Standing Committee on Planning.
- AASHTO Subcommittee on Construction.
- AASHTO Subcommittee on Traffic Engineering.
- TRB Committee on Geometric Design.
- TRB Committee on Operational Effects of Geometrics.
- TRB Committee on Construction Management.
- TRB Committee on Transportation Safety Management.

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The purpose of the surveys was to learn about the experiences of professionals related to ramp design speed strategies, including:

- What factors most influence the choice of design speed for various ramp types.
- To what extent design practices conform to or differ from *Green Book* guidance.
- How policies and practices differ by ramp type or location along the ramp.
- Experiences related to design speed that led to a consideration of a change in policy or practice.
- Practices related to advisory speed signing on ramps.
- Tort liability concerns related to ramp design speed.
- Use of the HSM Chapter 19 (safety prediction methodology for ramps).
- Use of design exceptions for ramp design speed (or horizontal alignment) and experiences resulting from those exceptions.
- Identification of ramp design projects that could be evaluated in this research project.

The survey to state DOTs included 16 questions related to ramp design speed practices and experiences. The survey targeting the private sector included 11 questions related to ramp design speed practices. Both surveys also asked respondents for current or recent ramp or interchange design projects that could potentially be used as case studies for this research, as well as for contact information so the research team could follow up regarding such sites or any previous responses. Information about the type of respondents and a summary of their responses to the survey are presented next.

2.2.1.1 Respondents to Web-Based Survey

An email explaining the overall objective of research and a link to the survey was sent to one design or traffic engineer at each of the 50 state DOTs. Instructions invited recipients to forward the email with the survey link to a person in their organization qualified to respond to questions related to ramp design speed selection. Twenty-two responses were received from 21 state DOTs.

The survey developed for local agencies, university researchers, and private-sector practitioners was sent to 324 members of the TRB and AASHTO committees listed previously. Like the survey to state DOT officials, this survey allowed recipients to forward the email to others within their organization most qualified to provide answers. Thirty complete and two partial survey responses were received. Agency types (i.e., local agencies, university researchers, and private-sector practitioners) that responded to the survey included:

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Organization type	Number of responses
City or County	3
Consultant	12
University—US	5
University—International	2
DOT (retired staff)	2
None given	8
Total	**32**

2.2.1.2 Summary of Responses to Web-Based Survey

Responses to both surveys are summarized below. Each survey question is shown in italics, followed by a parenthetical indication of which survey(s) the question appeared. Responses to questions that were identical on both surveys are grouped together, but if there were notable differences in how the two groups of respondents answered, these differences are discussed. Questions that were similar, but not identical, between surveys are shown consecutively, with the question and response from the state DOT survey presented before the question and response from the local agencies, university researchers, and private-sector practitioners. The percentages of responses are provided in the tables, along with the number of respondents in parentheses.

Nearly every question on the survey provided a text box for respondents to further clarify or explain their answer, or simply provide additional detail. Many of those comments have been included here and only edited slightly to improve readability where needed.

Survey questions regarding potential case study projects and survey respondents contact information are not included in this summary.

Question: *What level of importance do the following factors have when your agency considers the selection of an appropriate ramp design speed?* (both surveys)

Exit Ramps								
Answer Options	**High Importance**		**Medium Importance**		**Low Importance**		**N/A**	
Ramp configuration (e.g., diamond, loop, directional)	66.0%	(33)	24.0%	(12)	2.0%	(1)	8.0%	(4)
Design/operating speed of highway	82.0%	(41)	14.0%	(7)	0.0%	(0)	4.0%	(2)
Design/operating speed of intersecting roadway	48.0%	(24)	24.0%	(12)	22.0%	(11)	6.0%	(3)
Ramp grade	24.0%	(12)	50.0%	(25)	20.0%	(10)	6.0%	(3)
Number of lanes of ramp	16.0%	(8)	44.0%	(22)	26.0%	(13)	14.0%	(7)
Use of spiral transition into control curve	10.0%	(5)	20.0%	(10)	28.0%	(14)	42.0%	(21)
Presence of ramp meter	11.8%	(6)	11.8%	(6)	13.7%	(7)	62.7%	(32)
Type of freeway mainline ramp terminal (e.g., parallel or taper)	24.0%	(12)	28.0%	(14)	38.0%	(19)	10.0%	(5)
Type of crossroad ramp terminal (e.g., signalized, unsignalized, free-flow)	38.0%	(19)	32.0%	(16)	20.0%	(10)	10.0%	(5)
Degree of physical constraints at site	38.0%	(19)	42.0%	(21)	12.0%	(6)	8.0%	(4)
Degree of environmental constraints at site	40.0%	(20)	42.0%	(21)	12.0%	(6)	6.0%	(3)
Right-of-way costs	44.0%	(22)	36.0%	(18)	14.0%	(7)	6.0%	(3)

Red text highlights answers where 75% or more of respondents indicated a combination of high or medium importance.
Shaded cells highlight answers where 25% or more of respondents indicated a combination of low importance or not applicable.

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Entrance Ramps								
Answer Options	**High Importance**		**Medium Importance**		**Low Importance**		**N/A**	
Ramp configuration (e.g., diamond, loop, directional)	56.3%	(27)	27.1%	(13)	8.3%	(4)	8.3%	(4)
Design/operating speed of highway	77.1%	(37)	18.8%	(9)	0.0%	(0)	4.2%	(2)
Design/operating speed of intersecting roadway	38.8%	(19)	28.6%	(14)	26.5%	(13)	6.1%	(3)
Ramp grade	28.6%	(14)	42.9%	(21)	22.4%	(11)	6.1%	(3)
Number of lanes of ramp	18.4%	(9)	36.7%	(18)	30.6%	(15)	14.3%	(7)
Use of spiral transition into control curve	8.2%	(4)	18.4%	(9)	30.6%	(15)	42.9%	(21)
Presence of ramp meter	18.0%	(9)	24.0%	(12)	18.0%	(9)	40.0%	(20)
Type of freeway mainline ramp terminal (e.g., parallel or taper)	32.7%	(16)	24.5%	(12)	32.7%	(16)	10.2%	(5)
Type of crossroad ramp terminal (e.g., signalized, unsignalized, free-flow)	28.6%	(14)	24.5%	(12)	36.7%	(18)	10.2%	(5)
Degree of physical constraints at site	34.7%	(17)	44.9%	(22)	12.2%	(6)	8.2%	(4)
Degree of environmental constraints at site	36.7%	(18)	44.9%	(22)	12.2%	(6)	6.1%	(3)
Right-of-way costs	42.9%	(21)	36.7%	(18)	14.3%	(7)	6.1%	(3)

Red text highlights answers where 75% or more of respondents indicated a combination of high or medium importance.
Shaded cells highlight answers where 25% or more of respondents indicated a combination of low importance or not applicable.

Freeway-to-Freeway Ramps								
Answer Options	**High Importance**		**Medium Importance**		**Low Importance**		**N/A**	
Ramp configuration (e.g., diamond, loop, directional)	66.7%	(32)	18.8%	(9)	8.3%	(4)	6.3%	(3)
Design/operating speed of highway	77.1%	(37)	16.7%	(8)	2.1%	(1)	4.2%	(2)
Design/operating speed of intersecting roadway	57.1%	(28)	20.4%	(10)	8.2%	(4)	14.3%	(7)
Ramp grade	32.7%	(16)	40.8%	(20)	20.4%	(10)	6.1%	(3)
Number of lanes of ramp	22.4%	(11)	32.7%	(16)	32.7%	(16)	12.2%	(6)
Use of spiral transition into control curve	10.2%	(5)	22.4%	(11)	26.5%	(13)	40.8%	(20)
Presence of ramp meter	10.0%	(5)	16.0%	(8)	10.0%	(5)	64.0%	(32)
Type of freeway mainline ramp terminal (e.g., parallel or taper)	32.7%	(16)	14.3%	(7)	36.7%	(18)	16.3%	(8)
Type of crossroad ramp terminal (e.g., signalized, unsignalized, free-flow)	22.4%	(11)	18.4%	(9)	18.4%	(9)	40.8%	(20)
Degree of physical constraints at site	28.6%	(14)	46.9%	(23)	16.3%	(8)	8.2%	(4)
Degree of environmental constraints at site	32.7%	(16)	42.9%	(21)	18.4%	(9)	6.1%	(3)
Right-of-way costs	38.8%	(19)	36.7%	(18)	18.4%	(9)	6.1%	(3)

Red text highlights answers where 75% or more of respondents indicated a combination of high or medium importance.
Shaded cells highlight answers where 25% or more of respondents indicated a combination of low importance or not applicable.

Twenty state DOT representatives and 32 local agencies, university researchers, and private-sector practitioners provided at least partial responses to the question. Some respondents replied for some factors but not for others. Responses between the two groups were similar and are reported jointly here, although key differences are noted. In addition, responses were similar, in most cases, across ramp types. Notable differences are discussed.

The design speed or operating speed of the highway was most frequently selected as having high importance when determining ramp design speed for all three of the ramp types. Over 90 percent of respondents indicated it had high or medium importance. This was followed closely by ramp configuration, which was categorized as high or medium importance by over 80 percent of respondents. Other factors frequently reported as having high or medium importance included:

- Design/operating speed of intersecting roadway.
- Ramp grade.
- Degree of physical constraints at site.
- Degree of environmental constraints at site.
- Right-of-way costs.

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The factor most frequently identified as being not applicable to the selection of ramp design speed was the presence of a ramp meter, although ramp metering was considered to be slightly more important on entrance ramps than exit or freeway-to-freeway ramps. The use of spiral transition into the controlling curve was also considered to have little importance to the selection of ramp design speed by most respondents.

The type of crossroad ramp terminal was generally considered to have high or medium importance for exit ramps and little to no importance on freeway-to-freeway ramps, and responses were split for entrance ramps, with about half of respondents considering it important and half not.

The number of lanes on the ramp was more often considered to be high or medium importance to local agencies, university researchers, and private-sector practitioners (between 62 and 73 percent for all ramps types) than to the state DOT respondents (between 40 and 45 percent for all ramp types).

The question provided a comment box labeled "other" to allow respondents to input other factors they may consider in the determination of ramp design speed. One state DOT respondent commented that *"our guidelines do not specifically address or list requirements on the above considerations and each designer considers all of the above with different importance on each element depending on the project scope and conditions, highway, etc."* Comments from local agencies, university researchers, and private-sector practitioners included the following:

- Ramp terminal speed change from freeway to ramp.
- These factors, and others, must be viewed in terms of available trade-offs. For example, right-of-way costs must be viewed in terms of expected safety and operational outcomes, i.e., don't build something unsafe or which will operate poorly just to save money. Maybe the project should not be built if the necessary factors cannot be included.
- There are significant differences in service interchange requirements and system interchange requirements for design. Also, depending on interchange form, location (urban, suburban or rural), and interchanging facility types.
- Should consider needs of large vehicles.

Question: *What type of policy does your agency follow in selecting ramp design speeds?* (State DOT survey only)

Use the policy in the 2011 AASHTO *Green Book*	61.9%	(13)
Use our own agency's design policy which is very similar to the policy in the 2011 AASHTO *Green Book*	38.1%	(8)
Use our own agency's design policy which differs from the policy in the 2011 AASHTO *Green Book*	0.0%	(0)

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Question: *Specifically, does your agency use the Table 10-1 from the 2011 AASHTO Green Book in selecting ramp design speeds?* (State DOT survey)

Yes	80.0%	(16)
No	20.0%	(4)

Five respondents provided further comment, as follows:

- We use the state roadway design manual, but the table is almost identical.
- We use the information on pages 10-89 and 10-90 in the *Green Book* and provide additional guidance for direct and semidirect connections.
- The table is used as a guide, a range of values if you will, but is not stringently adhered to. For example, while design and posted speeds on rural freeways have increased in our state, the superelevation and radius for controlling curves we use on diamond exit and entrance ramps are still set at 55 and 50 mph respectively.
- There is direction to use the table within our Design Manual; however, it is not clear what the table means or how to apply it. It is generally recognized that the design speed varies through the ramp and design elements are controlled by insuring they meet the acceleration/deceleration table values. The overall length of the ramp is checked, and then the lengths between the design features are checked (i.e., if a 40 mph horizontal curve is needed on an exit ramp, there must be enough distance to decelerate to 40 mph prior to the curve).

Question: *Is Table 10-1 from the 2011 AASHTO Green Book sufficient for selecting ramp design speed?* (local agencies, university researchers, and private sector survey)

Yes	46.7%	(14)
No	53.3%	(16)

Several respondents provided an explanation or further comment, including the following:

- I feel that a ramp should not have a set design speed. Each end is controlled by the roadway that it is tying into—so it is really a transition from a mainline speed to a crossroad or between two mainlines. I do believe that all ramps should have a minimum design speed for the tightest curve on the ramp—say 50 mph for a directional and 30 for a diamond ramp.
- Large truck turnover potential.
- I never liked this table. There needs to be extensive description as to when the upper, middle, or lower ranges apply. Exit and entrance ramps should be handled separately.
- It should take into consideration freeway-to-freeway vs freeway to secondary road configurations, since in the first case rather increased speed values are implemented.
- The current table is adequate for non-loop ramps, but there is inadequate guidance for loop ramps as it is not possible to know how to design for lower design speeds that is

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typical of an urban loop ramp. It would also be desirable to expand the highway design speeds shown to higher values, as many states now have a 75 mph speed limit.

- Based on the relatively conservative nature of the AASHTO policy, designing a ramp based on the 50th or even 70th percentile speeds does not seem appropriate. Additional justification for using speeds within these ranges should be required before utilizing them in design.
- See *ITE Freeway and Interchange Geometric Design Handbook* (Leisch, 2005), Chapter 6, pages 238-241.
- The curve radius and superelevation should be the primary considerations for ramp design speeds rather than the highway design speed.
- Information regarding the effect of ramp type on design speed would be helpful. In addition, the table provides a very wide range for each highway design speed. More guidance on when to select the upper range, middle range, and lower range design speeds would be helpful. Finally, the presence of a stop-controlled intersection at an exit ramp terminal is also an important factor.
- Additional guidance of the recommended practice for choosing from the upper, medium, and lower ranges would be helpful. Either theoretical or empirical information would be helpful and deciding which range would provide the best fit for a particular project.
- Some of the agencies we design for have strict design speeds (often higher in speed). They provide a minimum design speed, not a selection of speeds like this table shows.
- This table provides no real guidance for different ramp configurations for example a tight loop. One could look at the table and say the upper range must be 60 mph or middle 50 mph when that may not be attainable because of other constraints. Most tight loops are 25-35 mph for a majority of their length.
- Read the text and apply judgment for the specific interchange design.
- It's sufficient as a guide. The engineer needs to consider the context of the site and use good judgment.
- Grades and other geometric features need to be considered.
- The table does not explain any factors involved.

Question: *To what portion of the ramp is the ramp design speed applicable?* (both surveys)

Answer Options	State DOT Responses		Local Agencies, University Researchers, and Private Sector Responses		Combined Responses	
Freeway mainline ramp terminal	65.0%	(13)	61.3%	(19)	62.7%	(32)
Ramp proper	80.0%	(16)	83.9%	(26)	82.4%	(42)
Crossroad ramp terminal	25.0%	(5)	41.9%	(13)	35.3%	(18)

Between the two surveys, responses were similar for the ramp proper and freeway mainline ramp terminal. However, only 25 percent of state DOT respondents indicated that the ramp design speed applied the crossroad ramp terminal compared to 42 percent of respondents from the local agency, university, and private sector survey.

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Additional comments from state DOT respondents include:

- Ramp proper design speed should be from 50 to 85 % of mainline design speed (70%-85% is preferred). Loops are the exception - we use a minimum of 25 mph (36 degree curve); 30 mph is preferred (but rarely attainable) for exit loops.
- Exit ramps are typically going from the freeway speed to a stop condition, while on the entrance ramp it is typical to assume to go from a 15 mph start speed to the freeway speed.

Additional comments from local agencies, university researchers, and private-sector practitioners include:

- The ramp proper is the only area not controlled by the roadway on either end of the ramp. Probably should just be a minimum for the type of ramp. Say 30 mph for a diamond ramp, for example.
- Storage for freeway meter and crossroad stop.
- The speed at the mainline/ramp gore areas is most applicable and important for both entrance and exit ramps. Sufficient distance to the terminal and corresponding design speed must be provided on entrance ramps for acceleration to main line and deceleration on the exit ramps to the crossroad.
- It is assumed that the crossroad ramp terminal design speed is a function of the stopping distance needed and traffic control at the crossroad. The freeway mainline ramp terminal is a function of the deceleration distance needed to reach the design speed for the ramp.
- I've seen ramp terminal merge points for 65 mph freeways that dump vehicles into mainline traffic with only the 30:1 taper length and no merging length. Vehicles need distance to get up to speed and then distance to merge as well.
- The ramp speed is dependent on the function of the ramp. Freeway ramp terminals should have a design speed more indicative of the freeways speed, while the ramp proper of the same ramp may be reduced after allowing for appropriate distance to slow down.
- The ramp design needs to be developed as a system of all the three elements above as well as the design of the roadway from which the ramp exits and onto which the ramp enters and the type of traffic control device.
- Ramp speed should ideally apply throughout the whole ramp. Stop-controlled intersection at exit ramp may allow for lower design speed.
- We verify the speed transition areas on the ramp are designed to the highest anticipated speed in that location. We establish a design speed profile to do this. The profile is based on both horizontal and vertical geometrics. One of our client's has a very specific procedure we are to follow to do this. We have applied this to other projects to improve our design.
- The speed transition must be considered. We use different speeds for different portions of the ramp. i.e., Usually 10 mph less than mainline for the first curve, transitioning to 30 mph if there is a stop condition at the crossroad terminal. Maintain higher speed for system ramps.

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Question: *Does your agency's policy for selection of ramp design speed differ between ramp types, ramp configurations, or some other factor?* (State DOT survey)

Answer Options	Yes	No
Ramp types (exit ramp vs. entrance ramp vs. freeway-to-freeway ramp)	15% (3)	85% (17)
Ramp configurations (diamond ramp vs. loop ramp vs. directional ramp)	50% (10)	50% (10)
Other (Please specify in comment box below)	10% (2)	N/A

Seven respondents provided further comment, mostly related to design speed practices on loop ramps. Comments included:

- Loop ramp design speeds are generally 25-30 mph. This does not meet our guidelines (50%-85% of mainline design speed). We use longer deceleration lengths (sometimes adjusted when there are heavy truck volumes) and longer spirals on exit loops.
- Loop ramps utilize 25-30 mph design.
- Reference the text rather than chart for loop ramps (min 25 mph).
- Physical/environmental constraints and right-of-way can have an impact.
- Loop ramp design speed selection is based more on engineering judgment and site conditions; diamond ramps generally adhere to standard layouts and design criteria. Standard exit and entrance ramps have a 5 mph design speed difference. Additionally, freeway-to-freeway ramp speed differential would be limited, whereas other types would allow for more difference between mainline and the ramp.
- We always start with Table 10-1 then adapt as necessary as funding dictates. (hard to design a loop ramp at 60 mph).

Question: *Should the practice for selection of ramp design speed differ between ramp types, ramp configurations, or some other factor?* (local agencies, university researchers, and private sector survey)

Answer Options	Yes	No
Ramp types (exit ramp vs. entrance ramp vs. freeway-to-freeway ramp)	69.0% (20)	31.0% (9)
Ramp configurations (diamond ramp vs. loop ramp vs. directional ramp)	73.3% (22)	26.7% (8)
Other (Please specify in comment box below)	20.0% (6)	N/A

Eleven respondents provided further comment, including:

- Large truck turn over potential.
- Exit and entrance ramps should be handled differently. Diamonds vs. loops vs. directional are clearly different in configuration and design speed expectation from motorists.
- Only freeway-to-freeway cases since increased speed values are found.
- Need to consider design guidance for non-traditional ramp designs, e.g., diverging diamonds.

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- Freeway-to-freeway ramps are often intended to maintain mainline speeds, while entrance or exit ramps are near mainline speeds. Exit and entrance ramps also may have a perceived design speed much higher than intended, creating safety concerns. Ramps would benefit from having perceived design speeds lower or at most equal to the posted mainline speed.
- The design of the ramp as previously mentioned also depends on the interchange form, location, and type of interchanging facilities.
- Design speeds on the freeway and crossroad should also be considered. Traffic control on crossroad is also a factor. Freeway-to-freeway ramps should typically have higher design speeds. Ramp configuration helps to form driver expectations (for example drivers expect low speeds on a loop ramp).
- Higher driver speed expectations for outer ramps as opposed to loop ramps.
- Type of control at the crossroad.
- Freeway-to-freeway is typically a fully directional movement with higher speeds. Loop ramps will typically have the lowest speeds. The geometry of a diamond interchange can allow for higher speeds; however, you need to consider the length, queuing, and stop condition.
- Grades and other horizontal restrictions.

Question: *Does your agency's policy specify a single design speed for a ramp as a whole or is the design speed allowed to vary along a ramp?* (State DOT survey)

Single design speed for entire ramp	30.0%	(6)
Design speed may vary along a ramp	60.0%	(12)
Depends on situation (Please explain below)	10.0%	(2)

Six respondents provided further comment, including:

- The design speed of the initial part of the ramp (ramp proper) is to be 50%-85% of the mainline, and the speed of the ramp approaching the crossroad (the terminal curve) should be 50%-85% of the main ramp speed. The intent is to provide a "stepped down approach" on exits, where we think this is most crucial.
- Physical/environmental constraints and right-of-way (ROW) can have an impact.
- Diverging diamond interchanges use different design speeds for the on-ramps and off-ramps depending on alignments.
- If approach road terminal is stop condition, vary from ramp proper to stop condition.
- For example, at a stop condition at the end of a diamond exit ramp, an intermediate speed will be used between the controlling exit curve and the stop condition (as vehicles are decelerating) that would control the geometrics. This is a much more economical approach.

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Question: *Should a single design speed be specified for an entire ramp or should the design speed vary along the ramp?* (local agencies, university researchers, and private sector survey)

Single design speed for entire ramp	24.1%	(7)
Design speed may vary along a ramp	51.7%	(15)
Depends on situation (Please explain below)	24.1%	(7)

Twelve respondents provided further comment, including:

- Design speed may differ, depending on the design speed of the termini. (e.g., free-flow termini should have high design speed, stopping termini may have lower design speed...the ramp design speed should be utilized to accommodate this transition).
- The design speed refers to the main body of the ramp. At the beginning and ending sections, the design should be consistent in terms of speed.
- For short ramps a single design speed may be ok, but for others it should be variable.
- Both safety and operations are compromised when the design of a ramp near merge/diverge points are not close to the mainline operating speeds. Cloverleaf ramps create insufficient weaving distances for adequate design speeds.
- Ideally one design speed is used, but site constraints and intersecting roadways will alter the selection. A freeway to local road ramp will require a higher design speed when exiting the roadway, but may have a reduced design speed near the terminus of the ramp that connects to the lower-speed local roadway.
- The physical and environmental constraints may dictate the geometry of the ramp which may require a variable design speed.
- For simplicity I would say that the design speed should be the same for the entire ramp. However, there are special cases such as a stop condition on a crossroad. Ramps are typically short in length so varying the design speed along the length does not seem beneficial.
- In situations where drivers will be expected to alter their speeds (i.e., approaching a stop-controlled intersections), a progression in design speeds is advisable to encourage this behavior.
- See previous comments about calculating design speed profiles along the ramp.
- Transition curves are often used near the gore areas that have a higher speed; however, I would recommend a single speed for the ramp proper.
- Length of ramp and geometric features may suggest two or more ramp speeds are appropriate.
- Depends on length of ramp.

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Question: *Is your agency considering any change in its policy for selection of ramp design speed?* (State DOT survey)

Yes	5.0%	(1)
No	95.0%	(19)

Only one of the twenty respondents indicated that their agency is considering a change in its policy for selection of ramp design speed. That respondent indicated that the state DOT was updating its design policy to include ramps. No other specifics about the policy change were given.

The survey included a follow up question that asked if there was any performance information or empirical data that has led the agency to consider changing its ramp design speed policy. All respondents answered "no".

Question: *Do you believe any changes to the AASHTO Green Book policy for selecting ramp design speed are needed?* (both)

	State DOT Responses		Local Agencies, University Researchers, and Private Sector Responses	
Yes	35.0%	(7)	64.3%	(18)
No	65.0%	(13)	35.7%	(10)

Comments from state DOT respondents included:

- Additional explanation regarding exit ramps and the need for using "stepped down" approach in selecting speeds.
- There are minor conflicts between the table and the text so greater clarification and guidance is needed.
- Better explanation of applicable design speed at terminals. Louisiana requires a minimum of 700 ft auxiliary lane (where ramp and mainline meet). We have seen operational problems for the 300-500 gap acceptance for entrance ramps.
- The information in the *Green Book* could be clearer.
- More considerations should be listed as guidance for considering alternative or lower speeds then are listed in Table 10-1.
- Possible changes/clarity: Current publications are indicating that design speeds are not operating speeds see FHWA Memorandum 20151007, HIPA-20, Relationship Between Design Speed and Posted Speed. Do the acceleration/deceleration tables reflect highway speed or design speed?

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Comments from local agencies, university researchers, and private-sector practitioners included the following:

- Generally do not use this information from the *Green Book*.
- I teach highway design, and have always had difficulty effectively conveying the ramp design speed procedure to students, largely because I'm not sure if I fully understand it myself.
- Selecting design speed for freeway-to-freeway vs freeway to secondary roads ramps.
- Additional notes and consideration should be included for weaving length and the actual ability to accelerate/decelerate to and from the design speeds.
- Allow the designer more discretion to change design speeds along a ramp.
- See *ITE Freeway and Interchange Geometric Design Handbook* (Leisch, 2005), Chapter 3, pages 99-103.
- More clarity for how the ramp design speeds were calculated and how to go about addressing unique situations such as ROW constraints.
- More guidance on when to select low, middle, or high range of values is needed. Special designs such as the diverging diamond interchange should also be considered.
- I recommend investigating the design speed profile calculations in use. It would be interesting to see how a design speed profile on a ramp compares to operating speed profiles on a built ramp.
- Ramp configuration and type should factor into design speed.
- More guidance should be given on designing the ramp to fit the context of the area.
- It appears there is inherent flexibility in design parameters, but it may be good to give the designer some discussion regarding other factors that may way their respective design decisions.
- If there is recent research that justifies it sure; however, I think it will be difficult to be more specific and engineers need to consider the context and use the flexibility provided.
- I feel that the "ramp" needs to be better described in the *Green Book*. What part of the ramp is controlled by the main lanes if any, and similarly for the intersecting street. Sometimes there is confusion on what curve radii to choose, and if it meets the design speed, based on where the curve is located along the ramp.

Question: *Does your agency's policy for signing advisory speeds on ramps relate directly to the design speed of the ramp?* (State DOT survey)

Yes	33.3%	(6)
No	38.9%	(7)
In some cases	27.8%	(5)

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Comments included the following:

- Sometime it is based on a field check.
- We utilize the MUTCD and PA Publication 236.
- Districts will ball bank the curves.
- Advisory speed signing, if necessary, on our ramps is determined by running a ball bank indicator test and signing appropriately.
- Signing for advisory ramp speed is based on the curve geometry of the ramp proper from the initial design, adhering to the *Green Book* Superelevation Tables (3-10b). If later in the life of the ramp the adequacy of the signing is in question, a ball bank indicator will be used to determine the appropriate advisory speed.
- Typically use MUTCD guidance related to ball bank values and posting advisory speeds when needed.
- May be based on ball-banking (e.g., loops).

Question: *Has your agency encountered any specific tort liability concerns on ramps whose design speeds were selected with your existing design policy that this research should address?* (State DOT survey)

All 20 respondents answered “no” to this question.

Question: *Are you aware of any specific tort liability concerns related to ramp design speeds that this research should address?* (local agencies, university researchers, and private sector survey)

Yes	16.7%	(5)
No	83.3%	(25)

The following additional comments were provided:

- Change ramp design speed and ramp length.
- Varying ramp speeds.
- I am not aware of any. Some old (50 year or so) designs did not anticipate the volumes or speeds associated with today's traffic.
- Use of improper superelevation.
- I would try to address weave length as part of the ramp design speed.
- Design speed vs. posted (advisory) speed.
- Yes, not all of our clients have tort immunity or a low dollar cap. The design speed impacts the clear zones and where we can then place roadside objects. Some clients are requiring level III barrier warrant analysis using RSAP software (the prior version as the new windows based version has too many bugs) to verify the benefit cost has been explored and the option to be built will be defendable in court.

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Question: *Has your agency/organization used the safety predictive models from Chapter 19 of the 2014 Supplement to the AASHTO HSM in the design of ramps and/or the selection of ramp design speeds?* (both surveys)

	State DOT Responses		Local Agencies, University Researchers, and Private Sector Responses	
Yes	10.0%	(2)	24.1%	(7)
No	90.0%	(18)	75.9%	(22)

State DOT respondents reported the following:

- Guidance on the applicability and use of the HSM is being developed. However, the HSM can be and has been utilized to analyze project specific decisions occasionally.
- Design of ramps. Some limitations to HSM.
- Not yet, but may in the future.

Comments from local agencies, university researchers, and private-sector practitioners included the following:

- Very good experience to date, but there are subtle issues to always consider what the HSM predictive models bring out.
- Agency recently introduced HSM analysis into design. Awaiting early results.
- Need additional research.
- We have not used it in the design of ramps or selection of ramp design speeds, but we have used the HSM for a variety of other tasks. For some designers, the predictive method can be too complicated and/or difficult to follow. Learning the predictive method becomes too overwhelming so they choose not to apply it.
- Yes, for some clients; however, the client we utilize design speed profiles and RSAP software to aid in the design of the ramps does not permit us to use HSM and its methodologies. I find this very odd.
- The results are not always intuitive. Not taking grade into account is a significant issue with the crash prediction for ramps.

Question: *In the past 5 years, how many design exceptions for ramp design speed (or design exceptions for horizontal alignment that are equivalent to a design exception for ramp design speed) has your agency initiated and approved?* (State DOT survey)

In the past 5 years, how many ramp/interchange design projects has your organization been involved with that required design exceptions for ramp design speed (or design exceptions for horizontal alignment that are equivalent to a design exception for ramp design speed)? (local agencies, university researchers, and private sector survey)

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Answer Options	State DOT Responses		Local Agencies, University Researchers, and Private Sector Responses	
None	38.9%	(7)	65.4%	(17)
1 or 2	38.9%	(7)	11.5%	(3)
3 to 5	5.6%	(1)	11.5%	(3)
6 to 10	5.6%	(1)	7.7%	(2)
More than 10	11.1%	(2)	3.8%	(1)

State DOT respondents provided the following additional information:

- We are reconstructing a full cloverleaf interchange (in phases) into a partial cloverleaf with a flyover ramp. As part of the final phase, we are rebuilding an existing loop ramp with improved geometry. The connection to the loop uses a "stepped down" approach to transition from a 60 mph to 25 mph condition, including an exception from our standard superelevation.
- Physical/environmental constraints and ROW have impacted.
- We do not have that information statewide, but I have not seen any in my responsible area (about half the state).
- This question is not able to be accurately answered because we do not have the data.
- Historic Property Impacts.
- Believe there have been 2-3 due to horizontal sight restrictions around barrier wall on inside.
- Sight distance over the barrier rail on a flyover. Typically for on- and off-ramps we design parallel so we can add or subtract length as necessary.
- Loop ramp where we had topography and geometric constraints. Existing ramp had multiple curve radius and superelevation. An effort was made to create a more consistent superelevation and curve radius that did not fit the *Green Book* values.
- Specifically for ramp loops where there are existing ROW constraints.

Comments from local agencies, university researchers, and private-sector practitioners included:

- One of two issues: 1) An Interstate to Interstate ramp (system ramp) should be designed for 50 mph, but because of ROW constraints ends up being a 30 mph loop. 2) Most of the deviations are for diamond ramps where the curve closest to the crossroad cannot meet the 30 mph minimum set for all service interchange ramps.
- None that I can recall.
- Typically dictated by ROW constraints, which would have significant impacts to the overall cost and schedule of the projects.
- System interchanges in an urban area where the agency had a guideline of 50 mph minimum design speed which was impossible to achieve.

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- Our office has not been involved with projects that required a federal design exception from the AASHTO criteria for ramp design speed. Several projects have required exceptions to the project criteria particularly as it relates to horizontal stopping sight distance.
- We do not want design exceptions for this. We avoid whenever possible and have been successful doing so thus far.
- None that I am aware of.
- Interchanges in tight urban areas have required design exceptions. We do not get design exceptions for design speed, we get design exceptions for each element that does not meet the design speed, i.e., horizontal curve, vertical curve, superelevation, etc.

Question: *Are there any other factors that affect the determination of ramp design speed that have not been addressed in this survey*? (both surveys)

Respondents from state DOTs provided the following feedback:

- Strengthened discussion on considerations at depressed interchanges (exit ramp sight lines, crossroad sight lines, etc.,) would be beneficial. Also, there should be discussion of the effects of vehicle queues at exit ramps.
- The design speeds are the same as the answers already provided, but maybe in other States the ramp operation (i.e., freeway-to-freeway direct connectors or "drop ramps" in the medians for HOV or HOT lanes) might have different design speeds.
- For ramp termini other than free-flow the *Green Book* is not real clear on the design speed of ramps (e.g., stop controlled, signal, or roundabouts). Would like to see continuous deceleration and/or acceleration discussed or addressed.
- We have moved to 80 mph posted speeds on much of our Interstate system. These speeds are off the ramp acceleration/deceleration charts. We are having questions as to whether the design speed of the Interstate and thus the ramp terminal should be moved to 80 mph. Note that the entire mainline route ball banks at 80 mph or greater. Should we be keeping the design speed of 70 mph for our main line or moving to 80 mph (which may give higher ball bank values and operational speeds)? Should we be doing design exceptions for routes that do not meet the 80 mph design values of the *Green Book*, but ball bank at 85+ mph? Does not meeting the ramp deceleration/acceleration lengths constitute a design exception for the design speed of the ramp? The grade factors seem questionable. Currently working through an IC design where entrance ramp that intersect a freeway section has a 5% grade. *Green Book* values are giving us ramp lengths of excess of 3000 ft to meet a 75 mph freeway posted speeds. Seems excessive in that the current ramps are about 1200 ft with no identifiable operational issues (i.e., mostly due to lower volume Interstate, 12000 ADT, crossroad at about 3000 ADT, and freeway speeds appear to be operating slower then posted speed). Truck rollover—it is rare here, but at what point does it become a determining factor in relationship to design speed?

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Comments from local agencies, university researchers, and private-sector practitioners included:

- Toll plazas.
- Design exceptions.
- There are non-technical factors (you have included two; environmental and ROW costs), such as political engineering (the other PE) where some developer/unit of government wants an interchange “squeezed in” where it should not be allowed. There are also public involvement factors (often personal interests) that may influence interchange design.
- Urban vs. rural design considerations. More could be presented relating to conceptual design approaches especially for off-ramps. Revisit difference in taper length estimate process—*Green Book* vs. MUTCD. Many subtle ramp design issues cannot be readily evaluated in terms of safety performance using ISATe. Look at differences in stopping sight distance between underpass ramp alignments vs. overpass ramps.
- Resulting weaving length and resulting speed determined by ramp length.
- Innovative geometrics such as diverging diamond or roundabout interchanges should be considered. I answered this survey based on my prior 14 years of experience as a Highway Engineer.
- Toll plazas need to be considered. They have deceleration and acceleration in advance and after the plaza. The plazas may also have differing speed lanes. Some lanes are 5 or 15 mph, and others are stop controlled. Some are just open gantries with all electronic toll collection and would not impact the speeds. We have some ramps that connect to collector-distributor (CD) roads instead of directly to the mainline. The CD roads are usually about 10 mph lower design speed of the mainline. We design the ramps to tie to the appropriate road's design speed (we would use the CD design speed in our design speed profile). I also find it interesting that some agencies specify an e_{max} for the entire ramp, but other agencies mix e_{max} rates. The one that mixes e_{max} has a larger value near the higher-speed road and along the ramp proper, but as the ramp's curves approach a slower crossroad ramp terminal, the e_{max} may drop by 2% to 4% for those curves. I don't know why some agencies do this and others do not. Making the switch to a lower e_{max} would certainly help the designer meet rollover criteria in the gores. Some of our ramps have barrier along them. Ramps are usually curved so it may be difficult to get higher design speeds in the tighter curves due to sight distance issues with the sight line hitting the barriers. Same could apply to a ramp just downstream or starting just in advance of a bridge where the bridge support could be in the sight line. Are our ramps being over designed due to having better pavement surface and better tires (more friction) than what have been included as the basis for the design resources we follow? One more for you. With future autonomous vehicles, do we need as much room for the ramps? Can we eliminate shoulders and eliminate sight lines? Will the vehicles be able to sense through or around barriers? Will they know a back of queue is ahead? Can we shorten acceleration/merging distances because the vehicles can talk to each other and set gaps?
- Sight distance. Lane drop or lane add. Climate. Terrain.
- Crash history.

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2.2.1.3 Discussion Points from Web-Based Survey

Several key points that can be drawn from the web-based survey are as follows:

- Committee members, local agencies, university researchers, and private-sector practitioners overwhelmingly think policy should differ by facility type and ramp configuration, but most states do not have policies that vary by those factors.
- Most state DOTs use *Green Book* or similar policies and have no plans to change, but most academics and private-sector practitioners would like to see changes to the policy.
- The factors most frequently selected as having high or medium importance when determining ramp design speed include:
  - The design speed or operating speed of the highway.
  - Ramp configuration.
  - Design/operating speed of intersecting roadway.
  - Ramp grade.
  - Degree of physical constraints at site.
  - Degree of environmental constraints at site.
  - ROW costs.
- Over 50 percent of respondents indicated that Table 10-1 from the 2011 AASHTO *Green Book* is insufficient for selecting ramp design speed.
- There is no consensus on what portion of the ramp to which the ramp design speed is applicable.
- A majority of the respondents think that the practice for selection of ramp design speed should differ between ramp types and ramp configurations.
- There is no consensus on whether a single design speed should be specified for an entire ramp or whether the design speed should vary along the ramp.
- No state DOTs that responded indicated any specific tort liability concerns on ramps whose design speeds were selected with existing design policies.
- Other factors to consider when providing guidance on the selection of an appropriate ramp design speed include:
  - Toll plazas.
  - 75+ mph speed limits.
  - Innovative geometrics such as diverging diamond or roundabout interchanges.

2.2.2 Interviews with Selected Professionals

As a follow up to the online survey, the research team interviewed four survey respondents by telephone to explore issues covered by the survey in greater depth. Key discussion items from the interviews are summarized as follows:

- **Green Book Table 10-1 is insufficient. Suggested modifications or changes included:**
  - The table needs to be expanded to show highway design speeds of 80 mph and 85 mph, perhaps even higher as speed limits are being raised to 75, 80, and 85 mph.

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- The table needs to be expanded to show different ramp types (exit, entrance, freeway-to-freeway) and ramp configurations (loops, diamonds).
- The table does not detail the three main areas of the ramp and should be expanded to show design speed (or range of design speeds) for:
  - The ramp terminal at the crossroad by intersection control (stop condition, signal, roundabout, etc.).
  - The ramp terminal at highway or freeway.
  - The ramp proper based on interpolation of speeds between the two terminals.
- Ramp metering should be a major consideration in designing entrance ramps, noting an existing metered ramp in Utah that is approximately 4,000 ft long with a 45 mph curve near the freeway entrance. The ramp was designed to accommodate queuing during ramp meter operations; but when the meters are not on, drivers perceive the ramp to have a much higher design speed due to the long, tangent ramp proper preceding the curve.
- There should be a varying ramp design speed for service interchanges. Freeway-to-freeway ramps should have a single design speed equal to that of the facilities to which they connect. For service interchanges, the design speed at the freeway mainline ramp terminal should be 50 mph (approximately 20 mph lower that the design speed of the freeway).

- **Factors to consider when selecting a ramp design speed:**
  - Additional guidance on ramp design for diverging diamonds and single point urban interchanges (SPUIs) is needed. Both normally utilize signalized control at the crossroad intersection.
  - Vertical alignment should be taken into consideration along with the type of design vehicle, especially trucks on significant up or down grades.
  - Concerns about use of adverse superelevation on loop ramps should factor into the selection of ramp design speed.
  - Additional guidance is necessary for weaving lengths at auxiliary lanes and on CD roads used in cloverleaf interchange design. Utah DOT is trying to eliminate all cloverleaf designs due to extremely short weaving lengths (500 ft for some), noting there are a high number of crashes on those short weaving lengths, and crash frequency diminishes greatly when weaving lengths are greater than 1,000 ft. Utah DOT uses a lot of SPUIs.
  - Illinois DOT is trying to eliminate all cloverleaf interchanges due to problems with large truck turnover potential. Where it is cost prohibitive to reconstruct the interchange and truck turnover crash data is high, they increase the superelevation rate on the loop ramp (up to 10% superelevation) to reduce those types of crashes. They considered using spirals, but generally there's not enough ROW. Part of the problem is that the trucks disregard the posted advisory speed limit on the ramp.
  - The use of speed profiles is employed based on Illinois Tollway Traffic Barrier Guidelines procedure. These are completely different than what Illinois DOT uses.

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The vertical alignments are also checked against the horizontal speed profiles to ensure consistency in vertical/horizontal design.

- **Ramp safety and use of the HSM:**
  - When using the HSM in ramp design, a few anomalies have been identified that do not seem logical. In comparing the "no build" and "build" alternatives, it was found that when the design speed increased, the number of crashes or severity of crashes also increased (or vice versa). The HSM also showed an improved travel time which seemed odd because the ramp was such a small portion of the overall design.
  - Both Ohio and New Mexico DOTs use AASHTO's HSM.
  - Several respondents noted using FHWA's Interchange Safety Analysis Tool for interchange safety analysis.
  - Safety related to the speed differential between the ramp and mainline should be addressed.
  - The HSM is not used in designing ramps. The *Green Book* and the Highway Capacity Manual are used to design ramps, and the HSM may be used to check the design for federal funding purposes; but the HSM is idealistic and not really effective.
- **Design exceptions:**
  - Design exceptions for ramp design speed are most often at the curve closest to the crossroad and are related to the superelevation of the curve. Ramp design superelevation is 8 percent maximum, and frequently there is not enough tangent distance to transition through the curve and back to the grade of the crossroad if the superelevation is set at 8 percent. To address this issue, the design criteria has recently been revised to allow for a 6 percent maximum superelevation rate and 25 mph design speed for the curve closest to the crossroad.
  - Designers should not seek too many design exceptions. It can lead to problems. For example, in the design of the first designated Interstate through an urban area in the late 50's, Illinois DOT obtained design exceptions for several slip ramps and short ramps where ROW was limited. Increased traffic and speeds 10-20 years later caused crash rates to rise, and it was extremely costly and difficult to remedy the situation.

2.3 Summary of Key Issues and Limitations of AASHTO's Current Policy on Ramp Design Speed

Based upon the literature review, the survey results, and interviews with selected professionals, this section summarizes the key issues and limitations of AASHTO's current policy on selection of an appropriate ramp design speed. The key issues and limitations are not presented in any particular prioritized order:

- ***Green Book* Table 10-1 is insufficient for selecting ramp design speed:** *Green Book* Table 10-1 is the centerpiece of AASHTO's current guidance on selecting an appropriate ramp design speed for the design of all ramp types (i.e., entrance, exit, and freeway-to-freeway ramps) and ramp configurations (e.g., diagonal, loop, semidirect, direct). However, more than 50 percent of respondents to the surveys indicated that *Green Book*

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Table 10-1 is insufficient for selecting ramp design speeds. Rather than *Green Book* Table 10-1 addressing entrance and exit ramps separately and the various types of ramp configurations, the table provides upper, middle, and lower range values of ramp design speeds. AASHTO policy then provides further guidance regarding desired, practical, and/or minimum ramp design speeds corresponding to the upper, middle, and lower speed ranges for different ramp configurations. Suggested revisions to the table included the need to address different ramp configurations and entrance and exit ramps.

- **No consensus on the portion of the ramp to which the ramp design speed applies:** Existing guidance states that the guide values for ramp design speed apply to the sharpest, or controlling, ramp curve, usually on the ramp proper, and the ramp design speeds do not pertain to the ramp terminals. However, there is no consensus among practitioners as to what portion of the ramp the ramp design speed applies. Approximately 82 percent of survey respondents indicated that the ramp design speed is applicable to the ramp proper; approximately 63 percent of survey respondents indicated that the ramp design speed is applicable to the freeway mainline ramp terminal, and approximately 35 percent of survey respondents indicated that the ramp design speed is applicable to the crossroad ramp terminal.
- **Little guidance is provided on how the selected ramp design speed is to be used to develop a balanced design among all components of a ramp:** AASHTO policy states that the ramp terminals are to be properly transitioned and provided with speed-change facilities adequate for the speed of the highway being considered. First, let us consider how this statement applies to freeway mainline ramp terminals. In addition to *Green Book* Table 10-1, *Green Book* Tables 10-3 and 10-5 provide the minimum acceleration lane lengths for entrance ramps and minimum deceleration lane lengths for exit ramps, respectively. The minimum acceleration and deceleration lane lengths are tied to the highway design speed and initial speeds at the beginning of the SCL and final speeds at the end of the SCL. Consistent with the guidance provided for using *Green Book* Table 10-1, *Green Book* Tables 10-3 and 10-5 do not specify a corresponding design speed for freeway mainline ramp terminals, nor is there guidance tying the conditions specified in *Green Book* Tables 10-3 and 10-5 for acceleration and deceleration lane lengths to the ramp design speed values in *Green Book* Table 10-1 for the controlling curve. Beyond AASHTO policy stating that the ramp terminals are to be properly transitioned and provided with speed-change facilities adequate for the speed of the highway being considered, there is no real guidance on how this is to be achieved for freeway mainline ramp terminals.

  The other type of ramp terminal is the crossroad ramp terminal. For exit ramps, *Green Book* Figure 9-2 (Figure 7) provides some guidance regarding how the functional area of an intersection, consisting of three basic elements (i.e., perception-reaction decision distance, maneuver distance, and queue storage distance), relates to speed transitions approaching a crossroad ramp terminal. For entrance ramps there is not any corresponding guidance in the *Green Book*.

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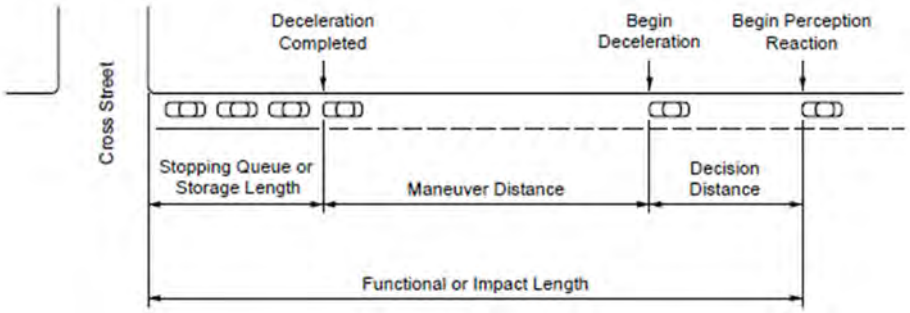


Figure 7. Elements of the Functional Area of an Intersection (AASHTO, 2011)

The guidance in Chapter 10 implies the use of the vehicle performance curves for acceleration and deceleration distances for passenger cars for coordinating the design of the ramp terminals with the controlling curve on the ramp proper. However, use of the performance curves provided in Chapter 2 of the *Green Book* is not explicitly stated or recommended in Chapter 10 as it relates to the use of *Green Book* Tables 10-1, 10-3, and 10-5. Also, associated with the need to provide transitions between the controlling curve and the ramp terminals, there may be connecting portions of the ramp that are tangent or have a radius larger than the controlling curve that have to be accounted for in the design of the entire ramp. Little guidance is provided within the *Green Book* on how the ramp terminals and adjoining sections are to be properly transitioned to connect with the controlling curve over the entire length of the ramp for a balanced ramp design that meets driver expectations.

- **No consensus on whether the ramp design speed should vary along the ramp:** Approximately 24 percent of survey respondents indicated that a single design speed should be specified for the entire ramp, and approximately 52 percent of survey respondents indicated that the ramp design speed may vary along the ramp. In many ways this relates to the need provide proper transitions between the ramp terminals, adjoining sections, and the controlling curve.
- **Important factors to consider when selecting ramp design speed:** The factors most frequently identified by survey respondents as having high or medium importance when determining ramp design speed include:
  - The design speed or operating speed of the highway.
  - Ramp configuration.
  - Design/operating speed of intersecting roadway.
  - Degree of physical constraints at site.
  - Degree of environmental constraints at site.
  - ROW costs.
- **Guidelines needed for higher ramp design speeds:** Many states are posting freeway speed limits as high as 75, 80, and 85 mph. Guidelines for ramp design speeds to address such conditions are necessary as the current guidelines only address highway design speeds up to 75 mph.
- **No specific tort liability concerns with current approach to selecting ramp design speed:** None of the state DOTs that responded to the survey indicated any specific tort

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liability concerns related to ramps designed using existing design policy for selecting ramp design speeds.

These key issues and limitations were considered in developing the Phase II work plan for this research and the guidelines that were developed.

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Section 3.
Observational Field Study

This section describes an observational field study of vehicle speeds on interchange ramps. The purpose of the study was to gather supplemental speed data along interchange ramps that could be used in combination with speed data collected along interchange ramps as part of previous research to further develop and refine speed prediction models for use in a spreadsheet-based RSPM tool that designers and traffic engineers may use to estimate vehicle speeds along interchange ramps. The research team had access to speed data collected along ramps as part of previous research (Torbic et al., 2012) and collected additional speed data at ramps to fill in gaps in these previous data sets to develop speed prediction models for interchange ramps.

3.1 Study Site Selection

As part of previous research (Torbic et al., 2012; and Torbic et al., 2016), speed data had been collected at service interchanges at both entrance and exit ramps and diagonal and loop ramps. To supplement these datasets, priority was given to collecting speed data at ramps of system interchanges and ramps along freeways with posted speeds limits of 75 mph or greater. The goal was to collect speed data at a one diagonal ramp, one loop ramp, one semidirect connection ramp, and one direct connection ramp for both entrance and exit ramps, for a total of four entrance ramps and four exit ramps.

Vehicle speeds operating in the normal traffic stream were collected at selected study locations in the metropolitan and surrounding areas of Kerrville, Texas, and Kansas City, Missouri. These areas provided a suitable variety of urban and rural ramps, high-speed ramps, and ramp configurations to satisfy the priorities and goals identified by the research team and project panel. Speeds were collected at three exit ramps and four entrance ramps.

To select study sites, research team members used aerial mapping tools to identify potential candidate locations based on ramp configuration and freeway posted speed limit, then they contacted colleagues at state DOTs to gain additional information as necessary on ramp design criteria and supplemental site characteristics.

3.2 Study Locations

The speed data along ramps were collected at selected study locations in the metropolitan and surrounding areas of Kerrville, Texas, and Kansas City, Missouri. Speed data were collected along four ramps in Texas and three ramps in Missouri. The study sites in Texas were located along Interstate I-10 between the towns of Sonora and Kerrville. The study sites in Missouri were all located near Kansas City along Interstates I-29 and I-635. Aerial images of the ramps where speed data were collected are provided in Figures 8 through 14.

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Figure 8. Loop Ramp – Westbound Exit at I-10 and Harper Rd
(Image Credit: Google Earth™ Mapping Service)



Figure 9. Loop Ramp – Eastbound Entrance at I-10 and Harper Rd
(Image Credit: Google Earth™ Mapping Service)

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Figure 10. Diagonal Ramp – Eastbound Entrance at I-10 and Martinez St
(Image credit: Google Earth™ Mapping Service)



Figure 11. Diagonal Ramp – Eastbound Exit at I-10 and US-277
(Image credit: Google Earth™ Mapping Service)

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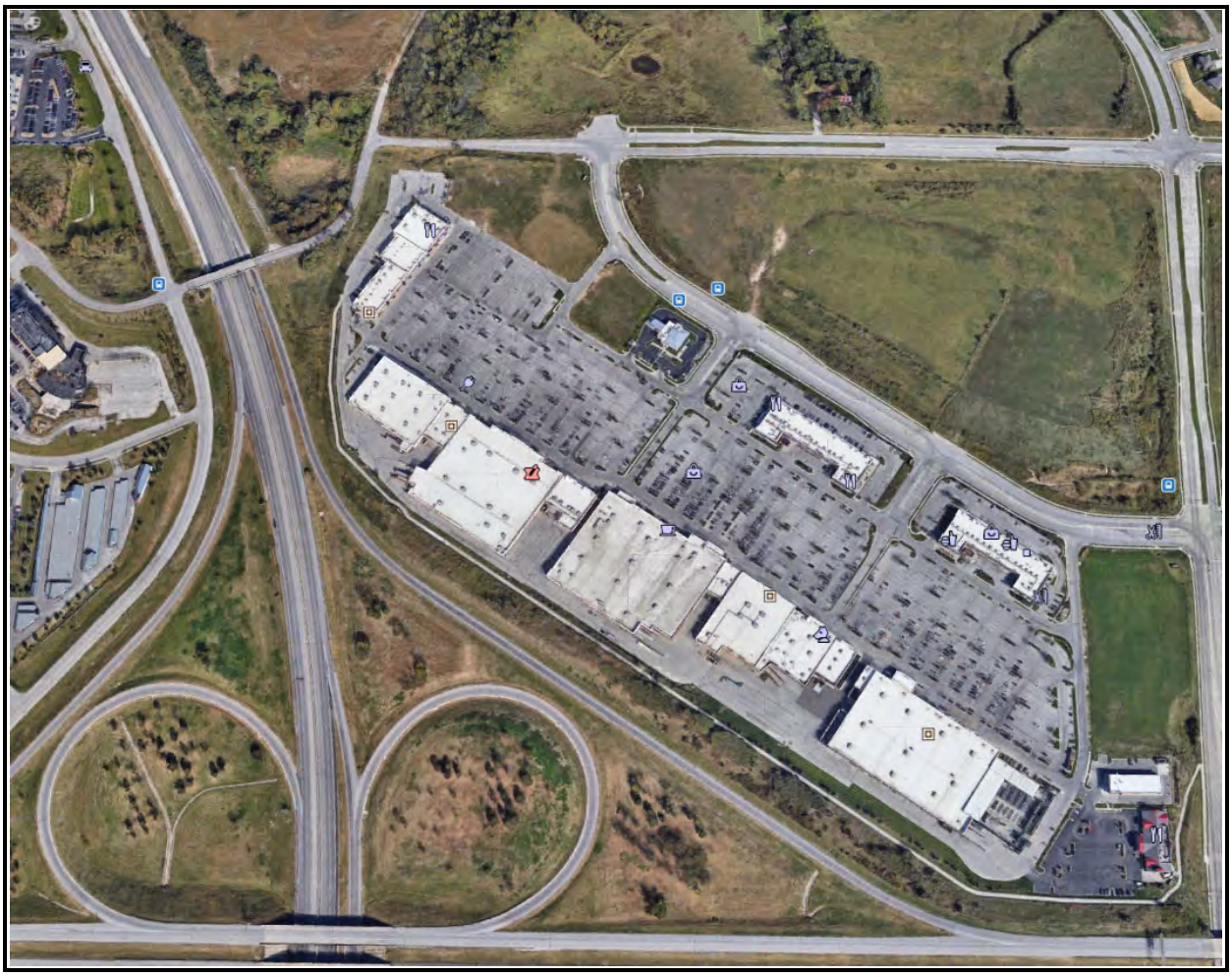


Figure 12. Outer Connection Ramp – Ramp from Westbound Missouri Route 152 to Northbound I-29 (Image credit: Google Earth™ Mapping Service)

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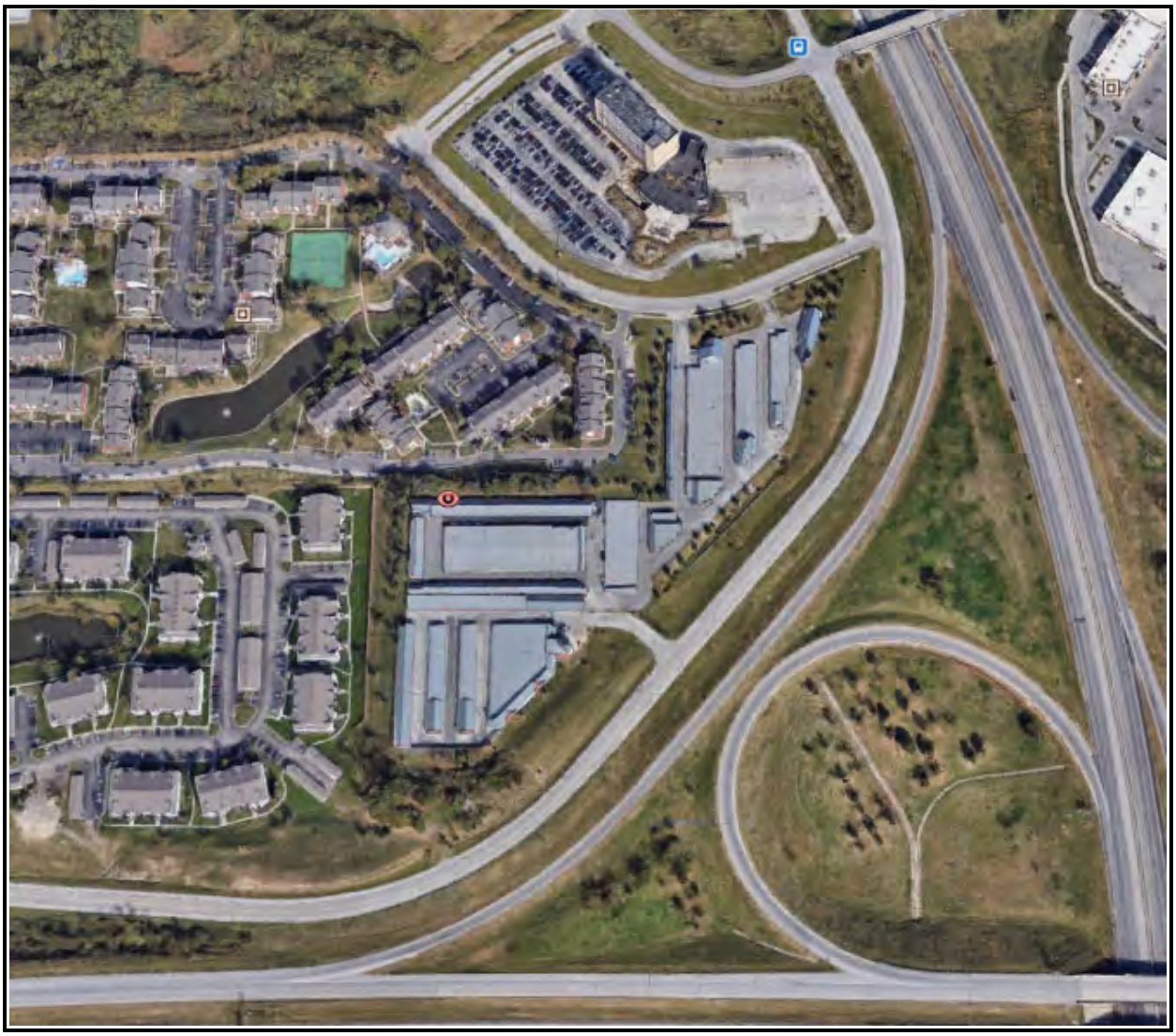


Figure 13. Outer Connection Ramp – Ramp from Southbound I-29 to Westbound Missouri Route 152 (Image credit: Google Earth™ Mapping Service)

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Figure 14. Direct Connection Ramp – Ramp from Northbound Missouri Route 9 and Southbound US-69 to Northbound I-635

(Image credit: Google Earth™ Mapping Service)

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3.3 Data Collection

Site characteristic data for the ramps were collected through a combination of reviewing plans, profile sheets, and aerial images, while other data were collected in the field or estimated based on available information. The following site characteristics were collected for the ramp proper and freeway mainline ramp terminals:

- Ramp configuration.
- Area type (dense urban, urban mixed-use, rural).
- Type of ramp (entrance, exit).
- Design speed, posted speed, and advisory speed of ramp.
- Curve radius.
- Curve length.
- Superelevation.
- Tangent length.
- Number of lanes.
- Lane and shoulder widths.
- Presence of barriers.
- Presence of compound curves.
- Grade (upgrade / downgrade).
- Presence of ramp meters.
- Type of diverge/merge.
- Taper length.
- Length from end of taper to gore nose.
- Length from gore nose to end of acceleration lane.

Speed data were collected at study sites on weekdays during non-peak hours. Data were collected using Pro Laser 3 laser guns. The data collected from the laser gun were the speeds and distances (i.e., locations) of subject vehicles along the ramps. Laser guns were hand-held and operated by a data collector on the roadside, located in a position chosen based on several criteria, including:

- Safety of data collectors and equipment.
- Minimal impact of presence of data collectors and equipment on driver behavior or desired speeds.
- View of as much of the ramp as possible.
- Minimum angle between the laser gun and the vehicles being tracked.

Speeds of free-flow vehicles were collected. Two laser guns were used to collect speeds and distances of vehicles along the ramps. The operator of the first laser gun tracked a vehicle along the ramp and then communicate by radio to the operator of the second laser gun the description (make, model, and color) of the vehicle being tracked. The operator of the second laser gun then tracked the subject vehicle when it came into view along the ramp. The operators then confirmed that the speeds and distances of the subject vehicle were successfully recorded. Speeds of vehicles were collected at critical locations along the ramp, such as the beginning (PC) and end (PT) of the controlling curve in the direction of travel on the ramp. Vehicles were classified as passenger vehicles (including light trucks and sport-utility vehicles) or trucks (including single-unit and tractor-semitrailer trucks). Table 16 indicates the posted speed limit of the freeway at each of the study ramps and the number of vehicles for which speed data were collected.

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Table 16. Summary of data collected

Freeway	Freeway/Crossroad	Ramp Configuration	Entrance or Exit Ramp	Freeway Posted Speed Limit (mph)	Number of Vehicles Observed
I-10	Harper Rd.	Loop	WB exit	75	130
I-10	Harper Rd.	Loop	EB entrance	75	128
I-10	Martinez St.	Diamond	EB entrance	80	21
I-10	US-277	Diamond	EB exit	80	94
I-29	Missouri Rt 152	Outer connection	WB MO-152 to NB I-29	70/65	125
I-29	Missouri Rt 152	Outer connection	SB I-29 to WB MO-152	70/65	110
I-635	Missouri Rt 9/US-69	Direct connection	NB MO-9/SB US-69 to NB I-635	65/55	115

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Section 4.
Development of Ramp Speed Prediction Models

Based upon a review of the speed data collected as part of this project (see Section 3) and speed data collected along ramps as part of previous research (Torbic et al., 2012), a database was assembled to develop ramp speed prediction models that could be incorporated into a spreadsheet-based RSPM tool to aid in ramp design. This section describes the database and statistical analysis used to develop the ramp speed prediction models. Separate statistical analyses were conducted to develop ramp speed prediction models for curves and tangents.

4.1 Ramp Speed Prediction Models for Curves

The objective of the statistical analysis was to develop ramp speed prediction models for curves using a database that included speed data from ramps located in Kansas, Missouri, and Texas. Several models were fit to predict vehicle speeds at the midpoint (MC) and end (PT) of curves on entrance ramps, exit ramps, and connector ramps. Only the best fitting models, based on both statistical criterion (BIC) and engineering judgment, are provided below. These are the models incorporated into the spreadsheet-based RSPM tool. Due to limited data on freeway-to-freeway ramps at system interchanges, no models were found that provided statistically reliable results that appeared valid from an engineering perspective. Therefore, only the models developed to predict vehicle speeds at the midpoint (MC) and end (PT) of curves on entrance and exit ramps at service interchanges are provided below. The number of curves and observations included in the final database used to develop models to predict vehicle speeds at the midpoint (MC) and end (PT) of curves on entrance and exit ramps is summarized in Table 17.

Table 17. Number of curves and observations in database to model speeds at midpoint and end of curves

	Entrance Ramps		Exit Ramps	
	Number of Curves	Number of Observations	Number of Curves	Number of Observations
Database Used to Model Vehicle Speeds at Midpoint of Curve (V_{MC})	11	1615	17	2086
Database Used to Model Vehicle Speeds at End of Curve (V_{PT})	11	1614	5	474

The dependent variables in the analyses were vehicle speed at the midpoint of the curve (MC) and vehicle speed at the end of the curve (PT). The candidate predictor variables included:

- Vehicle speed at the beginning of the curve (V_{PC}).
- Curve radius (R).
- Curve length (L_C).
- Ramp length (L_r).
- Gradient (level, upgrade, downgrade).
- Posted speed limit of the freeway ($V_{SL\ Frwy}$).

Table 18 provides summary statistics of the dependent and predictor variables in the ramp speed database for curves.

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Table 18. Descriptive statistics of ramp speed database for curves

	Variable	Descriptive	Entrance Ramp	Exit Ramp
Dependent Variables	Vehicle speed at midpoint of curve (V_{MC}) (mph)	Maximum	86.1	69.5
		Mean	46.3	33.8
		Standard deviation	7.3	10.1
		Median	45.9	32.0
		Minimum	17.3	12.0
	Vehicle speed at end of curve (V_{PT}) (mph)	Maximum	87.0	69.3
		Mean	47.0	43.5
		Standard deviation	8.2	11.6
		Median	46.7	46.1
		Minimum	11.7	10.0
Predictor Variables	Vehicle speed at beginning of curve (V_{PC}) (mph)	Maximum	85.1	69.6
		Mean	44.5	40.6
		Standard deviation	7.5	8.9
		Median	44.8	40.8
		Minimum	15.5	15.0
	Curve radius (R) (ft)	Maximum	12101	1848
		Mean	2376	475
		Standard deviation	2957	475
		Median	1531	264
		Minimum	158	106
	Curve length (L_C) (mi)	Maximum	0.12	0.20
		Mean	0.04	0.11
		Standard deviation	0.03	0.05
		Median	0.04	0.09
		Minimum	0.01	0.01
	Ramp length (L_r) (mi)	Maximum	0.70	0.37
		Mean	0.26	0.21
		Standard deviation	0.17	0.07
		Median	0.21	0.19
		Minimum	0.13	0.14
	Gradient (count/proportion)	Level	1306 (0.17)	873 (0.30)
		Upgrade	122 (0.07)	1165 (0.40)
		Downgrade	300 (0.76)	855 (0.30)
	Posted speed limit of the freeway ($V_{SL\ Frwy}$) (mph)	Maximum	75	80
		Mean	61	62
		Standard deviation	4.9	5.5
		Median	60	60
		Minimum	55	50

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4.1.1 Entrance Ramps

Several models were fit to predict vehicle speeds at the midpoint (MC) and end (PT) of curves on entrance ramps. Only the best fitting models are provided here based on both statistical criterion (BIC) and engineering judgment. In the final models, vehicle speed at the beginning of the curve (V_{PC}), curve radius (R), curve radius squared (R^2), and the posted speed limit of the freeway ($V_{SL\ Frwy}$) were treated as fixed effects, while the Curve ID was treated as a random effect.

The final model to predict vehicle speeds at the midpoint of curves on entrance ramps is provided as Equation 33. It is based on data from 1615 vehicles observed on 11 curves. Table 19 provides a summary of the statistical results of the final model to predict vehicle speeds at the midpoint of curves on entrance ramps, and Figure 15 provides a plot of the actual to predicted values.

$$V_{MC} = -8.7255 + 1.0125V_{PC} + 4.7053R - 2.0183R^2 + 0.1316V_{SL\ Frwy} \quad (33)$$

where:

V_{MC} = Predicted vehicle speed at the MC, mph.
V_{PC} = Vehicle speed at the beginning of a curve (i.e., initial point of horizontal curve – PC), mph.
R = Curve radius, mi.
$V_{SL\ Hwy}$ = Regulatory speed limit on the highway (or freeway) mainline, mph.

Table 19. Statistical parameters of final model to estimate vehicle speed at midpoint of curve on an entrance ramp

Summary of Fit						
RSquare	0.983638					
RSquare Adj	0.983598					
Root Mean Square Error	0.932094					
Mean of Response	46.29867					
Observations (or Sum Wgts)	1615					
Parameter Estimates						
Term		**Estimate**	**Std Error**	**DFDen**	**t Ratio**	**Prob>|t|**
Intercept		-8.725519	2.755146	7.15	-3.17	0.0153
V_{PC} (mph)		1.012546	0.004218	1609	240.05	<.0001
R (mi)		4.705275	0.967322	7.084	4.86	0.0018
R^2 (mi)		-2.018305	0.388952	7.018	-5.19	0.0013
$V_{SL\ Frwy}$ (mph)		0.131626	0.046524	7.104	2.83	0.0250
REML Variance Component Estimates						
Random Effect	**Var Ratio**	**Var Component**	**Std Error**	**95% Lower**	**95% Upper**	**Pct of Total**
Curve_ID	0.226312	0.1966198	0.1087033	-0.016435	0.4096744	18.455
Residual		0.8687993	0.0306873	0.811656	0.9322278	81.545
Total		1.0654191	0.1128967	0.874650	1.3265609	100.000
-2 LogLikelihood = 4405.6999379						
Note: Total is the sum of the positive variance components.						
Total including negative estimates = 1.0654191						
Fixed Effect Tests						
Source	**Nparm**	**DF**	**DFDen**	**F Ratio**	**Prob > F**	
V_{PC} (mph)	1	1	1609	57624.11	<.0001*	
R (mi)	1	1	7.084	23.6607	0.0018*	
R^2 (mi)	1	1	7.018	26.9266	0.0013*	
$V_{SL\ Frwy}$ (mph)	1	1	7.104	8.0044	0.0250*	

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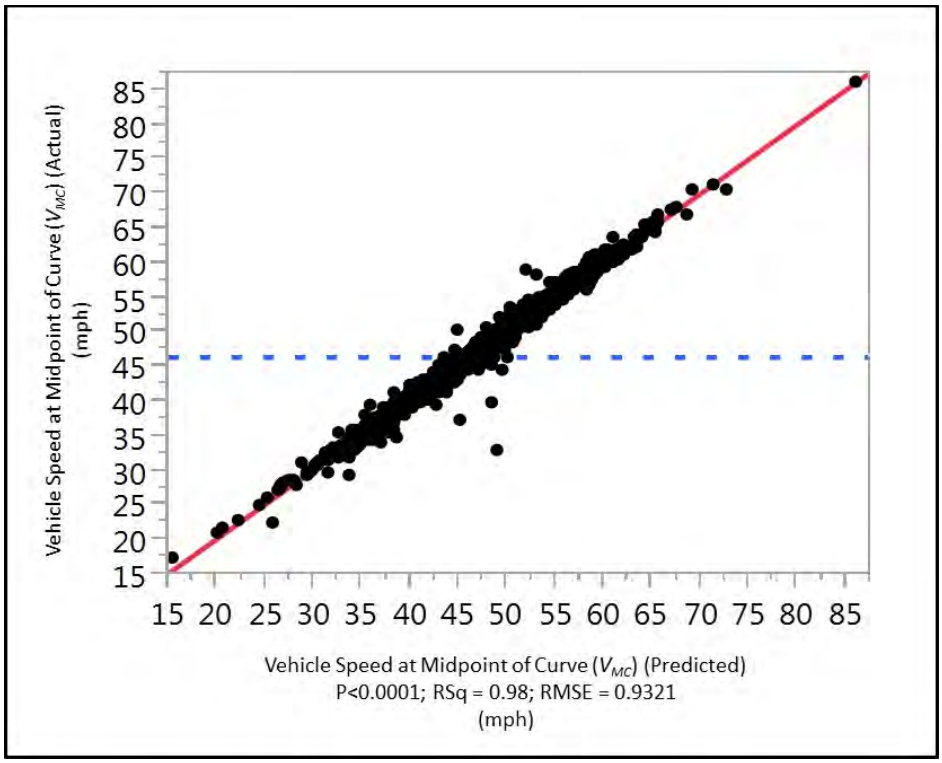


Figure 15. Plot of actual versus predicted values of final model to estimate vehicle speed at midpoint of curve on an entrance ramp

The final model to predict vehicle speeds at the end of curves on entrance ramps is provided as Equation 34. It is based on data from 1614 vehicles observed on 11 curves. Table 20 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of curves on entrance ramps, and Figure 16 provides a plot of the actual to predicted values.

$$V_{PT} = -12.1179 + 1.0127V_{PC} + 8.3558R - 3.5183R^2 + 0.1882V_{SL\,Frwy} \quad (34)$$

where:

V_{PT} = Predicted vehicle speed at the end of a curve (PT), mph.

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Table 20. Statistical parameters of final model to estimate vehicle speed at end of curve on an entrance ramp

Summary of Fit	
RSquare	0.943958
RSquare Adj	0.943818
Root Mean Square Error	1.835069
Mean of Response	47.25775
Observations (or Sum Wgts)	1614

Parameter Estimates

Term	**Estimate**	**Std Error**	**DFDen**	**t Ratio**	**Prob>|t|**
Intercept	-12.11791	3.976288	7.247	-3.05	0.0179
V_{PC} (mph)	1.01265	0.008338	1596	121.44	<.0001
R (mi)	8.35582	1.393358	7.127	6.00	0.0005
R^2 (mi)	-3.51828	0.559098	7.002	-6.29	0.0004
$V_{SL\ Frwy}$ (mph)	0.18820	0.067053	7.152	2.81	0.0257

REML Variance Component Estimates

Random Effect	**Var Ratio**	**Var Component**	**Std Error**	**95% Lower**	**95% Upper**	**Pct of Total**
Curve_ID	0.1168546	0.3935055	0.2247703	-0.047036	0.8340472	10.463
Residual		3.3674784	0.1189807	3.1459274	3.6134073	89.537
Total		3.7609839	0.2539168	3.3090162	4.3128333	100.000

-2 LogLikelihood = 6578.4878322
Note: Total is the sum of the positive variance components.

Total including negative estimates = 3.7609839

Fixed Effect Tests

Source	**Nparm**	**DF**	**DFDen**	**F Ratio**	**Prob > F**
V_{PC} (mph)	1	1	1596	14748.62	<.0001
R (mi)	1	1	7.127	35.9627	0.0005
R^2 (mi)	1	1	7.002	39.5991	0.0004
$V_{SL\ Frwy}$ (mph)	1	1	7.152	7.8781	0.0257

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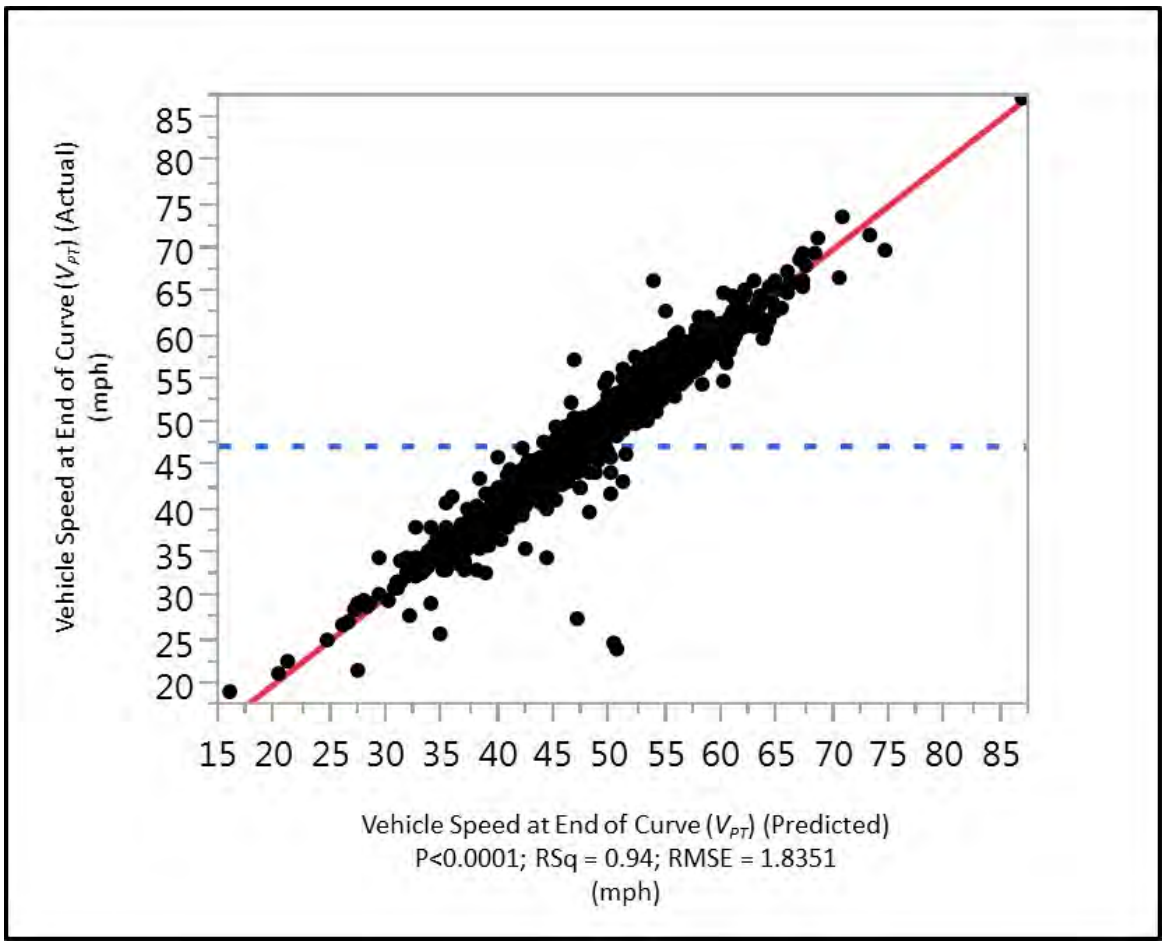


Figure 16. Plot of actual versus predicted values of final model to estimate vehicle speed at end of curve on an entrance ramp

4.1.2 Exit Ramps

Several models were fit to predict vehicle speeds at the midpoint (MC) and end (PT) of curves on exit ramps. Only the best fitting models are provided here based on both statistical criterion (BIC) and engineering judgment. In the final models, vehicle speed at the beginning of the curve (V_{PC}), curve radius (R), curve radius squared (R^2), curve length (L_C), gradient, and the posted speed limit of the freeway ($V_{SL\ Hwy}$) were treated as fixed effects, while the Curve ID was treated as a random effect.

The final model to predict vehicle speeds at the midpoint of curves on exit ramps is provided as Equation 35. It is based on data from 2086 vehicles observed on 17 curves. Table 21 provides a summary of the statistical results of the final model to predict vehicle speeds at the midpoint of curves on exit ramps, and Figure 17 provides a plot of the actual to predicted values.

$$V_{MC} = -13.4726 + 0.5951V_{PC} + 208.5633R - 521.3073R^2 + 0.2361V_{SL\ Hwy} + 2.1981I_{DN} + 0.7507\ I_{LV} - 2.9488\ I_{UP} \quad (35)$$

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where:

I_{DN} = Indicator variable for downward slope (= 1 if the overall ramp slopes downward, 0 otherwise).
I_{LV} = Indicator variable for level slope [= 1 if the overall ramp is level (i.e., ± 2 percent), 0 otherwise].
I_{UP} = Indicator variable for upward slope (= 1 if the overall ramp slopes upward, 0 otherwise).

Table 21. Statistical parameters of final model to estimate vehicle speed at midpoint of curve on an exit ramp

Summary of Fit

RSquare	0.931439
RSquare Adj	0.931241
Root Mean Square Error	2.645201
Mean of Response	33.75379
Observations (or Sum Wgts)	2086

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	-13.47259	8.7789	11.08	-1.53	0.1529
V_{PC} (mph)	0.59513	0.0126	2075	47.13	<.0001
R (mi)	208.56333	44.2822	11.08	4.71	0.0006
R^2 (mi)	-521.30730	134.9935	11.07	-3.86	0.0026
$V_{SL\ Hwy}$ (mph)	0.23606	0.1343	11.06	1.76	0.1064
I_{DN}	2.19809	1.4102	11.15	1.56	0.1470
I_{LV}	0.75067	1.6935	11.15	0.44	0.6661
I_{UP}	-2.94880				

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Curve_ID	1.1879258	8.31202	3.5684894	1.31791	15.306132	54.295
Residual		6.99709	0.2175978	6.58945	7.443952	45.705
Total		15.30911	3.5750283	10.15883	25.690158	100.000

-2 LogLikelihood = 10040.753648
Note: Total is the sum of the positive variance components.

Total including negative estimates = 15.30911

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F
V_{PC} (mph)	1	1	2075	2221.6480	<.0001
R (mi)	1	1	11.08	22.1828	0.0006
R^2 (mi)	1	1	11.07	14.9129	0.0026
$V_{SL\ Hwy}$ (mph)	1	1	11.06	3.0886	0.1064
Gradient	2	2	11.10	5.0777	0.0272

Effect Details (Gradient)
Least Squares Means Table

Level	Least Sq Mean	Std Error
I_{DN}	36.578765	1.4599695
I_{LV}	35.131343	2.1113467
I_{UP}	31.431912	1.0579163

LSMeans Differences Tukey HSD
α = 0.050

Level			Least Sq Mean
I_{DN}	A		36.578765
I_{LV}	A	B	35.131343
I_{UP}		B	31.431912

Levels not connected by same letter are significantly different.

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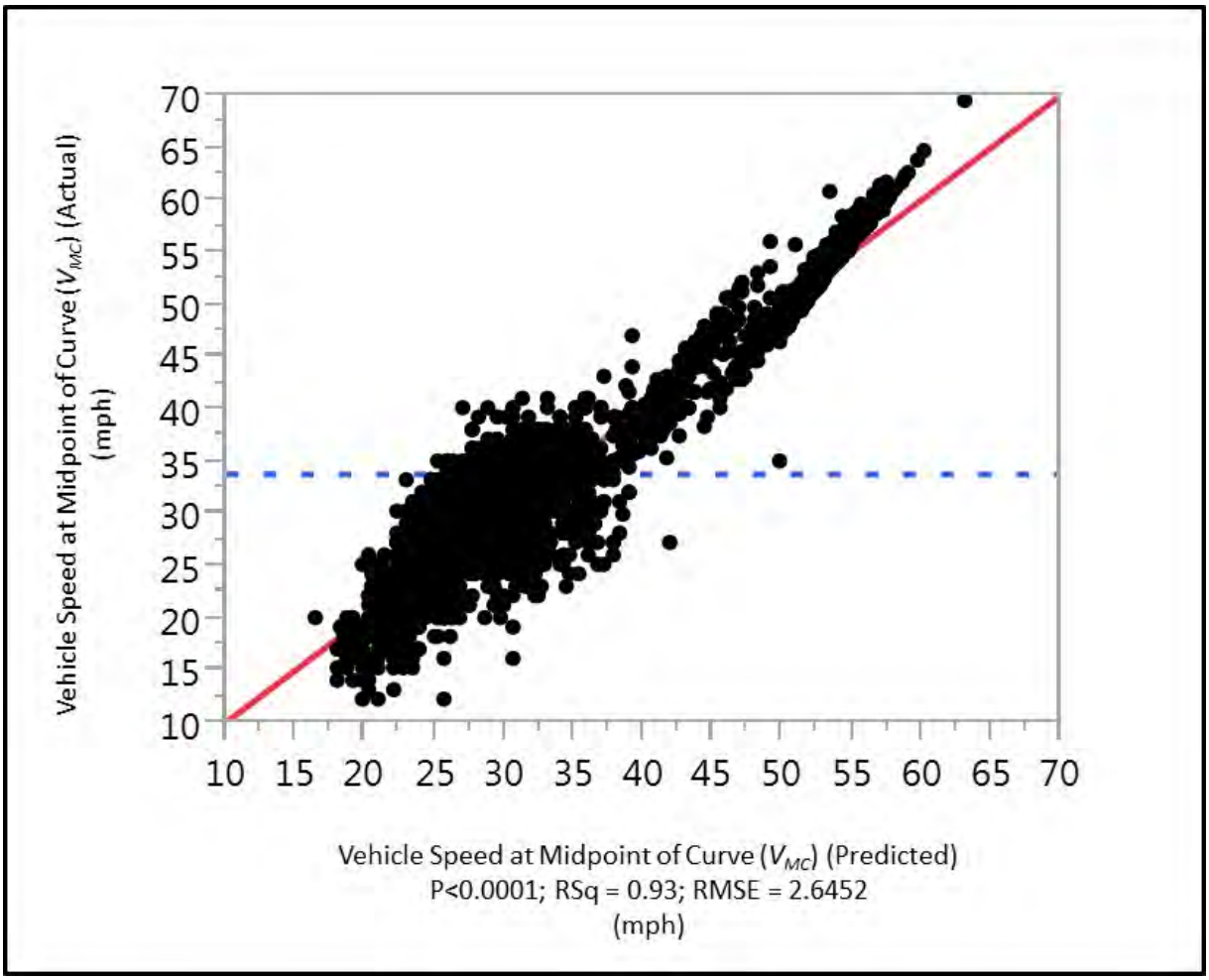


Figure 17. Plot of actual versus predicted values of final model to estimate vehicle speed at midpoint of curve on an exit ramp

The final model to predict vehicle speeds at the end of curves on exit ramps is provided as Equation 36. It is based on data from 474 vehicles observed on five curves. Table 22 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of curves on exit ramps, and Figure 18 provides a plot of the actual to predicted values.

$$V_{PT} = -0.6272 + 0.8637V_{PC} + 108.0929R - 265.9747R^2 - 190.89.8941L_C \quad (36)$$

where:

L_c = Curve segment length, mi.

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Table 22. Statistical parameters of final model to estimate vehicle speed at end of curve on an exit ramp

Summary of Fit	
RSquare	0.95242
RSquare Adj	0.95201
Root Mean Square Error	2.55227
Mean of Response	43.54403
Observations (or Sum Wgts)	474

Parameter Estimates

Term	**Estimate**	**Std Error**	**DFDen**	**t Ratio**	**Prob>|t|**
Intercept	-0.6272	2.50145	1.396	-0.25	0.8339
V_{PC} (mph)	0.8637	0.02292	441.2	37.69	<.0001
R (mi)	108.0929	23.70501	0.999	4.56	0.1376
R^2 (mi)	-265.9747	62.03266	0.971	-4.29	0.1514
L_C (mi)	-190.8941	14.25417	1.007	-13.39	0.0467

REML Variance Component Estimates

Random Effect	**Var Ratio**	**Var Component**	**Std Error**	**95% Lower**	**95% Upper**	**Pct of Total**
Curve_ID	0.0571489	0.3722732	0.6345339	-0.87139	1.615937	5.406
Residual		6.5140974	0.4258399	5.75387	7.436460	94.594
Total		6.8863706	0.7613931	5.60761	8.661051	100.000

-2 LogLikelihood = 2223.6163292
Note: Total is the sum of the positive variance components.

Total including negative estimates = 6.8863706

Fixed Effect Tests

Source	**Nparm**	**DF**	**DFDen**	**F Ratio**	**Prob > F**
V_{PC} (mph)	1	1	441.2	1420.360	<.0001
R (mi)	1	1	0.999	20.793	0.1376
R^2 (mi)	1	1	0.971	18.384	0.1514
L_C (mi)	1	1	1.007	179.350	0.0467

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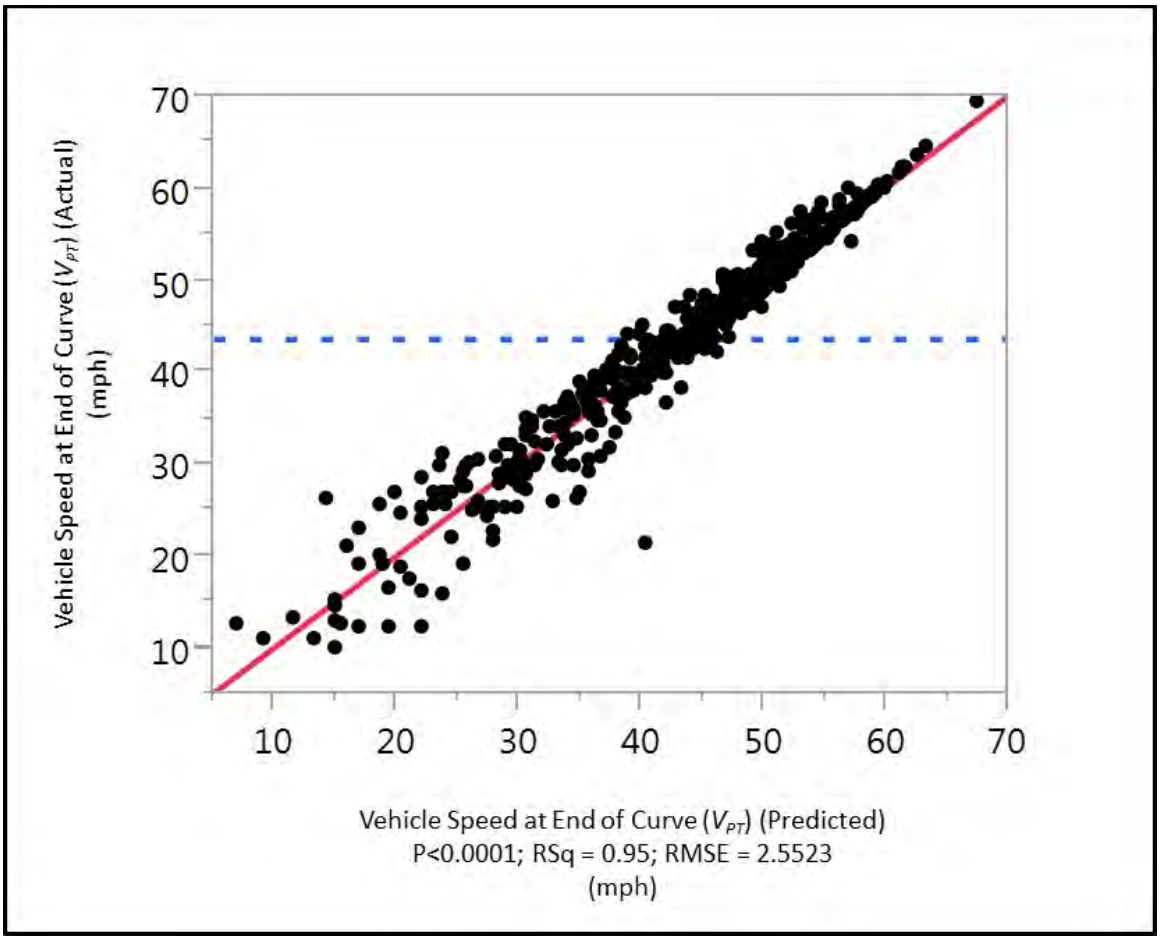


Figure 18. Plot of actual versus predicted values of final model to estimate vehicle speed at end of curve on an exit ramp

4.2 Ramp Speed Prediction Models for Tangents

The objective of this statistical analysis was to develop ramp speed prediction models for tangents using an assembled database that included speed data from ramps located in Kansas, Missouri, and Texas. Speed data consisted of observations at the beginning of each tangent, at a point one-third of the tangent length, at a point two-thirds of the tangent length, and at the end of the tangent. Several models were fit, and it was decided to focus on predicting vehicle speeds at the end of the tangent for tangents with subsequent curves and for tangents without subsequent curves on entrance ramps, exit ramps, and connector ramps. Only the best fitting models, based on both statistical criterion (BIC) and engineering judgment, are provided below. These are the models incorporated into the spreadsheet-based RSPM tool. Due to limited data for tangents on freeway-to-freeway ramps, no models were found that provided statistically reliable results that appeared valid from an engineering perspective. Therefore, only the models developed to predict vehicle speeds at the end of the tangent, for tangents with and without subsequent curves on entrance and exit ramps at service interchanges are provided below. The number of tangents and observations included in the final database used to develop models to predict vehicle speeds at the end of the tangent for tangents with subsequent curves and without subsequent curves on entrance and exit ramps is summarized in Table 23.

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Table 23. Number of tangents and observations in database to model speeds at the end of the tangent

	Entrance Ramps		Exit Ramps	
	Number of Tangents	**Number of Observations**	**Number of Tangents**	**Number of Observations**
Database used to model vehicle speed at end of Tangent with subsequent curve ($V_{t,e,c}$)	5	546	6	840
Database used to model vehicle speed at end of tangent without subsequent curve ($V_{t,e,nc}$)	5	753	1	77

The dependent variables in the analyses were vehicle speed at the end of the tangent with a subsequent curve ($V_{t,e,c}$) and without a subsequent curve ($V_{t,e,nc}$). The candidate predictor variables included:

- Vehicle speed at the beginning of the tangent ($V_{Tan\text{-}Begin}$).
- Curve radius of subsequent curve (R_{next}).
- Tangent length (L_T).
- Ramp length (L_r).
- Gradient (level, upgrade, downgrade).
- Posted speed limit of the freeway ($V_{SL\ Hwy}$).

Table 24 provides summary statistics of the dependent and predictor variables in the ramp speed database for tangents.

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Table 24. Descriptive statistics of ramp speed database for tangents

	Variable	Descriptive	Entrance	Exit
Dependent Variable	Vehicle speed at end of tangent ($V_{Tan\text{-}End}$) (mph)	Maximum	72.4	64.3
		Mean	47.4	42.6
		Standard deviation	8.0	9.1
		Median	47.3	42.9
		Minimum	14.4	5.7
Predictor Variables	Vehicle speed at beginning of tangent ($V_{Tan\text{-}Begin}$) (mph)	Maximum	69.7	68.8
		Mean	41.0	53.0
		Standard deviation	10.3	5.4
		Median	42.9	53.0
		Minimum	10.7	21.3
	Curve radius of subsequent curve (R_{next}) (mi)	Maximum	18.9	18.9
		Mean	11.1	2.0
		Standard deviation	9.1	5.6
		Median	18.9	0.10
		Minimum	0.13	0.02
	Tangent length (L_T) (mi)	Maximum	0.1	0.18
		Mean	0.06	0.07
		Standard deviation	0.02	0.03
		Median	0.07	0.07
		Minimum	0.03	0.02
	Ramp length (L_r) (mi)	Maximum	0.70	0.19
		Mean	0.27	0.16
		Standard deviation	0.17	0.02
		Median	0.21	0.17
		Minimum	0.13	0.14
	Gradient (count/proportion)	Level	897 (0.69)	727 (0.78)
		Upgrade	216 (0.17)	200 (0.22)
		Downgrade	186 (0.14)	4 (0.00)
	Posted speed limit of the freeway ($V_{SL\ Hwy}$) (mph)	Maximum	65	80
		Mean	60.1	61.6
		Standard deviation	3.0	2.6
		Median	60	60
		Minimum	55	60

4.2.1 Entrance Ramps

Several models were fit to predict vehicle speeds at the end of a tangent for tangents with subsequent curves and for tangents without subsequent curves on entrance ramps. Only the best fitting models are provided here based on both statistical criterion (BIC) and engineering judgment. In the final models, vehicle speed at the beginning of the tangent ($V_{Tan\text{-}Begin}$), curve radius of subsequent curve (R_{next}), tangent length (L_T), ramp length (L_r), gradient, and the posted speed limit of the freeway ($V_{SL\ Hwy}$) were treated as fixed effects, while the Tangent_ID was treated as a random effect.

The final model to predict vehicle speeds at the end of a tangent followed by a subsequent curve on entrance ramps is provided as Equation 37. It is based on data from 546 vehicles observed on five tangents.

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Table 25 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of tangents followed by a subsequent curve on entrance ramps, and Figure 19 provides a plot of the actual to predicted values.

$$V_{Tan\text{-}End,C} = 0.9667V_{Tan\text{-}Begin} + 143.9664L_T - 5.3122I_{55} - 2.6028I_{60} - 7.9150I_{65} \quad (37)$$

where:

$V_{Tan\text{-}End,C}$ = Predicted vehicle speed at the end of a tangent if the tangent is followed by a curve, mph.
$V_{Tan\text{-}Begin}$ = Predicted vehicle speed at the beginning of a tangent, mph.
L_T = Tangent length, mi.
I_{55} = Indicator variable for 55-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 55 mph, 0 otherwise).
I_{60} = Indicator variable for 60-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 60 mph, 0 otherwise).
I_{65} = Indicator variable for 65-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 65 mph, 0 otherwise).

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Table 25. Statistical parameters of final model to estimate vehicle speed at end of tangent with subsequent curve on an entrance ramp

Summary of Fit

RSquare	0.82122
RSquare Adj	0.82023
Root Mean Square Error	3.21828
Mean of Response	43.67047
Observations (or Sum Wgts)	546

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
$V_{Tan\text{-}Begin}$ (mph)	0.96669	0.031027	504.7	31.16	<.0001
L_T (mi)	143.96639	32.3598	3.016	4.45	0.0209
I_{55}	-5.31221	3.32688	2.137	-1.60	0.2435
I_{60}	-2.60280	2.30603	2.016	-1.13	0.3754
I_{65}	7.91500				

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Tangent_ID	1.4832228	15.362187	15.612177	-15.23712	45.961492	59.730
Residual		10.357303	0.6303256	9.2249577	11.71288	40.270
Total		25.71949	15.624279	10.310942	140.05874	100.000

-2 LogLikelihood = 2836.5425462
Note: Total is the sum of the positive variance components.

Total including negative estimates = 25.71949

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F
$V_{Tan\text{-}Begin}$ (mph)	1	1	504.7	970.7041	<.0001
L_T (mi)	1	1	3.016	19.7930	0.0209
$V_{SL\ Hwy}$	2	2	2.15	3.1259	0.2310

Effect Details ($V_{SL\ Frwy}$)
Least Squares Means Table

Level	Least Sq Means	Std Error
I_{55}	38.466183	4.4690497
I_{60}	41.175590	2.3541431
I_{65}	51.693413	3.5073697

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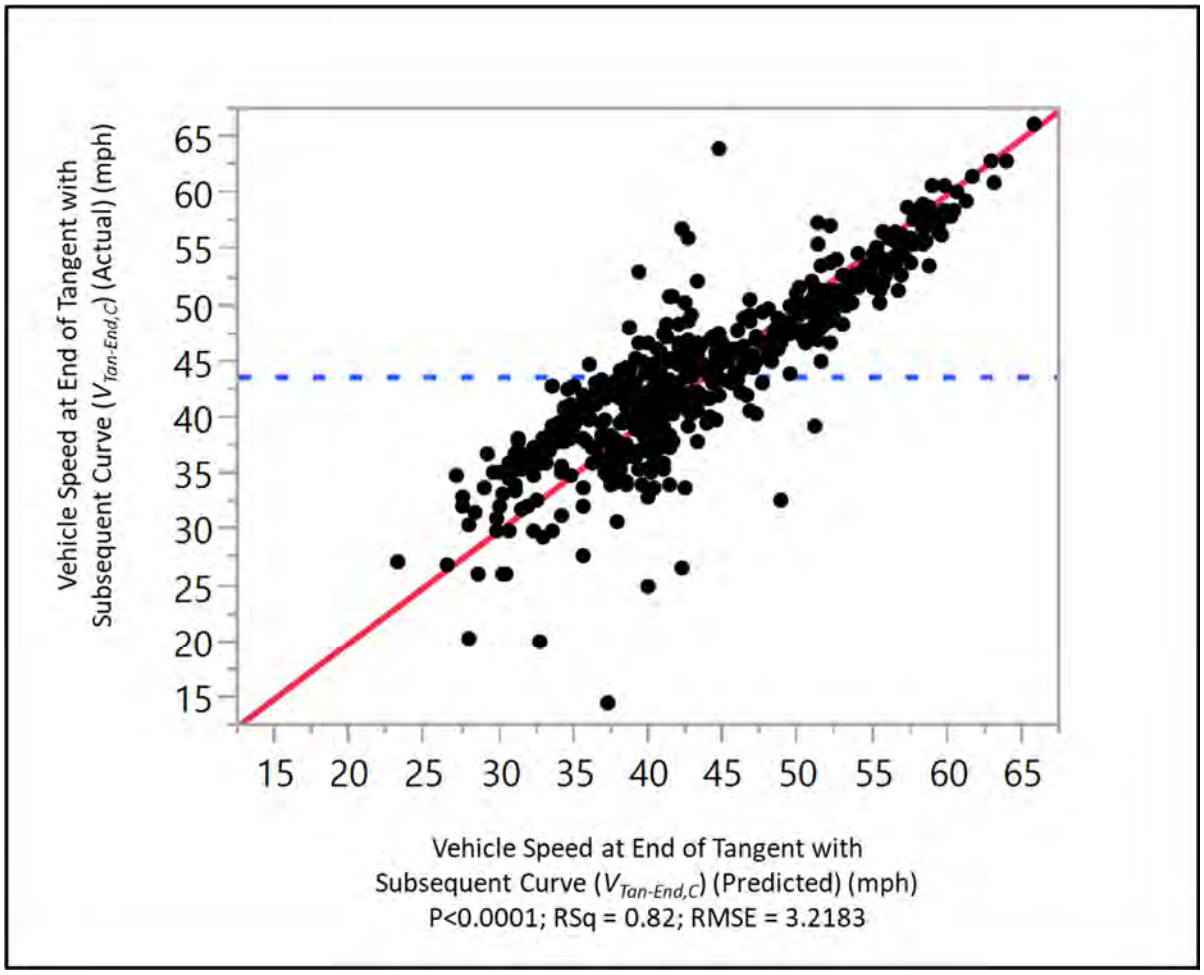


Figure 19. Plot of actual versus predicted values of final model to estimate vehicle speed at end of tangent followed by subsequent curve on an entrance ramp

The final model to predict vehicle speeds at the end of a tangent without a subsequent curve on entrance ramps is provided as Equation 38. It is based on data from 753 vehicles observed on 5 tangents. Table 26 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of tangents without a subsequent curve on entrance ramps, and Figure 20 provides a plot of the actual to predicted values.

$$V_{Tan\text{-}End,NC} = 1.0118V_{Tan\text{-}Begin} + 78.3087L_T \quad (38)$$

where:

$V_{Tan\text{-}End,NC}$ = Predicted vehicle speed at the end of a tangent if the tangent is not followed by a curve, mph.

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Table 26. Statistical parameters of final model to estimate vehicle speed at end of tangent without subsequent curve on an entrance ramp

Summary of Fit

RSquare	0.89381
RSquare Adj	0.89366
Root Mean Square Error	2.36062
Mean of Response	50.12304
Observations (or Sum Wgts)	753

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
$V_{Tan\text{-}Begin}$ (mph)	1.011792	0.01672	330	60.52	<.0001
L_T (mi)	78.308715	19.42207	6.994	4.03	0.0050

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Tangent_ID	0.8901097	4.9601523	3.7301438	-2.350795	12.2711	47.093
Residual		5.5725182	0.2883932	5.0478176	6.183946	52.907
Total		10.532671	3.7393175	5.8301529	24.49974	100.000

-2 LogLikelihood = 3451.9758141
Note: Total is the sum of the positive variance components.

Total including negative estimates = 10.532671

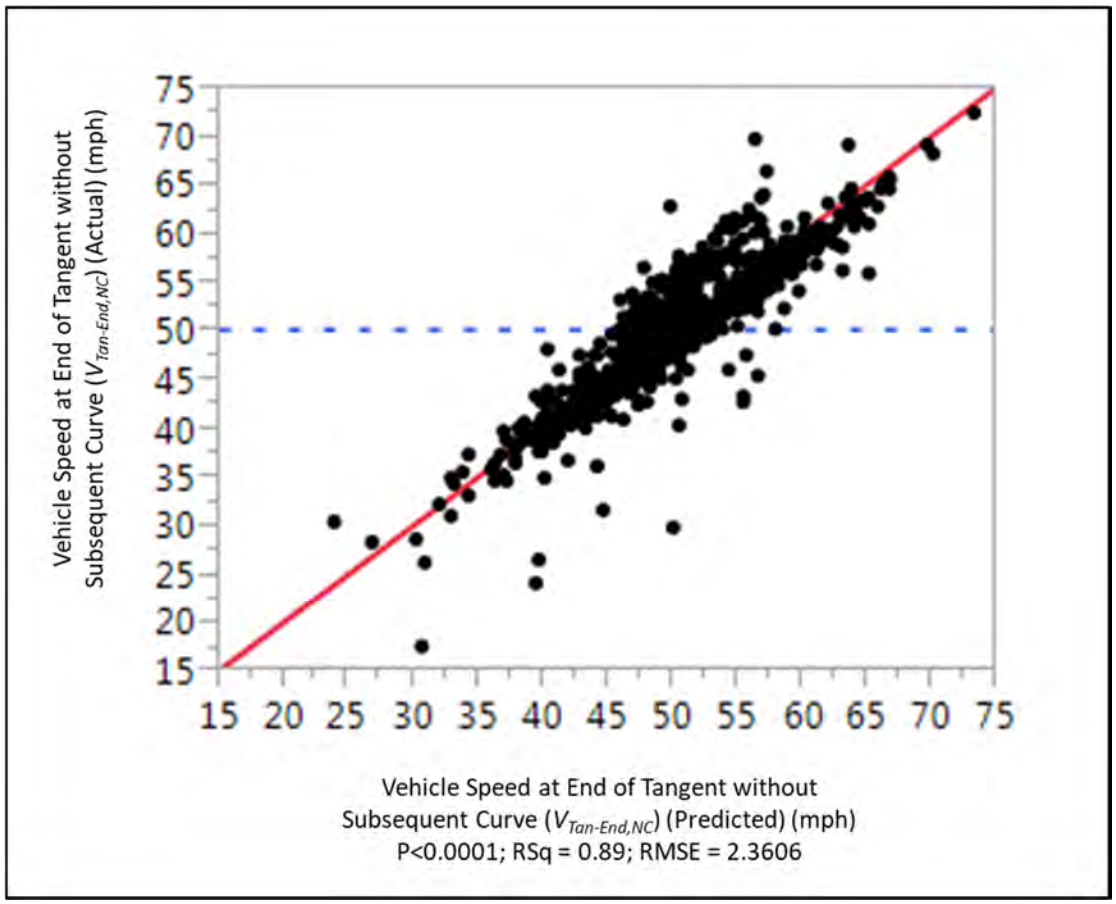


Figure 20. Plot of actual versus predicted values of final model to estimate vehicle speed at end of tangent without subsequent curve on an entrance ramp

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4.2.2 Exit Ramps

Several models were fit to predict vehicle speeds at the end of a tangent for tangents with subsequent curves and for tangents without subsequent curves on exit ramps. Only the best fitting models are provided here based on both statistical criterion (BIC) and engineering judgment. In the final models, vehicle speed at the beginning of the tangent ($V_{Tan\text{-}Begin}$), curve radius of subsequent curve (R_{next}), and tangent length (L_T) were treated as fixed effects, while the tangent ID was treated as a random effect.

The final model to predict vehicle speeds at the end of a tangent followed by a subsequent curve on exit ramps is provided as Equation 39. It is based on data from 840 vehicles observed on six tangents. Table 27 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of tangents followed by a subsequent curve on exit ramps, and Figure 21 provides a plot of the actual to predicted values.

$$V_{Tan\text{-}End,C} = 0.7980V_{Tan\text{-}Begin} + 228.6196R_{next} - 575.2145R_{next}^2 - 645.4397L_T + 6733.5203L_T^2 \quad (39)$$

where:

R_{next} = Radius of subsequent curve, mi.

Table 27. Statistical parameters of final model to estimate vehicle speed at end of tangent with subsequent curve on an exit ramp

Summary of Fit						
RSquare	0.885613					
RSquare Adj	0.885066					
Root Mean Square Error	2.482274					
Mean of Response	44.39558					
Observations (or Sum Wgts)	840					
Parameter Estimates						
Term		**Estimate**	**Std Error**	**DFDen**	**t Ratio**	**Prob>|t|**
$V_{Tan\text{-}Begin}$ (mph)		0.7978	0.0183	833.8	43.71	<.0001
R_{next} (mi)		228.6196	45.8453	4.228	4.99	0.0065
R_{next}^2 (mi)		-575.2145	146.1954	4.002	-3.93	0.0170
L_T (mi)		-645.4397	164.7479	4.636	-3.92	0.0130
L_T^2(mi)		6733.5203	2091.9690	4.172	3.22	0.0304
REML Variance Component Estimates						
Random Effect	**Var Ratio**	**Var Component**	**Std Error**	**95% Lower**	**95% Upper**	**Pct of Total**
Tangent_ID	0.2516656	1.5506842	1.5990759	-1.583447	4.6848154	20.106
Residual		6.1616863	0.3019219	5.6102467	6.7990578	79.894
Total		7.7123705	1.6268531	5.3048116	12.238543	100.000
-2 LogLikelihood = 3894.0927503						
Note: Total is the sum of the positive variance components.						
Total including negative estimates = 7.7123705						

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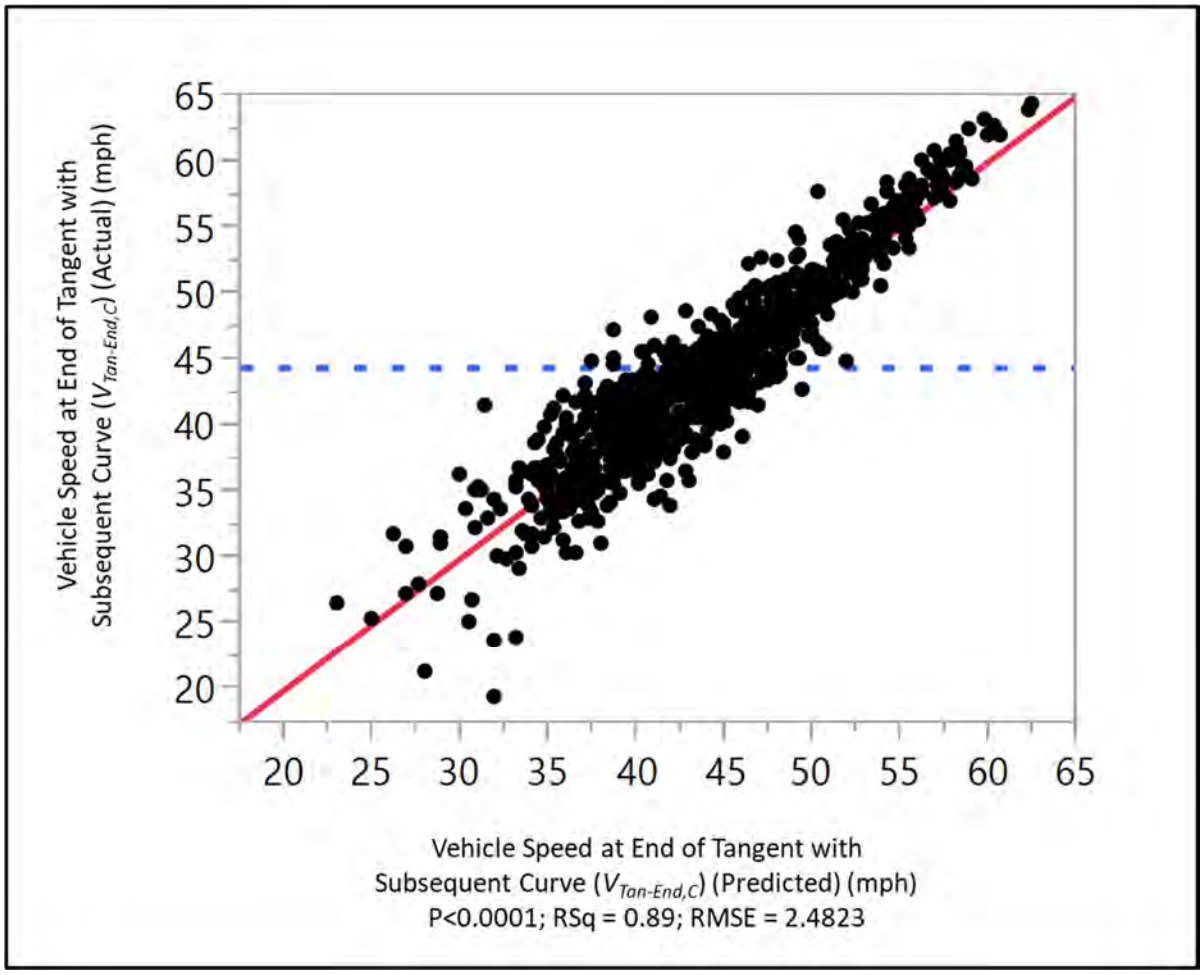


Figure 21. Plot of actual versus predicted values of final model to estimate vehicle speed at end of tangent with subsequent curve on an exit ramp

The final model to predict vehicle speeds at the end of a tangent without a subsequent curve on exit ramps is provided as Equation 40. It is based on data from 71 vehicles observed on one tangent. Table 28 provides a summary of the statistical results of the final model to predict vehicle speeds at the end of tangents without a subsequent curve on exit ramps, and Figure 22 provides a regression plot of the speed at the beginning of the tangent and speed at the end of the tangent and the fitted regression line.

$$V_{Tan\text{-}End,NC} = 0.4890 V_{Tan\text{-}Begin} \quad (40)$$

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Table 28. Statistical parameters of final model to estimate vehicle speed at end of tangent without subsequent curve on an exit ramp

Summary of Fit	
Root Mean Square Error	3.756292
Mean of Response	26.36756
Observations (or Sum Wgts)	77

Parameter Estimates

Term	**Estimate**	**Std Error**	**t Ratio**	**Prob>|t|**
$V_{Tan\text{-}Begin}$ (mph)	0.4890396	0.007894	61.95	<.0001

Analysis of Variance

Source	**DF**	**Sum of Squares**	Mean Square	F Ratio
Model	1	54149.423	54149.4	3837.737
Error	76	1072.339	14.1	**Prob > F**
C Total	77	55221.762		<.0001

Tested against reduced model: Y=0

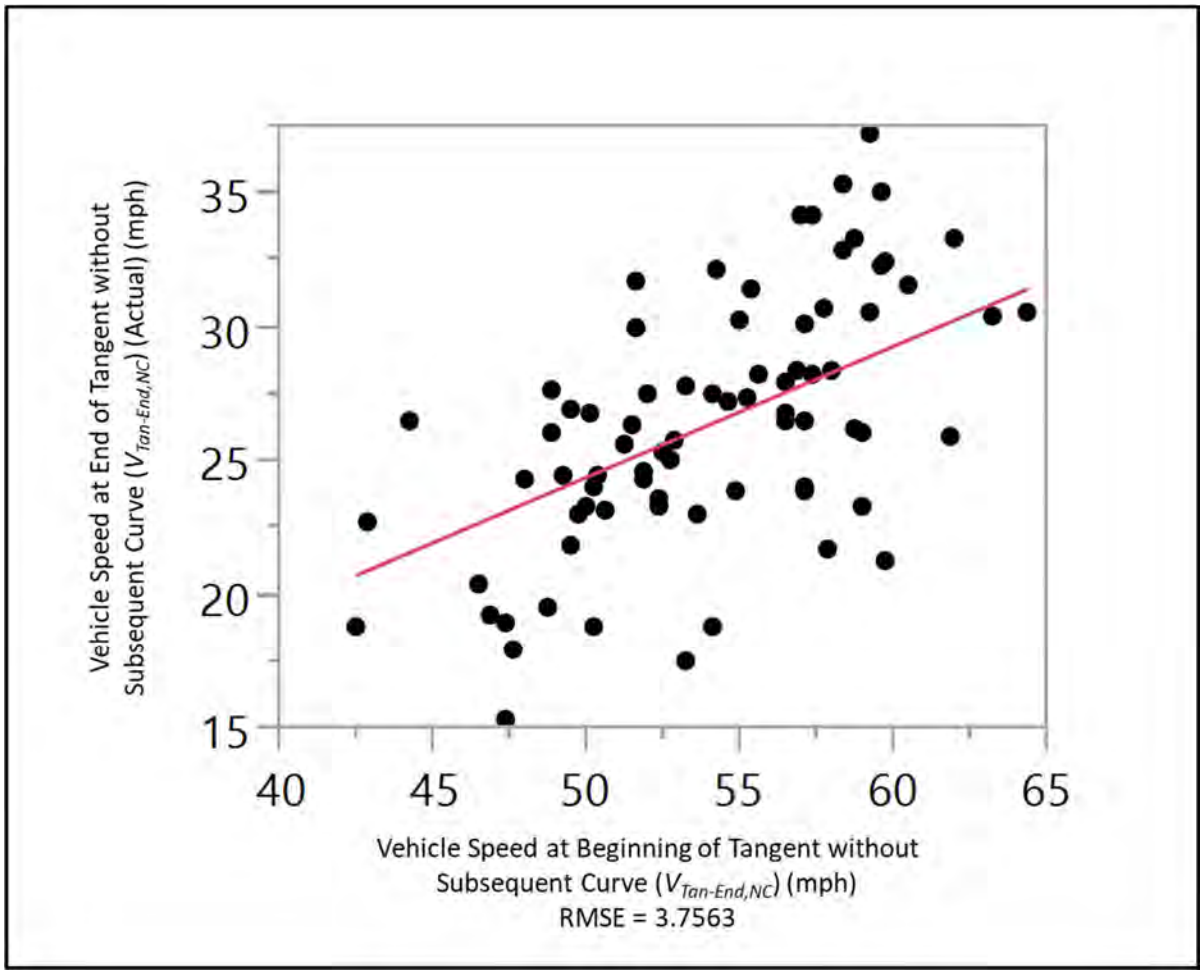


Figure 22. Regression plot of speed at beginning of tangent and speed at end of tangent for final model to estimate vehicle speed at end of tangent without subsequent curve on an exit ramp

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Section 5.
Summary of Design Guidelines

This section summarizes guidelines for selecting appropriate ramp design speeds. A companion document to this final report presents the enhanced guidelines in more detail (Torbic et al., 2021). The guidelines facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information. The guidelines are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads.

Ramp design speed is a selected speed used to determine the various geometric design features of a ramp. The selected ramp design speed should be a logical one with respect to the type of intersecting highway, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). Table 29 provides guide values for ramp design speeds for system and service interchanges. The guide values are presented as a range of values given the context, ramp configuration, and type of interchange. For the specified design condition, it is recommended that the ramp design speed for a given ramp be within the range of guide values. The lower guide values are generally associated with constrained conditions, and the higher guide values are associated with unconstrained conditions. The range of guide values in Table 29 were developed to be relatively consistent with guidance presented in the 2018 edition of the *Green Book*.

Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper (i.e., adjoining tangents and horizontal curves in the direction of travel) that connect to the controlling curve should be designed to allow for appropriate speed transitions as drivers travel from one section of the ramp to the next, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed.

Where the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp and applies to the tangential section of the ramp proper.

The ramp design speed is typically associated with an individual segment of a ramp near the upstream or downstream end of the ramp. For an entrance ramp, the ramp design speed is normally associated with a segment on the downstream portion of the ramp. With an exit ramp, the ramp design speed is usually associated with a segment on the upstream portion of the ramp.

Each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. The design speed of the controlling feature of the ramp (i.e., the ramp design speed) is selected

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Table 29. Range of Guide Values for Ramp Design Speed as Related to Highway Design Speed, Interchange Type, Ramp Configuration, and Contextual Considerations

Context		Ramp configuration	Highway design speed (mph)											
			30	35	40	45	50	55	60	65	70	75	80	85
			Ramp design speed (mph)											
System Interchanges														
Rural		Direct connections	-	-	-	-	40-45	40-50	40-55	45-60	45-65	50-70	50-75	50-80
		Semidirect connections	-	-	-	-	30-40	30-45	35-45	35-50	40-55	40-60	40-65	45-65
		Loop ramps	-	-	-	-	25-30	25-30	25-35	25-35	30-35	30-40	35-45	35-45
Urban		Direct connections	-	-	-	-	35-40	35-40	35-50	40-50	40-55	40-60	-	-
		Semidirect connections	-	-	-	-	30-35	30-40	30-45	30-45	35-50	35-55	-	-
		Loop ramps	-	-	-	-	20-25	20-30	25-30	25-30	25-35	25-40	-	-
Service Interchanges														
Rural:	Primary or major arterial	Direct connections	-	-	-	35-40	35-45	40-45	40-45	40-50	45-50	45-55	50-60	50-65
		Diagonal ramps[1]	20-25	20-25	25-30	25-35	30-40	30-45	30-50	30-55	35-55	40-60	45-65	45-70
		Loop ramps	15-20	20-25	20-25	20-25	20-25	25-30	25-30	25-30	30-35	30-40	35-45	35-45
	Minor arterial or collector	Diagonal ramps[1]	20	20-25	25	25-30	25-35	30-40	30-45	30-50	35-50	40-55	45-60	45-65
		Loop ramps	15-20	15-20	15-20	15-25	20-25	20-30	20-30	25-30	25-35	30-40	30-45	30-45
Urban:	Primary or major arterial	Direct connections	-	-	-	30-35	35-40	35-40	35-40	40-45	40-50	40-55	-	-
		Diagonal ramps[1]	15-20	20-25	25-30	25-30	30-35	30-40	30-45	30-45	35-50	40-55	-	-
		Loop ramps	15-20	15-20	15-20	20-25	20-25	20-30	25-30	25-30	25-35	25-40	-	-
	Minor arterial or collector	Diagonal ramps[1]	15	20	20-25	20-30	25-35	25-40	25-45	30-45	30-50	35-55	-	-
		Loop ramps	15-20	15-20	15-20	15-25	15-25	20-30	20-30	20-30	20-35	25-40	-	-

[1] Where the horizontal alignment of a diagonal entrance ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the freeway mainline ramp terminal. For this situation, it is recommended that the ramp design speed be within 15 to 20 mph of the highway design speed.

Where the horizontal alignment of a diagonal exit ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the crossroad ramp terminal. The recommended ramp design speed for this situation is 15 or 20 mph, as it is expected that the crossroad ramp terminal at all diagonal ramps will either be stop-, yield-, or signal-controlled.

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first, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly. It is desirable (but not essential) that the design speeds of the contiguous segments along the ramp proper increase or decrease in a stepwise manner. The change in design speed between adjoining sections should be limited to no more than 10 to 15 mph. For entrance ramps, this means that the design speeds of the contiguous segments along the ramp proper should typically increase in a stepwise manner; and for exit ramps the design speeds of the contiguous segments along the ramp proper should decrease in a stepwise manner. Decreasing design speeds sequentially along a ramp in a stepwise manner is more critical for exit ramps than increasing design speeds along a ramp for entrance ramps, as unexpected and abrupt changes in design speeds along an exit ramp are more likely to be associated with potential lane keeping and/or loss of control issues.

The design speed of each individual curve and tangent along the ramp proper should be consistent with the speeds that drivers are likely to travel along the respective curve and tangent. If the design speed of an individual curve or tangent (or multiple curves and tangents) along the ramp proper is not consistent with the speeds that drivers are likely to be traveling along the respective curve and tangent, consider potential options so that the design speed would be more consistent with operating speeds. For example, if the anticipated operating speed exceeds the design speed, consider changing (e.g., raising or potentially decreasing) the design speed for the respective section. If changing the design speed of an individual section means that the design speeds of the contiguous segments along the ramp proper no longer increase or decrease in a stepwise manner, consider changing the ramp design speed. If the design speed or ramp design speed cannot be changed (e.g., due to physical site constraints, environmental or social impacts, economic constraints, etc.), consider implementing speed-control measures so that the design speed or ramp design speed would be more consistent with operating speeds.

The freeway mainline ramp terminal portion of a ramp does not have an associated design speed. However, the operational and design conditions of the freeway mainline ramp terminal are related to and coordinated with the operational and design conditions of the adjoining freeway and ramp proper.

Each component of the ramp should be designed such that a vehicle has sufficient distance to accelerate or decelerate within the performance capabilities of the vehicle and driver expectations given the design speeds and features of the individual components of the ramp. Several design controls and criteria may be used in the design of ramps so that the individual components and features of the ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next. *Green Book* Figures 2-33 and 2-34 (presented here as Figures 23 and 24) may be used as a guide for determining minimum practical lengths of individual tangent and curve sections along the ramp proper, and *Green Book* Tables 10-4, 10-5, and 10-6 may be used to determine minimum acceleration and deceleration lengths for the freeway mainline ramp terminal portion of the ramp.

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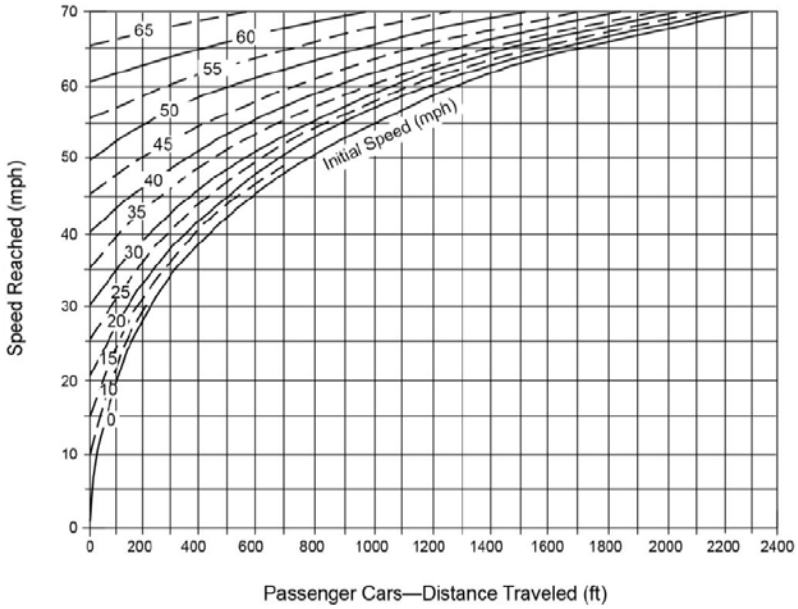


Figure 23. Acceleration of Passenger Cars, Level Condition (AASHTO, 2018)

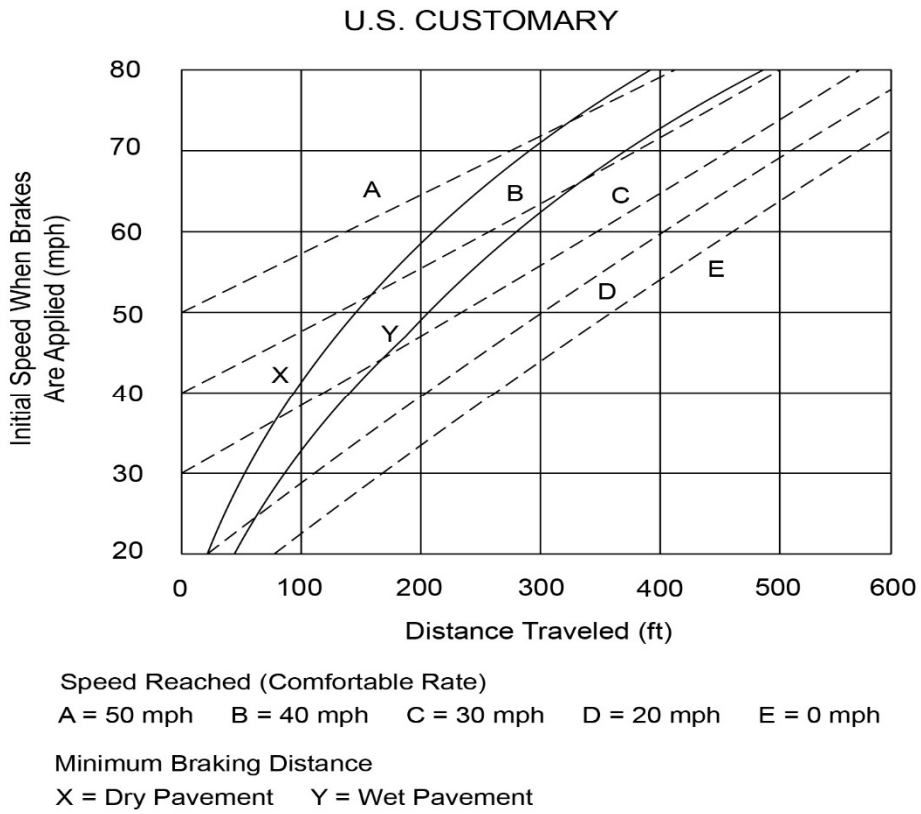


Figure 24. Deceleration Distances for Passenger Vehicles Approaching Intersections (AASHTO, 2018)

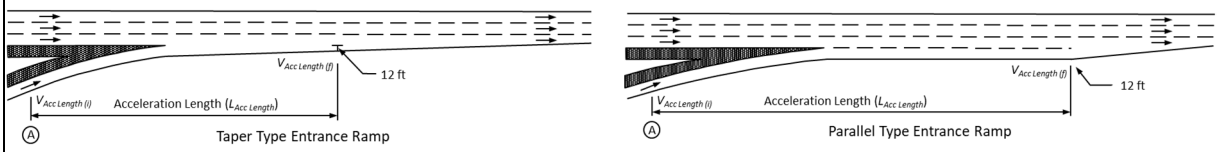
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Green Book Tables 10-4 and 10-6, presented here as Tables 30 and 31 respectively, have been adapted slightly based on this research. The column headings were adapted to be more consistent with the enhanced design guidelines. In addition, the guide values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds.

Table 30. Minimum Acceleration Lengths for Entrance Terminals with Flat Grades of Less Than 3 Percent (adapted from AASHTO, 2018)

Highway		Acceleration Length, $L_{Acc\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point)											
		Stop Condition	15	20	25	30	35	40	45	50	55[1]	60[1]	65[1]
Design Speed $V_{DS\ Hwy}$ (mph)	Merge Speed $V_{Acc\ Length(f)}$ (mph)	Average Running Speed (i.e., Initial Speed) Exiting the Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point), $V_{Acc\ Length(i)}$											
		0	14	18	22	26	30	36	40	44	48	52	56
30	23	180	140										
35	27	280	220	160									
40	31	360	300	270	210	120							
45	35	560	490	440	380	280	160						
50	39	720	660	610	550	450	350	130					
55	43	960	900	810	780	670	550	320	150				
60	47	1200	1140	1100	1020	910	800	550	420	180			
65	50	1410	1350	1310	1220	1120	1000	770	600	370	135		
70	53	1620	1560	1520	1420	1350	1230	1000	820	580	350	135	
75	55	1790	1730	1630	1580	1510	1420	1160	1040	780	550	300	
80	57	2000	1900	1800	1750	1680	1600	1340	1240	980	725	500	200
85[1]	59	2230	2090	1985	1930	1865	1790	1525	1455	1250	940	700	500

NOTE: Uniform 50:1 to 70:1 tapers are recommended where acceleration lengths exceed 1,300 ft.



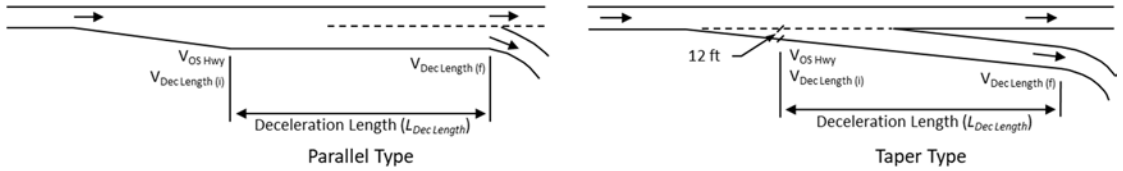
[1] Acceleration lengths estimated based on extrapolation.

Appendix A presents recommended changes to the next edition of the *Green Book*, based upon the findings and conclusions of this research.

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Table 31. Minimum Deceleration Lengths for Exit Terminals with Flat Grades of Two Percent or Less (adapted from AASHTO, 2018)

Highway Design Speed, $V_{DS\ Hwy}$ (mph)	Diverge Speed, $V_{OS\ Hwy}$ $V_{Dec\ Lenth\ (i)}$ (mph)	Deceleration Length, $L_{Dec\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point)								
		Stop Condition	15	20	25	30	35	40	45	50
		Average Running Speed at the End of the Deceleration Length, $V_{Dec\ Length\ (f)}$ (mph), Entering the Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point)								
		0	14	18	22	26	30	36	40	44
30	28	235	200	170	140	-	-	-	-	-
35	32	280	250	210	185	150	-	-	-	-
40	36	320	295	265	235	185	155	-	-	-
45	40	385	350	325	295	250	220	-	-	-
50	44	435	405	385	355	315	285	225	175	-
55	48	480	455	440	410	380	350	285	235	-
60	52	530	500	480	460	430	405	350	300	240
65	55	570	540	520	500	470	440	390	340	280
70	58	615	590	570	550	520	490	440	390	340
75	61	660	635	620	600	575	535	490	440	390
80	64	705	680	665	645	620	580	535	490	440
85[1]	**67**	**750**	**725**	**710**	**690**	**665**	**625**	**580**	**540**	**485**



[1] Deceleration lengths estimated based on extrapolation.

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Section 6.
Project Summary and Future Research Needs

The design speed concept has been the centerpiece of AASHTO's geometric design policy since the 1930s. Designing extended sections of highway based on the design speed process is relatively straightforward. However, when applied to interchange ramps where high-speed facilities meet low-speed facilities and drivers are expected to accelerate or decelerate over short distances, application of the design speed process is more complex.

The design guidance presented in the 2018 *Green Book* related to application of the design speed process for ramps and the appropriate selection of a ramp design speed have several overarching challenges that lead to confusion and inconsistent interpretation of the guidance. In particular, the *Green Book* does not specifically define ramp design speed. *Green Book* Table 10-1, which is the centerpiece of AASHTO's guidance on selecting an appropriate ramp design speed for the design of all ramp types, does not provide sufficient detail for selecting ramp design speeds for specific ramp types and configurations. Little guidance is provided on how the selected ramp design speed is to be used to develop a balanced design among all components of a ramp, nor is there consensus on the portion of the ramp to which the ramp design speed applies or whether the ramp design speed should be a single value or vary along the ramp. Therefore, enhanced design guidelines are necessary for selecting appropriate ramp design speeds so ramps are designed in a consistent manner based on the selected ramp design speed, accounting for sequential speed transitions from one component or section to the next, consistent with performance capabilities of vehicles and driver expectations.

The enhanced guidelines developed as part of this project [summarized here and presented in more detail in the companion document (Torbic et al., 2021) to this final report] address the overarching limitations of AASHTO's current policy related to selection and application of ramp design speeds. The enhanced guidelines define ramp design speed and related terminology and discuss in detail the most relevant elements that affect selecting an appropriate ramp design speed. A new table developed as part of the guidelines provides guide values for ramp design speeds for system and service interchanges, and the guide values are presented as a range of values given the context, ramp configuration, and type of interchange. The guidelines also address how to apply design controls and criteria so that individual components and features of ramps are designed in a consistent manner, accounting for sequential speed transitions from one component or section to the next. In addition to the guidelines, a design tool was developed that can be used to estimate vehicle speeds along interchange ramps to help designers and traffic engineers conduct operational analyses and make design decisions.

The enhanced guidelines are intended to facilitate the selection of an appropriate ramp design speed based on a combination of contextual considerations and quantitative information and are expected to result in ramp designs consistent with driver expectations and behaviors over a range of traffic conditions and the functional classification of the two intersecting roads. In addition, the new table (i.e., Table 29) of guide values for ramp design speeds was developed to be relatively consistent with guidance presented in the 2018 edition of the *Green Book*. In the *Green Book* further guidance is provided beyond the design table (i.e., *Green Book* Table 10-1) for selecting a ramp design speed based on various conditions and ramp types. For example, text in the *Green Book* indicates that ramp design speeds for semidirect connections are typically 30 to

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40 mph, and ramp design speeds less than 30 mph should not be used. Similarly, text in the *Green Book* indicates that the minimum ramp design speed for direct connections preferably should be 40 mph. For a majority of the conditions and guidance presented in the *Green Book* related to appropriate ramp design speeds for specific ramp configurations, the range of guide values for ramp design speeds in Table 29 are consistent with such guidance.

In some cases, the range of guide values in Table 29 extend slightly beyond the range of values suggested in the *Green Book*. In some cases, this was done intentionally. A table of guide values should not be too prescriptive that it limits flexibility and inhibits designers from exercising engineering judgment. In other cases, extending beyond the range of values suggested in the *Green Book* was necessary to address the fact that many states are posting freeway speed limits as high as 75, 80, and 85 mph, and guidelines for selecting ramp design speeds to address such conditions is necessary.

Because the range of guide values for ramp design speeds in Table 29 are relatively consistent with guidance presented in the *Green Book*, the enhanced guidelines are not expected to introduce unforeseen safety or operational issues, create the need for more design exceptions, or increase or decrease construction costs. The enhanced guidelines should result in the design of ramps that would be similar to ramps designed using guidance from the 2018 edition of the *Green Book* as intended. The enhanced guidelines simply provide more clarity on the primary elements that factor into selecting an appropriate ramp design speed and how the ramp design speed is used to coordinate the design of adjacent sections of the ramp to accommodate sequential speed transition from one component or section on the ramp to next.

The enhanced guidelines may help to improve safety and operational issues related to ramp design by reducing the number of ramps designed and built in an inconsistent manner due to confusion or inconsistent interpretation of existing AASHTO policy. Ramps designed in an inconsistent manner may not sufficiently accommodate sequential speed transitions from one component to the next. As such, driver expectations may be violated, resulting in abrupt changes in speed and/or creating safety and operational concerns. The enhanced guidelines may also benefit an agency's overall program budget by reducing the number of ramps that need to be reconstructed at a later date because they were originally designed and built in an inconsistent manner.

The enhanced guidelines are not expected to create any specific tort liability concerns related to ramp design. The enhanced guidelines developed through this research are relatively consistent with guidance presented in the *Green Book* and simply provide more clarity on selecting an appropriate ramp design speed and how the ramp design speed is used to coordinate the design of adjacent sections of the ramp. There is no indication that guidance presented in the *Green Book* has created any tort liability concerns related to the selection and application of ramp design speed, so there is not an apparent need for the enhanced guidelines to alleviate tort liability concerns based on use of current design guidance. (Note, adherence to standards or design guidance does not in itself provide tort immunity.) Terms such as "must", "shall", or "required" that could create obvious tort liability concerns are not used in the enhanced guidelines; and in general, if engineering decisions are made sensibly and are well documented, then no tort liability concerns should arise. Thus, the enhanced guidelines are not expected to create any specific tort liability concerns related to ramp design.

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Finally, throughout this project, several research needs were identified, namely:

- *Green Book* Table 10-4, which provides guide values for minimum acceleration lengths, was adapted (see Table 30). In addition to modifying the caption headings, the design values were expanded to include additional design conditions for higher ramp design speeds and higher highway (or freeway) design speeds. These new design values were estimated through numeric extrapolation of the minimum acceleration lengths for the lower-speed design conditions and to be consistent with the trends and acceleration rates of the design values in the original table. They were not developed based on field data or vehicle dynamics models that estimate vehicle performance. Similarly, *Green Book* Table 10-6, which provides guide values for minimum deceleration lengths, was adapted (see Table 31) for use with higher design speed facilities. Additional research should be conducted to verify and/or modify these design values for higher-speed facilities.
- A first version of a spreadsheet tool was developed as part of this research to estimate vehicle speeds along interchange ramps to help designers and traffic engineers conduct operational analyses and make design decisions. Additional research should be conducted to address limitations of this design tool and expand its capabilities.

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Section 8.
Abbreviations, Acronyms, Initialisms, and Symbols

Term	Description	Equation No.
AASHTO	American Association of State Highway and Transportation Officials.	
NCHRP	National Cooperative Highway Research Program.	
ITE	Institute of Transportation Engineers.	
HSM	*Highway Safety Manual.*	
SCL	Speed-change lane.	
SPUI	Single point urban interchanges.	
RSPM	Ramp speed profile model.	
PC	Point of curvature (i.e., point at which the curve begins).	
MC	Midpoint of a curve.	
PT	Point of tangency (i.e., point at which the curve ends).	
FI	Fatal-and-injury severity levels.	
PDO	Property-damage-only severity level.	
CMF	Crash modification factor.	
SPF	Safety performance function.	
N	Predicted crash frequency per year on the ramp.	1
L_r	Ramp length.	1
$AADT_r$	Average annual daily volume on the ramp.	1
a,b,c,d	Regression coefficients.	1
X_i	Milepost of the point of change from tangent to curve (PC) for curve *i*.	5,7,9
R_i	Radius of curve *i.*	2
$L_{C,i}$	Length of horizontal curve *i*.	4,5,6,8,9,10
V_{Frwy}	Average traffic speed on freeway during off-peak periods of the typical day.	3,4,5,6,7
V_{XRoad}	Vehicle speed at the crossroad ramp terminal.	3,7,8,9,10
vmax,i	Limiting speed for curve *i*.	2,4,6,8,10
$v_{ent,1}$	Average entry speed for Curve 1.	3,4,7,8
vext,1	Average exit speed for Curve 1.	4,8
vent,i	Average entry speed for curve *i* (*i* = 2, 3, ...).	5,6,9,10
vext,i	Average exit speed for curve *i* (*i* = 2, 3, ...).	5,6,9,10
$v_{ent,c,MC}$	Average passenger car speed at the midpoint of the entrance ramp controlling curve.	11
$v_{ent,c,PT}$	Average passenger car speed at the PT of the entrance ramp controlling curve.	12
$v_{ext,c,PC}$	Average passenger car speed at the PC of the exit ramp controlling curve.	13
$v_{ext,c,MC}$	Average passenger car speed at the midpoint of the exit ramp controlling curve.	14
I_{l2}	Indicator variable for lane 2 (= 1 if predicting speed in the outside lane, 0 otherwise).	11,12
$I_{l2\text{-}3}$	Indicator variable for lanes 2 and 3 (= 1 if predicting speed in the middle or outside lanes, 0 otherwise).	14
R	Radius (measured to the inside of the traveled way).	11,12,13,14
W_l	Lane width.	11
W_{os}	Outside (left) shoulder width.	11,12,14

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Term	Description	Equation No.
W_{is}	Inside (right) shoulder width.	11
I_{tk}	Indicator variable for trucks (= 1 if predicting truck speed, 0 otherwise).	11,12,13,14
I_{rs}	Indicator variable for curve radius type (= 1 if simple, 0 if compound).	14
I_d	Indicator variable for drop SCL (= 1 if present, 0 otherwise).	14
I_p	Indicator variable for parallel SCL (= 1 if present, 0 otherwise).	14
I_w	Indicator variable for weaving SCL (= 1 if present, 0 otherwise).	14
v_c	Average passenger car speed.	15
DC	Degree of horizontal curvature.	15
Z	Distance to the first downstream at-grade signalized or stop-controlled intersection.	15
v_{TZ}	Point speeds in the transition zone.	16
v_{FW}	Point speeds on the freeway mainline.	16
$v_{85,DL}$	85th-percentile operating speeds on the deceleration lane.	17
$v_{85,FW}$	85th-percentile operating speeds on the freeway mainline.	17,20
$v_{85,UR}$	85th-percentile operating speeds on the upstream ramp curve.	18
$v_{85,LR}$	85th-percentile operating speeds on the downstream ramp curve.	19
L_{DL}	Length of deceleration lane.	17,20
SL_{RP}	Speed limit on the ramp.	17
R_1	Radius of the upstream ramp curve.	18,21,22
R_2	Radius of the downstream ramp curve.	19,22
85 V_{FW-DL}	85th-percentile speed difference between freeway mainline and the deceleration lane.	20
85 V_{DL-UR}	85th-percentile speed difference between deceleration lane and the upstream ramp curve.	21
85 V_{UR-LR}	85th-percentile individual speed difference between upstream ramp curve and the downstream ramp curve.	22
D_1	Deflection angle of the upstream ramp portion.	21
$v_{c,PC}$	Average passenger car speed at the PC of the exit ramp controlling curve.	23
$v_{t,PC}$	Average truck speed at the PC of the exit ramp controlling curve.	24
$v_{c,MC}$	Average passenger car speed at the midpoint of the exit ramp controlling curve.	25
$v_{t,MC}$	Average truck speed at the midpoint of the exit ramp controlling curve.	26
R	Radius.	23,24,25,26
	Longitudinal gradient.	25,26
$v_{a,u,l}$	Average speed of unimpeded trucks at a level site.	27
D	Distance from end of scale.	27

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Term	Description	Equation No.
$v_{85Merge}$	85th-percentile speed of merging vehicles.	28,29,30
v_{85RL}	85th-percentile speed of vehicles in the rightmost lane of the freeway mainline.	30
Θ	Angle of convergence between entrance ramp and mainline.	28,29
L	Entrance ramp length.	28,29,31
$AADT_{SCL}$	Annual average daily traffic volume on entrance ramp SCL.	28
Q_{SCL}	Equivalent hourly traffic volume on entrance ramp SCL.	29
Q_{RL}	Equivalent hourly traffic volume in rightmost freeway mainline lane.	30
$D_{85Merge}$	85th-percentile passenger car merging distance.	31
$CMF_{enr\|agg}$	Aggregated ramp entrance CMF.	32
P_{enr}	Proportion of freeway segment length adjacent to a ramp entrance.	32
l_{enr}	Average ramp entrance length.	32
V_{MC}	Predicted vehicle speed at the MC.	33,35
V_{PC}	Vehicle speed at the beginning of a curve (i.e., initial point of horizontal curve – PC).	33,34,35,36
R	Curve radius.	33,34,35,36
$V_{SL\ Hwy}$	Regulator speed limit on the highway (or freeway) mainline.	33,34,35
V_{PT}	Predicted vehicle speed at the end of a curve (PT).	34,36
I_{DN}	Indicator variable for downward slope (= 1 if the overall ramp slopes downward, 0 otherwise).	35
I_{LV}	Indicator variable for level slope (= 1 if the overall ramp is level, 0 otherwise).	35
I_{UP}	Indicator variable for upward slope (= 1 if the overall ramp slopes upward, 0 otherwise).	35
L_C	Curve length.	36
$V_{Tan\text{-}End,C}$	Predicted vehicle speed at the end of a tangent if the tangent is followed by a curve.	37,39
$V_{Tan\text{-}End,NC}$	Predicted vehicle speed at the end of a tangent if the tangent is not followed by a curve.	38,40
$V_{Tan\text{-}Begin}$	Predicted vehicle speed at the beginning of a tangent.	37,39,40
L_T	Tangent length.	37,38,39
I_{55}	Indicator variable for 55-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 55 mph, 0 otherwise).	37
I_{60}	Indicator variable for 60-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 60 mph, 0 otherwise).	37
I_{65}	Indicator variable for 65-mph regulatory speed limit on the highway (or freeway) mainline (= 1 if the regulatory speed limit is 65 mph or above, 0 otherwise).	37
R_{next}	Radius of subsequent curve.	39

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Appendix A.
Suggested Text for Future Edition of AASHTO *Green Book*

This appendix provides potential changes recommended for consideration to the next edition of the AASHTO *Green Book*, based upon the findings and conclusions of this research. The recommendations are based upon a review of the 2018 edition of the *Green Book*. Recommended modifications are specified for selected sections in Chapter 10 as follows, using track changes.

Chapter 10 (Beginning on Page 10-102 of 2018 *Green Book*)

10.9.6 Ramps

10.9.6.1 Types and Examples

The term “ramp” includes all types, arrangements, and sizes of turning roadways that connect two or more legs at an interchange. The components of a ramp are a terminal at each leg and a connecting road. The geometry of the connecting road usually involves some curvature and a grade. Generally, the horizontal and vertical alignment of ramps is based on lower design speeds than the intersecting highways, but in some cases it may be equal.

Figure 10-60 illustrates several types of ramps and their characteristic shapes. Various configurations are used; however, each can be broadly classified as one of the types shown. Each ramp generally is a one-way roadway. Diagonal ramps (Figure 10-60A) are almost always one way but usually have both a left- and right-turning movement at the terminal on the minor intersecting road. A diagonal ramp may be largely tangent or wishbone in shape with a reverse curve. Diamond interchanges generally have four diagonal ramps.

A loop ramp may have single turning movements (left or right) or double turning movements (left and right) at either or both ends. Figure 10-60B shows the case where there are only single turns made at both ends of the ramp. With this loop pattern, a left-turning movement is made without an at-grade crossing of the opposing through traffic. Instead, drivers making a left-turn travel beyond the highway separation, turn to the right through approximately 270 degrees to enter the other highway. A loop ramp usually involves more indirect travel distance than any other type of ramp.

With a semidirect connection (Figure 10-60C), the driver exits to the right first, heading away from the intended direction, gradually reversing, and passing around other interchange ramps before entering the other road. This semidirect connection may also be used for right turns, but there is little reason for its use if the conventional diagonal can be provided. A descriptive term frequently associated with this type of ramp is “jug-handle,” the obvious plan shape. Travel distance on this ramp is less than that for a comparable loop and more than that for a direct connection. Figure 10-60D is termed an outer connection, while Figure 10-60E is referred to as a direct connection.

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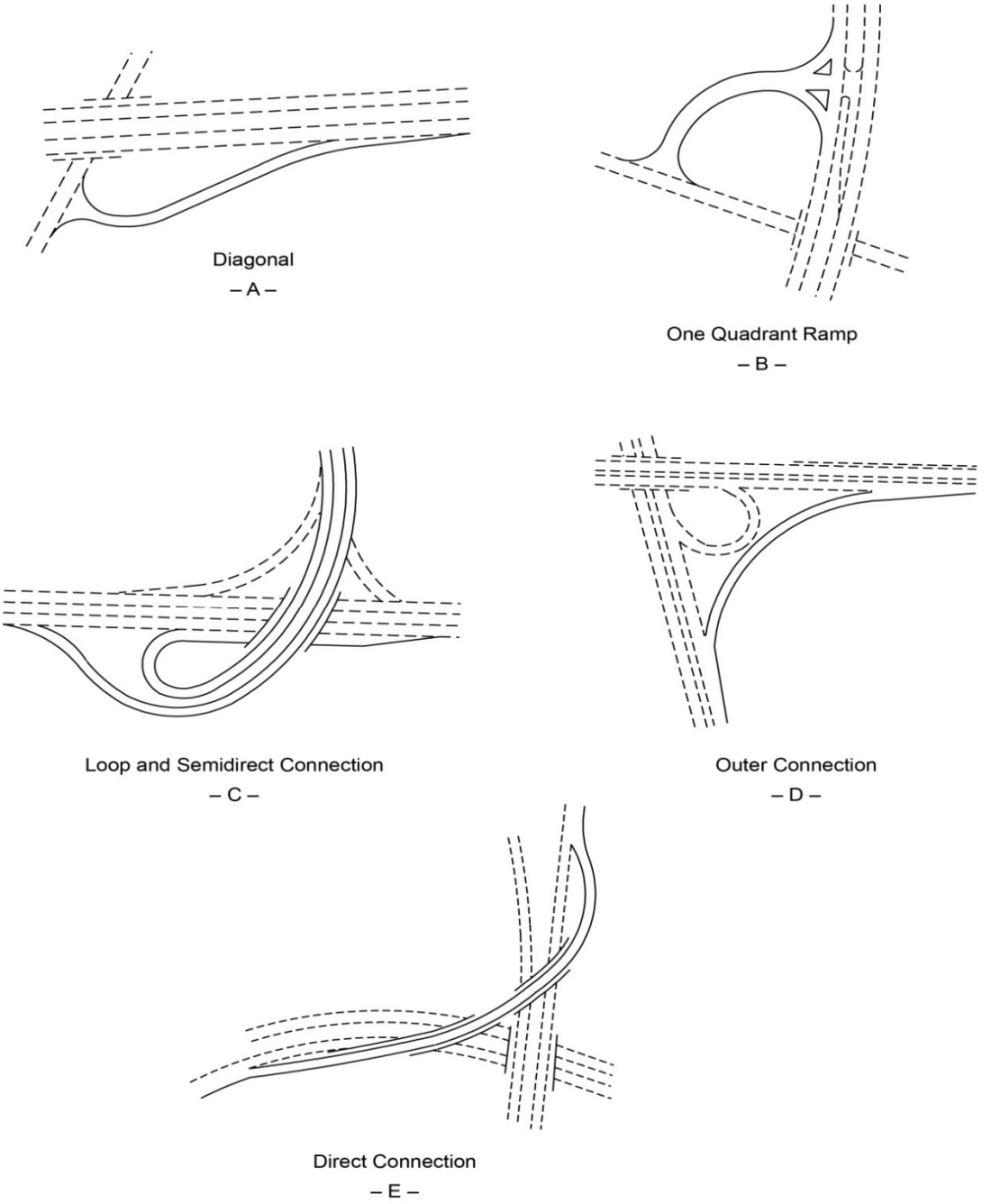


Figure 10-60. General Types of Ramps

The different ramp patterns of an interchange (i.e., the different types of interchange configurations) are comprised of various combinations of these types of ramps. For example, the trumpet configuration has one loop, one semidirectional ramp, and two right-turn directional or diagonal ramps.

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10.9.6.2 General Ramp Design Considerations

10.9.6.2.1 Ramp Design Speed

The ramp design speed is a selected speed used to determine the various geometric design features of a ramp. The ramp design speed should be a logical one with respect to the type of intersecting highways, area type (i.e., urban or rural), ramp configuration, and site constraints (including physical, environmental, and social). The amount of speed transition to be accommodated along a ramp is dictated by the design speeds of the intersecting freeways at a system interchange and the design speed of the freeway and operations near the crossroad ramp terminal at a service interchange.

The combination of the type of intersecting highways, area type, ramp configuration, and site constraints are the primary determinant elements and contextual considerations that should factor into the selection of an appropriate ramp design speed. Based on these key elements and contextual considerations, Table 10-1 provides a range of guide values for ramp design speeds for system and service interchanges. For the specified design condition, it is recommended that the ramp design speed for a given ramp be within the range of guide values. By providing a range of guide values for ramp design speeds, Table 10-1 provides flexibility in determining the contextual and design considerations that apply to a given ramp.

Conceptually, contextual considerations can be viewed from a qualitative perspective and classified as either creating constrained or unconstrained conditions. For example, where the physical site characteristics for a given ramp may be considered typical or less restrictive than normal for a given area, the site may be classified as unconstrained. On the other hand, where physical site characteristics exist beyond what may be considered as typical or normal for a given area, the site may be classified as constrained. Similarly, sites may be classified as unconstrained or constrained as they pertain to environmental and social impacts and economic factors. In general, the lower guide values in Table 10-1 are associated with constrained conditions, and the higher guide values are associated with unconstrained conditions.

With a system interchange, vehicles exit a high-speed facility and enter another high-speed facility, so it is desirable to provide a ramp between the two facilities that enables vehicles to maintain a speed as close as practical to freeway speeds, minimizing the need for deceleration or acceleration along the ramp. For situations where the design speeds of both intersecting freeways of a ramp are equivalent, the highest possible ramp design speed for the connecting ramp would be the design speed of the intersecting freeways. However, in most cases it is not practical for the ramp design speed to be equivalent to the design speed of the two intersecting freeways; therefore, in those cases, the ramp design speed may be less than the intersecting freeways' design speed. For the situation where two intersecting freeways do not have equivalent design speeds, the ramp design speed may be a speed between the two highway design speeds of the intersecting freeways; however, where this is not practical, the ramp design speed may be less than the lower design speed of the two intersecting freeways.

With a service interchange, vehicles either accelerate from a near-stop condition at the crossroad ramp terminal to a higher speed on the freeway (e.g., in the case of entrance ramps) or decelerate from a higher speed on the freeway to a stop or near-stop condition at the crossroad ramp terminal (e.g., in the case of exit ramps). For both entrance and exit ramps, ramps need to be designed to accommodate the necessary acceleration or deceleration. In rural or unconstrained areas where the intersecting crossroad is a primary highway or major street, minimal to moderate

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acceleration or deceleration along the ramp is expected by the driver; while if the intersecting crossroad is a minor arterial or collector, moderate acceleration or deceleration is expected by the driver along the ramp. In urban or constrained areas where the intersecting facility is a primary highway or major street, moderate to significant acceleration or deceleration along the ramp is expected by the driver; while if the intersecting road is a minor arterial or collector, significant acceleration or deceleration is expected by the driver along the ramp.

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Table 10-1. Range of Guide Values for Ramp Design Speed as Related to Highway Design Speed

U.S. Customary													
Context		Ramp configuration	Highway design speed (mph)											
			30	35	40	45	50	55	60	65	70	75	80	85
			Ramp design speed (mph)											
System Interchanges														
Rural		Direct connections	-	-	-	-	40-45	40-50	40-55	45-60	45-65	50-70	50-75	50-80
		Semidirect connections	-	-	-	-	30-40	30-45	35-45	35-50	40-55	40-60	40-65	45-65
		Loop ramps	-	-	-	-	25-30	25-30	25-35	25-35	30-35	30-40	35-45	35-45
Urban		Direct connections	-	-	-	-	35-40	35-40	35-50	40-50	40-55	40-60	-	-
		Semidirect connections	-	-	-	-	30-35	30-40	30-45	30-45	35-50	35-55	-	-
		Loop ramps	-	-	-	-	20-25	20-30	25-30	25-30	25-35	25-40	-	-
Service Interchanges														
Rural:	Primary or major arterial	Direct connections	-	-	-	35-40	35-45	40-45	40-45	40-50	45-50	45-55	50-60	50-65
		Diagonal ramps[1]	20-25	20-25	25-30	25-35	30-40	30-45	30-50	30-55	35-55	40-60	45-65	45-70
		Loop ramps	15-20	20-25	20-25	20-25	20-25	25-30	25-30	25-30	30-35	30-40	35-45	35-45
	Minor arterial or collector	Diagonal ramps[1]	20	20-25	25	25-30	25-35	30-40	30-45	30-50	35-50	40-55	45-60	45-65
		Loop ramps	15-20	15-20	15-20	15-25	20-25	20-30	20-30	25-30	25-35	30-40	30-45	30-45
Urban:	Primary or major arterial	Direct connections	-	-	-	30-35	35-40	35-40	35-40	40-45	40-50	40-55	-	-
		Diagonal ramps[1]	15-20	20-25	25-30	25-30	30-35	30-40	30-45	30-45	35-50	40-55	-	-
		Loop ramps	15-20	15-20	15-20	20-25	20-25	20-30	25-30	25-30	25-35	25-40	-	-
	Minor arterial or collector	Diagonal ramps[1]	15	20	20-25	20-30	25-35	25-40	25-45	30-45	30-50	35-55	-	-
		Loop ramps	15-20	15-20	15-20	15-25	15-25	20-30	20-30	20-30	20-35	25-40	-	-

[1] Where the horizontal alignment of a diagonal entrance ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the freeway mainline ramp terminal. For this situation, it is recommended that the ramp design speed be within 15 to 20 mph of the highway design speed.

Where the horizontal alignment of a diagonal exit ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the crossroad ramp terminal. The recommended ramp design speed for this situation is 15 or 20 mph, as it is expected that the crossroad ramp terminal at all diagonal ramps will either be stop-, yield-, or signal-controlled.

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Metric											
Context		Ramp configuration	Highway design speed (km/h)									
			50	60	70	80	90	100	110	120	130	140
			Ramp design speed (km/h)									
System Interchanges												
Rural		Direct connections	-	-	-	60-70	60-80	60-90	70-100	80-110	80-120	80-130
		Semidirect connections	-	-	-	50-60	50-70	60-70	60-80	60-100	60-110	70-110
		Loop ramps	-	-	-	40-50	40-50	40-60	40-60	50-60	50-70	50-70
Urban		Direct connections	-	-	-	60-70	60-70	60-80	60-80	60-100	-	-
		Semidirect connections	-	-	-	50-60	50-60	50-70	50-80	60-90	-	-
		Loop ramps	-	-	-	30-40	30-50	40-50	40-50	40-60	-	-
Service Interchanges												
Rural:	Primary or major arterial	Direct connections	-	-	60-70	60-70	60-70	60-70	70-80	70-90	80-100	80-110
		Diagonal ramps[1]	30-40	30-50	40-60	50-60	50-70	50-80	50-90	70-100	70-110	70-110
		Loop ramps	20-30	30-40	30-40	30-40	40-50	40-50	40-50	50-60	60-70	60-70
	Minor arterial or collector	Diagonal ramps[1]	30	30-40	40-50	40-60	50-60	50-70	50-80	60-90	70-100	70-110
		Loop ramps	20-30	20-30	20-40	30-40	30-50	30-50	40-50	50-60	50-70	50-70
Urban:	Primary or major arterial	Direct connections	-	-	50-60	60-70	60-70	60-70	60-80	60-90	-	-
		Diagonal ramps[1]	20-30	30-40	40-50	50-60	50-60	50-70	50-80	60-90	-	-
		Loop ramps	20-30	20-30	30-40	30-40	30-50	40-50	40-50	40-60	-	-
	Minor arterial or collector	Diagonal ramps[1]	20	30-40	30-50	40-60	40-60	40-70	50-80	60-90	-	-
		Loop ramps	20-30	20-30	20-40	20-40	30-50	30-50	30-50	40-60	-	-

[1] Where the horizontal alignment of a diagonal entrance ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the freeway mainline ramp terminal. For this situation, it is recommended that the ramp design speed be within 20 to 30 km/h of the highway design speed.

Where the horizontal alignment of a diagonal exit ramp is relatively straight and has little impact on vehicle speeds, the ramp design speed is dependent on the operational characteristics of the crossroad ramp terminal. The recommended ramp design speed for this situation is 20 or 30 km/h, as it is expected that the crossroad ramp terminal at all diagonal ramps will either be stop-, yield-, or signal-controlled.

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The application of values in Table 10-1 to various conditions and ramp types is discussed below.

10.9.6.2.2 Portion of Ramp to Which the Ramp Design Speed Is Applicable
The values in Table 10-1 apply to the controlling feature on the ramp proper. Where the horizontal alignment of the ramp is curvilinear, the ramp design speed applies to the controlling curve on the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For an exit ramp, the controlling curve is the first curve encountered along the ramp proper that significantly affects vehicle speed. For direct connection and outer connection ramps, the sharpest curve on the ramp proper is considered the controlling curve. The freeway mainline ramp terminal, the crossroad ramp terminal, and the adjoining sections of the ramp proper (i.e., adjoining tangents and horizontal curves in the direction of travel) that connect to the controlling curve should be designed to allow for appropriate speed transitions, consistent with driver behavior and expectations and vehicle performance capabilities, for the selected ramp design speed.

If the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, as with some configurations of diagonal ramps, the ramp design speed applies to the tangential section of the ramp proper and is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp.

The ramp design speed is typically associated with an individual segment of a ramp near the upstream or downstream end of the ramp. Each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. The design speed of the controlling feature of the ramp (i.e., the ramp design speed) is selected first, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly. The ramp terminals (i.e., freeway mainline ramp terminal and crossroad ramp terminal) do not have an associated design speed. However, the operational and design conditions of both terminals are related to and coordinated with the operational and design conditions of the adjoining sections of the freeway, ramp proper, and/or crossroad.

10.9.6.2.3 Diagonal Ramps
Diagonal ramps are used for service interchanges and typically have both a left- and right-turn movement at the crossroad ramp terminal. Diagonal ramps usually consist of a high-speed freeway mainline ramp terminal, a tangent or curved alignment on the ramp proper, and stop or yield conditions at the crossroad ramp terminal. Depending on the configuration of the crossroad, a free-merge right-turn maneuver may be provided.

For diagonal ramps where the horizontal alignment is curvilinear, the horizontal alignment will be the controlling feature of the ramp. For diagonal ramps where the horizontal alignment is relatively flat (i.e., where the horizontal alignment has little or no impact on the speeds of vehicles), the ramp design speed is based on the operational characteristics of the freeway mainline ramp terminal for an entrance ramp and the operational characteristics of the crossroad ramp terminal for an exit ramp. For an entrance ramp, the design speed of the tangent section of the ramp proper (i.e., the ramp design speed) should be consistent with the speed entering the freeway mainline ramp terminal. For an exit ramp, the design speed of the tangent section of the

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ramp proper (i.e., the ramp design speed) should be consistent with the operating conditions at the crossroad ramp terminal. Because diagonal ramps are used for service interchanges and because the controlling feature of a diagonal ramp may be either the horizontal alignment or the operational characteristics of one of the terminals, diagonal ramps may be designed with ramp design speeds as low as 15 to 20 mph (20 to 30 km/h) and potentially as high as 45 to 70 mph (70 to 110 km/h)]. Diagonal ramps are unique compared to other ramp configurations in that the controlling feature of a diagonal ramp may be either the horizontal alignment or the operational characteristics of one of the terminals. With the other ramp configurations, the horizontal alignment of the ramp will likely be the controlling feature.

10.9.6.2.4 Loop Ramps

Loop ramps may be used in both system and service interchanges. The design speed of a loop ramp is limited by the radius of the controlling curve. For important movements on highways with design speeds of 50 mph [80 km/h] and higher, the radii of controlling curves on loop ramps generally range from 150 to 250 ft [50 to 75 m] with corresponding ramp design speeds of about 25 to 30 mph [40 to 50 km/h]. Ramp design speeds above 30 mph [50 km/h] for loops involve large land areas that are rarely available in urban areas. The long loop ramps needed for higher design speeds are costly and require left-turning drivers to travel a considerable extra distance, but if less restrictive conditions exist, the loop ramp design speed and the radius may be increased.

For minor movements on highways with design speeds of less than 50 mph [80 km/h], ramp design speeds for loop ramps may be as low as 15 to 20 mph [20 to 30 km/h].

10.9.6.2.5 Two-Lane Loop Ramps

With development and additional traffic on freeways, the need for two-lane loop ramps has increased. The two-lane loop configuration should not be immediately preceded or followed by a loop ramp. The radius of the inner edge of the traveled way of the loop ramp normally should not be less than 180 to 200 ft [55 to 60 m]. For additional design details, see the ITE *Freeway and Interchange Geometric Design Handbook* (*15*).

10.9.6.2.6 Semidirect Connections

Semidirect connection ramps are commonly used at system interchanges and are intended to encourage high-speed, free-flow merging with freeway traffic. As such, high ramp design speeds are preferred but may not be practical due to the curvilinear alignment. Ramp design speeds for semidirect connections typically range from 30 to 40 mph [50 to 60 km/h]. A design speed less than 30 mph [50 km/h] should not be used. Generally, for short single-lane ramps, a design speed greater than 50 mph [80 km/h] is not practical. The horizontal alignment is commonly the controlling feature of semidirect connection ramps. For semidirect connection ramps, the sharpest curve on the ramp proper that significantly affects vehicle speed is considered the controlling curve.

10.9.6.2.7 Direct Connections

Direct connection ramps are commonly used at system interchanges. As such, they are intended to encourage high-speed, free-flow merging with freeway traffic and may be designed with ramp design speeds ranging from 35 to 80 mph [60 to 130 km/h]. Direct connection ramps are also

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used at service interchanges, and in these situations the ramp design speeds may range from 30 to 65 mph [50 to 100 km/h]. For both system and service interchanges, the horizontal alignment is the controlling feature of direct connection ramps. The sharpest curve on the ramp proper that significantly affects vehicle speed is considered the controlling curve.

10.9.6.2.8 Increasing and Decreasing Design Speeds on the Ramp Proper

The highway with the greater design speed should be the control in selecting the ramp design speed for the ramp as a whole. At the same time, each individual section of the ramp proper (i.e., tangent and curve) has its own design speed. The design speed for the controlling feature of the ramp (i.e., the ramp design speed) is selected first, and then the design speeds of the other elements or segments of the ramp proper should be determined accordingly. It is desirable (but not essential) that the design speeds of the contiguous segments along the ramp proper increase or decrease in a stepwise manner. The change in design speed between adjoining sections should be limited to no more than 10 to 15 mph [20 to 30 km/h]. For entrance ramps, this means that the design speeds of the contiguous segments along the ramp proper may increase in a stepwise manner; and for exit ramps the design speeds of the contiguous segments along the ramp proper may decrease in a stepwise manner. Decreasing design speeds sequentially along a ramp in a stepwise manner is more critical for exit ramps than increasing design speeds along a ramp for entrance ramps, as unexpected and abrupt changes in design speeds along an exit ramp are more likely to be associated with potential lane keeping and/or loss of control issues.

10.9.6.2.9 At-Grade Terminals

Where a ramp joins a major crossroad, frontage road, or street, forming an intersection at grade, Table 10-1 is not applicable to that portion of the ramp near the intersection because a stop sign or signal control is normally employed. This terminal design should be predicated on near-minimum turning conditions, as given in Section 9.6. In urban areas, where the land adjacent to the interchange is developed commercially, provisions for pedestrian and bicycle movements through the interchange area should also be considered.

Where the ramp joins a frontage road, the ramp design speed may vary with the portion of the ramp closer to the lower-speed frontage road being designed for the lower speed of that frontage road.

10.9.6.2.10 Curvature

The design guidelines for turning roadways at interchanges are discussed in Section 3.3.7. They apply directly to the design of ramp curves. Theoretically, when designing a horizontal curve based on a point mass and the basic curve equation, it is assumed that a vehicle travels at a constant speed. However, for an entrance ramp, it is assumed that vehicle speed will increase slightly over the length of the curve. Similarly, along an exit ramp, it is assumed that vehicle speeds will decrease slightly over the length of the curve. For the design of a horizontal curve on an interchange ramp, rather than designing a curve based on a constant speed, the speed at the MC may be used to design the curve rather than the speeds at the beginning or end of the curve.

Compound or spiral curve transitions are desirable to: (1) obtain the desired alignment of ramps, (2) provide for a comfortable transition between the design speeds of the through and turning roadways, and (3) fit the natural paths of vehicles. Caution should be exercised in the use of

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compound curvature to prevent unexpected and abrupt speed adjustments. Additional design information regarding the use of compound curves is presented in Section 3.3.7.

The general shape of a ramp evolves from the type of ramp selected, as previously described and shown in Figure 10-60. The specific shape, or curvature, of a ramp may be influenced by such factors as traffic pattern, traffic volume, design speed, topography, culture, intersection angle, and type of ramp terminal.

Several ramp shapes may be used for the loop and outer connection of a directional interchange, as shown in Figure 10-61A. Except for its terminals, the loop may be a circular arc or some other symmetrical or asymmetrical curve that is formed with spiral transitions. The asymmetrical arrangement may fit where the intersecting roads are not of the same importance and the ramp terminals are designed for different speeds, so that the ramp in part functions as a speed-change area. Similar shapes may be dictated by ROW controls, profile and sight distance conditions, and terminal location. The freeway terminal should normally be placed in advance of the structure. The most desirable alignment for an outer connection is on a continuous curve (line A). This arrangement, however, may involve extensive ROW. Another desirable arrangement has a central tangent and terminal curves (lines B-B and C-C). Where the loop is more important than the outer connection, reverse alignment on the outer connection may be used to reduce the area of ROW, as shown by line D-D. Any combination of lines B, C, and D may be used for a practical shape.

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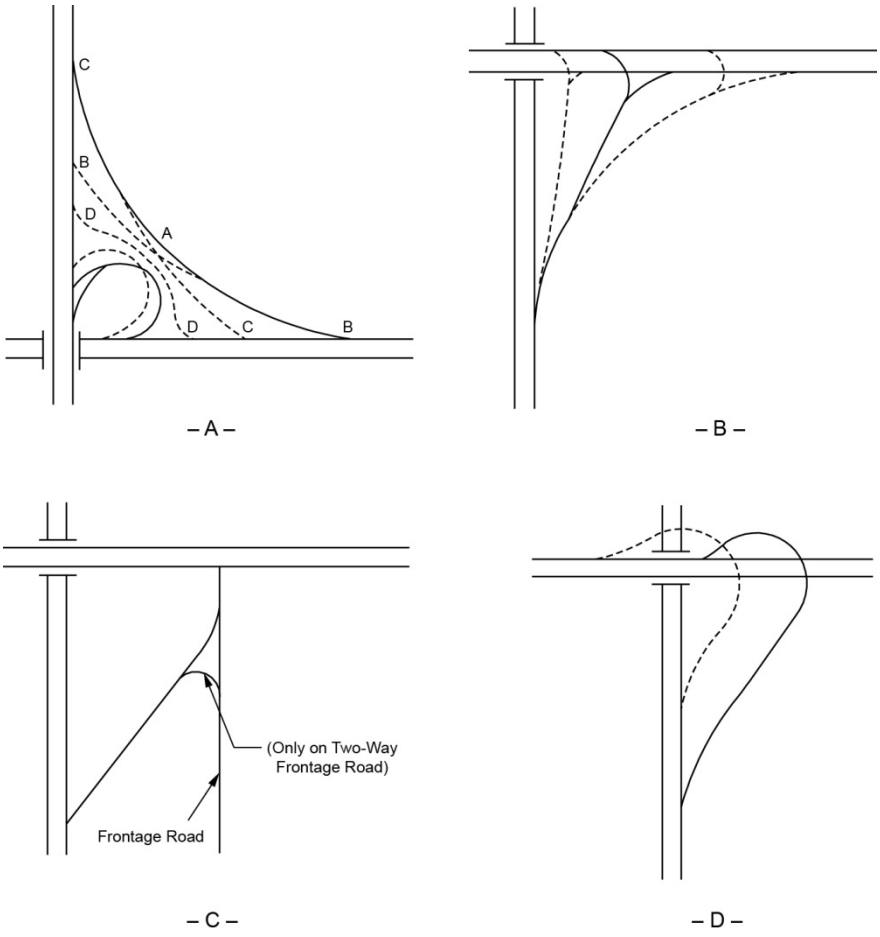


Figure 10-61. Ramp Shapes

In Figure 10-61A, the loop and the outer connection are separated, as is generally desirable. However, where the movements are minor and economy is desired, a portion of the two ramps may be combined into a single two-way roadway. Where this design is used, a barrier should separate the traffic in two directions. This design is generally discouraged.

Diagonal ramps may assume a variety of shapes, depending on the pattern of turning traffic and ROW limitations. As shown in Figure 10-61B, the ramp may be a diagonal tangent with connecting curves (solid line). To favor a right-turning movement, the ramp may be on a continuous curve to the right with a spur to the left for left turns. On restricted ROW along the major highway, it may be appropriate to use reverse alignment with a portion of the ramp being parallel to the through roadway.

Another variation of diagonal ramps, usually called “slip ramps,” connects with a parallel frontage road, as shown in Figure 10-61C. Where this design is used, it is desirable to have one-way frontage roads. Ramps to two-way frontage roads introduce the possibility of wrong-way entry onto the through lanes. If two-way frontage roads are used, special attention should be given in the design and signing of ramps to discourage the possibility of wrong-way entry.

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The shape of a semidirect connection (Figure 10-61D) is influenced by the location of the terminals with respect to the structures, the extent to which the structure is widened, and the curve radii needed to maintain a desired turning speed for an important left-turning movement. The angular position or the curvature may be dictated somewhat by the relative design speeds of the intersection legs and by the proximity of other roadways.

10.9.6.2.11 Sight Distance

Sight distance along a ramp should be at least as great as the design stopping sight distance. Sight distance for passing is not needed. There should be a clear view of the entire exit terminal, including the exit nose and a section of the ramp roadway beyond the gore.

The sight distance on a freeway preceding the approach nose of an exit ramp should exceed the minimum stopping sight distance for the through traffic design speed, desirably by 25 percent or more. Decision sight distance, as discussed in Section 3.2.3, is desired where practical. There should be a clear view of the entire exit terminal, including the exit nose. See Sections 3.2.2 and 3.3.12 for ranges in design values for stopping sight distance on horizontal and vertical curves for open road conditions and turning roadways.

10.9.6.2.12 Grade and Profile Design

The profile of a typical ramp usually consists of a central portion on an appreciable grade, coupled with terminal vertical curves and connections to the profiles of the intersection legs. The following references to ramp gradient pertain largely to the central portion of the ramp profile. Profiles at the terminals largely are determined by through-road profiles and are seldom tangent grades.

Ramp grades should be as flat as practical to minimize the driving effort needed in maneuvering from one road to another. Most ramps are curved, and steep ramp grades in combination with curves hamper traffic flow. The slowing down of vehicles on an ascending ramp is not as serious as on a through road, provided the speed is not decreased sufficiently to result in a peak-hour backup onto the through road. Most diamond ramps are only 400 to 1,200 ft [120 to 360 m] long, and the short central portion with the steepest gradient has only moderate operational effect. Accordingly, gradients on ramps may be steeper than those on the intersecting highways. For any one ramp, the gradient to be used is dependent on a number of factors unique to that site and quadrant. The flatter the gradient on a ramp, the longer it will be, but the effect of gradient on ramp length is not substantial. The conditions and designs at ramp terminals frequently have an effect equal to the effect of the gradient. For example, when the ramp profile is opposite in direction to that of the through highway, a fairly long vertical curve is needed because of the large algebraic difference in grade; this adds considerably to the length of ramp. As another example, additional length may be needed to warp the ramp profile to attain superelevation or to provide drainage.

In general, adequate sight distance is more important than a specific gradient control and should be favored in design. Usually, these two controls are compatible. Values in Table 10-2 provide general design criteria for ramp grades and the application of grade values is discussed below. On one-way ramps, a distinction should be made between ascending and descending gradients. For high-speed ramp designs, the values cited in the next paragraph apply. However, with proper

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ramp terminal facilities, short upgrades of 7 to 8 percent permit good operation without unduly slowing passenger cars. Short upgrades of as much as 5 percent do not unduly interfere with truck and bus operation. On one-way downgrade ramps, gradients of up to 8 percent do not cause undesirable operation due to excessive acceleration of passenger vehicles. However, there is a greater potential for heavy trucks to increase their speeds on downgrades. Therefore, downgrades should desirably be limited to 3 or 4 percent on ramps with sharp horizontal curvature and significant heavy truck or bus traffic. In many areas, consideration of snow and ice conditions may limit the choice of gradient regardless of the direction of the grade.

From the foregoing discussion, it can be seen that ramp grades are not directly related to ramp design speed; however, ramp design speed is a general indication of the quality of design being used, and the gradient for a ramp with a high ramp design speed should be flatter than for one with a low ramp design speed. Where ramp terminals are properly located and fit other design needs and where the curvature conforms to a reasonable design speed, ramps are generally long enough to attain the difference in elevation with grades that are level or, at least, not too steep. The cases in which grade is a determining factor in the length of the ramp are as follows: (1) for intersection angles of 70 degrees or less, the ramp may need to be located farther from the structure to provide a ramp of sufficient length with reasonable grade; (2) where the intersection legs are on appreciable grade, with the upper road ascending and the lower road descending from the structure, the ramp will have to attain a large difference in elevation that increases with the distance from the structure; (3) where a ramp leaves the lower road on a downgrade and meets the higher road on a downgrade, longer-than-usual vertical curves at the terminals may need a long ramp to meet grade limitations. For these reasons, alignment and grade of a ramp should be determined jointly.

Table 10-2. Guidelines for Maximum Ramp Grade

U.S. Customary		Metric	
Ramp Design Speed (mph)	**Maximum Grade for Upgrades and Downgrades (%)[1,2]**	Ramp Design Speed (km/h)	**Maximum Grade for Upgrades and Downgrades (%)[1,2]**
45 or greater	3 – 5	70 or greater	3 – 5
35 - 40	4 – 6	60	4 – 6
25 - 30	5 – 7	40 - 50	5 – 7
15 - 20	6 – 8	20 - 30	6 – 8

Notes:

(1) Where appropriate for topographic conditions, upgrades steeper than the desirable may be used.

(2) One-way downgrades on ramps should be held to the same general maximums, but in special cases they may be 2% greater.

10.9.6.2.13 Vertical Curves

Usually, ramp profiles assume the shape of the letter “S” with a sag vertical curve at the lower end and a crest vertical curve at the upper end. Additional vertical curves may be needed, particularly on ramps that overpass or underpass other roadways. Where a crest or sag vertical curve extends onto the ramp terminal, the length of curve should be determined by using a design speed between those on the ramp and the highway. See Section 3.4.6 for design values for open and turning roadway conditions.

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10.9.6.2.14 Superelevation and Cross Slope

The following guidelines should be used for cross-slope design on ramps:

1. Superelevation rates, as related to curvature and design speed on ramps, are given in Tables 3-8 through 3-12.
2. The cross slope on portions of ramps on tangent normally should be sloped one way at a practical rate ranging from 1.5 to 2 percent for high-type pavements.
3. In general, the rate of change in cross slope in the superelevation runoff section should be based on the maximum relative gradients (Δ) presented in Equation 3-23. The values listed in this table are applicable to single-lane rotation. The adjustment factors b_w listed in Table 3-15 allow for slight increases in the effective gradient for wider rotated widths. The superelevation development is started or ended along the auxiliary lane of the ramp terminal. Alternate profile lines for both edges should be studied so that all profiles match the control points and that no unsightly bumps and dips are inadvertently developed. Spline profiles are very useful in developing smooth lane/shoulder edges.
4. Another important control in developing superelevation along the ramp terminal is that of the crossover crown line at the edge of the through-traffic lane. The maximum algebraic difference in cross slope between the auxiliary lane and the adjacent through lane is shown in Table 9-18. At the ramp terminal the design of the cross slope should accommodate all roadway users.
5. Three segments of a ramp should be analyzed to determine superelevation rates that would be compatible with the design speed and the configuration of the ramp. The exit terminal, the ramp proper, and the entrance terminal should be studied in combination to ascertain the appropriate design speed and superelevation rates.

The guidelines in Item 5 can vary by the type of ramp configuration used. Three ramp configurations are described in the following paragraphs. The diamond ramp usually consists of a high-speed exit terminal, tangent or curved alignment on the ramp proper, and stop or yield conditions at the entrance terminal. Deceleration to the first controlling curve speed should occur on the auxiliary lane of the exit terminal and continued deceleration to stop or yield conditions should occur on the ramp proper. As a result, superelevation rate and radii used should reflect a decreasing sequence of design speeds for the exit terminal, ramp proper, and entrance terminal.

The loop ramp consists of a moderate-speed exit terminal connecting to a slow-speed ramp proper, which in turn connects to a moderate-speed acceleration lane. The curvature of the ramp proper may be a simple curve or a combination of curves, and such curvature determined by the design speed and superelevation rate used. Superelevation should be gradually developed into and out of the curves for the ramp proper, as detailed later in this discussion.

Direct and semidirect ramps generally are designed with a high-speed exit, a moderate- or high-speed ramp proper, and a high-speed entrance. As a result, the design speed and superelevation rates used are comparable to open-road conditions.

The method of developing superelevation at free-flow ramp terminals is illustrated in Figure 10-62.

Figure 10-62A shows a tapered exit from a tangent section with the first ramp curve falling beyond the design deceleration length. The normal cross slope is projected onto the auxiliary lane, and no superelevation is needed until the first ramp proper curve is reached.

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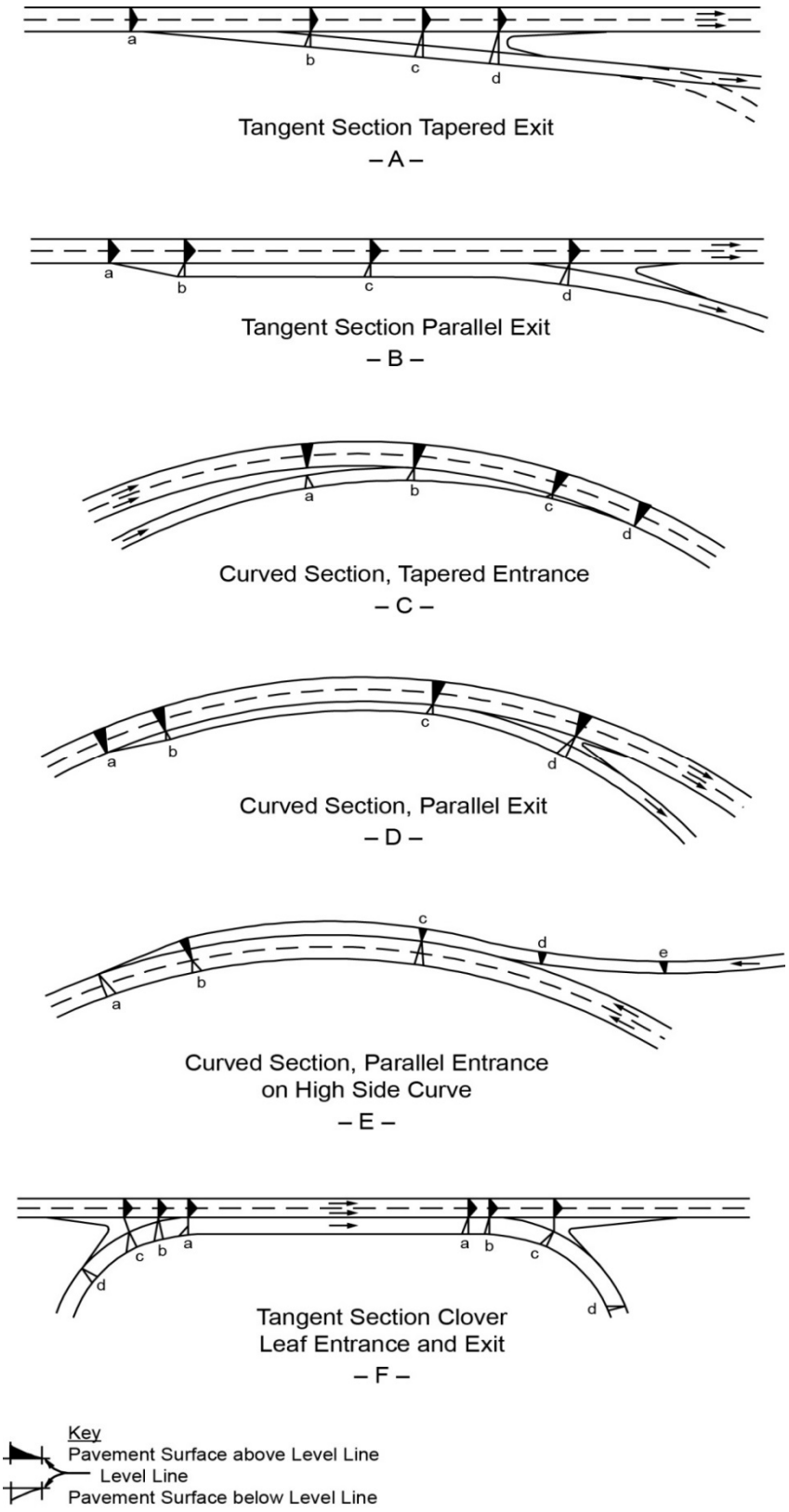


Figure 10-62. Development of Superelevation at Free-Flow Ramp Terminals

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Figure 10-62B shows a parallel-type exit from a tangent section that leads into a flat exiting curve. At point b, the normal cross slope of the through roadway is projected onto the auxiliary lane. At point c, the cross slope can be gradually changed to start the development of superelevation for the exiting curve. At point d, two breaks in the crossover crown line may be conducive to developing a full superelevation in the vicinity of the physical nose.

Figures 10-62C and 10-62D show ramp terminals on which the superelevation of the through roadway would be projected onto the auxiliary lane. Figure 10-62E shows a parallel entrance terminal on the high side of a curve. Wherever practical, a tangent section between the ramp and the main line should be provided to accommodate the superelevation transition. At point e, the superelevation on the ramp begins to decrease and is gradually decreased through the tangent section to point d. At point d, the cross slope is gradually rotated to eventually meet the superelevation rate of the main line at point c.

Figure 10-62F shows a parallel exit from a tangent section with sharp curvature developing in advance of the physical nose. This design is typical for cloverleaf loops. Part of the cross-slope transition can be accomplished over the length of the parallel lane with about half of the total superelevation being developed at point b. Full superelevation of the ramp proper is reached beyond the physical nose.

Care should be exercised to see that the rate of change in cross slope in the runoff section is based on the maximum relative gradients from Equation 3-23 and that the algebraic difference in cross slope does not exceed the values presented in Table 9-18.

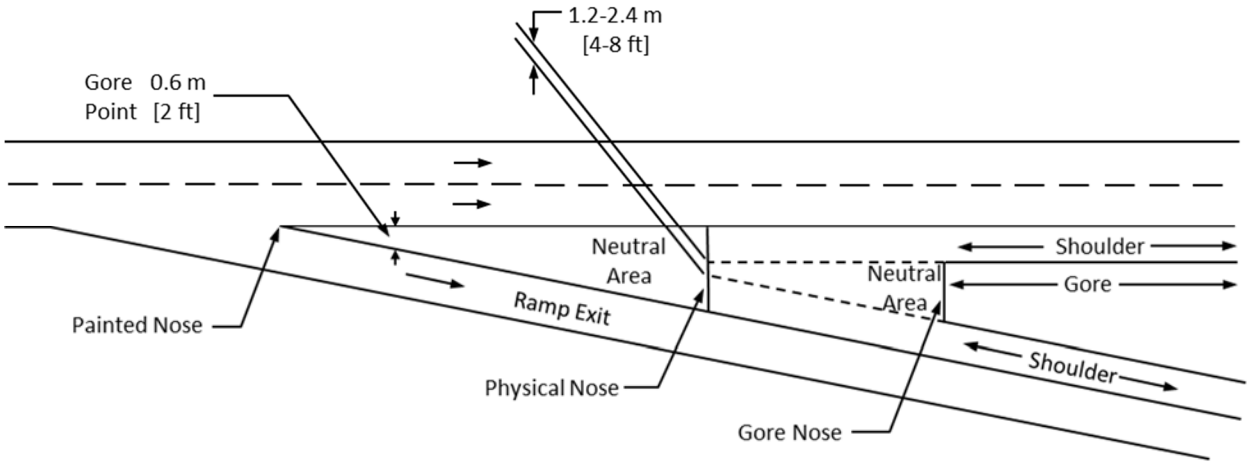


Figure 10-63. Typical Exit Gore Area Characteristics

10.9.6.2.15 Gores

The term “gore” indicates an area downstream from the shoulder intersection points as illustrated in Figure 10-63. The painted nose is a point, having no dimensional width, occurring at the separation of the roadways. The gore point is located where the pair of solid white pavement edge markings that separate the ramp from the intersecting roadway are 2 ft apart. If the markings do not extend to a point where they are 2.0 ft apart, then the gore point is found by extrapolating both markings until the extrapolated portion is 2.0 ft apart. The physical nose is a

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point upstream from the gore, having some dimensional width that separates the roadways. The neutral area refers to the triangular area between the painted nose and the gore nose and incorporates the physical nose. The geometric layout of these is an important part of exit ramp terminal design. It is the decision point area that should be clearly seen and understood by approaching drivers. Furthermore, the separating ramp roadway not only should be clearly evident but should also have a geometric shape appropriate for the likely speeds at that point. In a series of interchanges along a freeway, the gores should be uniform and have the same appearance to drivers.

The width at the gore nose is typically between 20 to 30 ft [6.0 to 9.0 m], including paved shoulders, measured between the traveled way of the main line and that of the ramp. This dimension may be increased if the ramp roadway curves away from the freeway immediately beyond the gore nose or if speeds in excess of 60 mph [100 km/h] are expected to be common.

The entire triangular area, or neutral area, should be striped to delineate the proper paths on each side and to assist the driver in identifying the gore area. The MUTCD (*9*) may be referenced for guidance on channelization. Standard or snow-plowable raised reflective markers can be employed for additional delineation.

Rumble strips may be placed in the neutral area but should not be located too close to the gore nose because such placement renders them ineffective for warning high-speed vehicles. In all cases, supplemental devices of this type should be placed to provide the driver with ample advance warning to make timely corrections in the vehicle's path.

The rate of crashes in gore areas is typically greater than the rate of run-off-the-road crashes at other locations. For this reason, the gore area, and the unpaved area beyond, should be kept as free of obstructions as practical to provide a clear recovery area. The unpaved area beyond the gore nose should be graded to be as nearly level with the roadways as practical so that vehicles inadvertently entering will not be overturned or abruptly stopped by steep slopes. Heavy sign supports, luminaire supports, and roadway structure supports should be kept well out of the graded gore area. In addition, yielding or breakaway supports should be employed for the exit sign, and concrete footings, where used, should be kept flush with the ground level.

Unfortunately, there will be situations where placement of a major obstruction in a gore is unavoidable. Gores that occur at exit ramp terminals on elevated structures are a prime example. Also, there are occasions when locating a bridge pier in a gore cannot be avoided. Guardrails and bridge rails are designed to handle angular impacts but are not effective in handling the kind of near head-on impacts that occur at these gores.

In recognition of the exposed position of fixed objects in gore areas, a considerable effort has been directed toward the development of cushioning or energy-dissipating devices for use in front of such fixed objects. At present, several types of crash cushions are being used. These devices substantially reduce the severity of fixed-object collisions. Thus, adequate space should be provided for the installation of a crash-cushion device whenever a major obstruction is present in a gore on a high-speed highway. Reference may be made to Section 4.10.4 and to the *Roadside Design Guide* (*3*) for details on the installation of crash-cushion devices.

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Although the term “gore” generally refers to the area between a through roadway and an exit ramp, the term may also be used to refer to the similar area between a through roadway and a converging entrance ramp. At an entrance terminal, the point of convergence (beginning of all paved area) is defined as the “merging end.” In shape, layout, and extent, the triangular maneuver area at an entrance terminal is much like that at an exit. However, it points downstream and separates traffic streams already in lanes; thus, it is less of a decision area. The width at the base of the paved triangular area is narrower, however, and is usually limited to the sum of the shoulder widths on the ramp and freeway plus a narrow physical nose 4 to 8 ft [1.2 to 2.4 m] wide.

Figure 10-64 illustrates typical gore designs for free-flow exit ramps. Figures 10-64A and 10-64B depict a recovery area adjacent to the outside through lane and moderate offset to the left of the ramp traveled way.

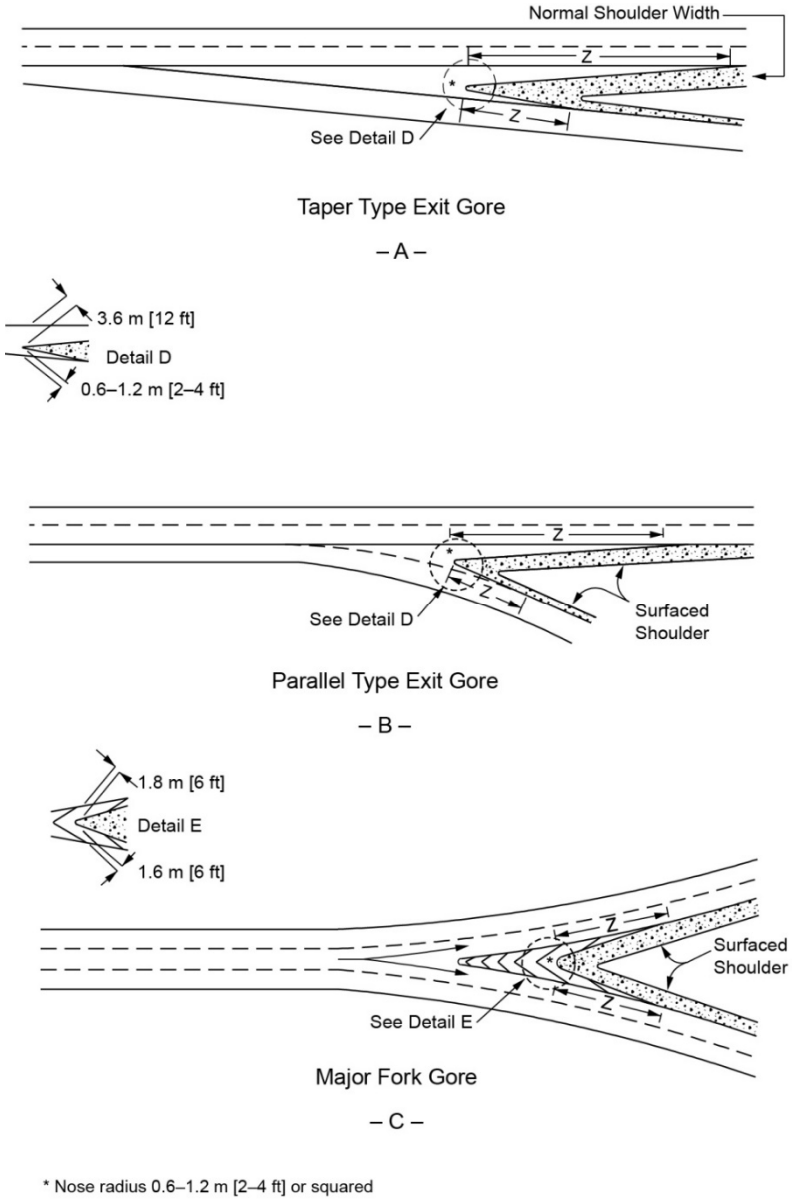


Figure 10-64. Typical Gore Details

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Figure 10-64C presents a major fork, with neither diverging roadway having priority. The offset is equal for each roadway, and striping or rumble strips are placed upstream from the physical nose. Desirably, curbs, utility poles, and sign supports should be omitted from the gore area, especially on high-speed facilities. When curbs are used, they should be low-profile, sloping designs, and the geometry of the gore area intersection points is usually curved. When curbs are not used, the geometry of the gore area intersection points can be squared or truncated.

Table 10-3 gives the minimum lengths for tapers beyond the offset nose (shown as length Z in Figure 10-64). However, another alternative for providing a recovery area is the use of the paved shoulder of the through lane.

Table 10-3. Minimum Length of Taper Beyond an Offset Nose

U.S. Customary		Metric	
Design Speed of Approach Highway (mph)	Length of Nose Taper (Z) per Unit Width of Nose Offset	Design Speed of Approach Highway (km/h)	Length of Nose Taper (Z) per Unit Width of Nose Offset
30	15.0	50	15.0
35	17.5	60	20.0
40	20.0	70	22.5
45	22.5	80	25.0
50	25.0	90	27.5
55	27.5	100	30.0
60	30.0	110	35.0
65	32.5	120	40.0
70	35.0	130	45.0
75	37.5		
80	40.0		

Figure 10-65 shows an entrance ramp, as at a cloverleaf loop, where a reduction in the ramp lane width is appropriate to maintain a single-lane entrance. Another option is to begin the reduction in the ramp lane width at the end of the ramp curvature.

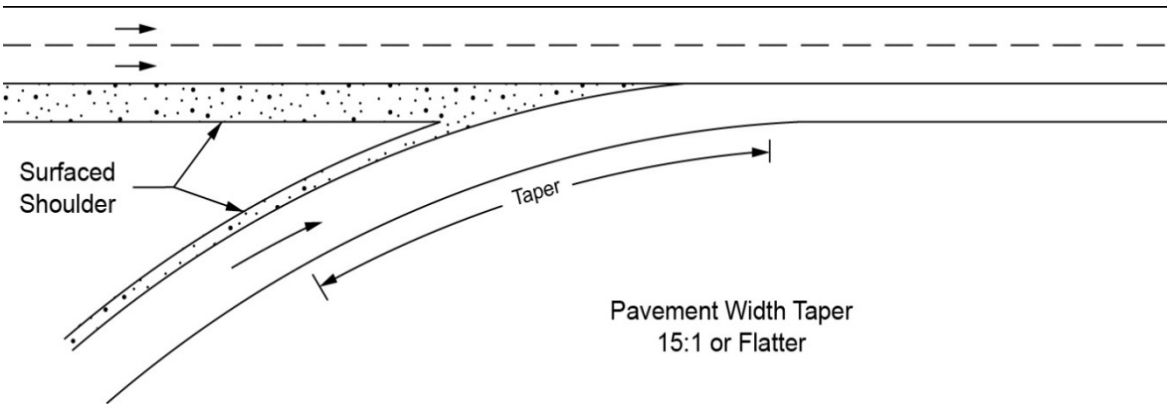


Figure 10-65. Traveled Way Narrowing on Entrance Ramps

Figure 10-66 presents a photograph of a single-lane exit. The striping, pavement reflectors, delineators, and fixed-source lighting help guide the exiting motorist.

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Figure 10-66. Gore Area, Single-Lane Exit ***Source: Arizona DOT***

Figure 10-67 shows a gore at a major fork between two freeways. The small angle of divergence results in the long, gradual split with a clear recovery area. Overhead signs are provided.



Figure 10-67. Gore Area, Major Fork ***Source: Georgia DOT***

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Whereas Figure 10-68 shows a gore at a two-lane exit from a freeway, Figure 10-69 shows a typical gore and ramp terminal for a ramp entering a freeway.



Figure 10-68. Gore Area, Two-Lane Exit ***Source: Virginia DOT***



Figure 10-69. Entrance Terminal ***Source: Virginia DOT***

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10.9.6.3 Ramp Traveled-Way Widths

10.9.6.3.1 Width and Cross Section

Ramp traveled-way widths are governed by the type of operation, curvature, and volume and type of traffic. It should be noted that the roadway width for a turning roadway includes the traveled-way width plus the shoulder width or equivalent offset outside the edges of the traveled way. Section 3.3.11 on "Widths for Turning Roadways at Intersections" may be referenced for additional discussion on the treatments at the edge of the traveled way.

Design widths of ramp traveled ways for various conditions are given in Table 3-27. Values are shown for three general design traffic conditions, as follows:

- **Traffic Condition A**—predominantly P vehicles, but some consideration for SU trucks
- **Traffic Condition B**—sufficient SU vehicles to govern design, but some consideration for semitrailer vehicles
- **Traffic Condition C**—sufficient buses and combination trucks to govern design

Traffic conditions A, B, and C are described in broad terms because design traffic volume data for each type of vehicle are not available to define these traffic conditions with precision in relation to traveled-way width. In general, traffic condition A has a small volume of trucks or only an occasional large truck, traffic condition B has a moderate volume of trucks (in the range of 5 to 10 percent of the total traffic), and traffic condition C has more and larger trucks.

10.9.6.3.2 Shoulder Widths and Lateral Offset

Design values for shoulders and lateral offsets on the ramps are as follows:

- Where paved shoulders are provided on ramps, they should have a uniform width for the full length of ramp. A paved shoulder width of 2 to 4 ft [0.6 to 1.2 m] is desirable for the left-side shoulder. A paved shoulder width of 6 to 10 ft [2.4 to 3.0 m] is desirable for the right-side shoulder. For one-way operation, the combined left- and right-side shoulders should be 10 to 14 ft [3.0 to 4.3 m].
- The left and right shoulder widths may be reversed if needed to provide additional sight distance.
- The ramp traveled-way widths from Table 3-27 for Case II and Case III should be modified when paved shoulders are provided on the ramp. The ramp traveled-way width for Case II should be reduced by the total width of both right and left shoulders. However, in no case should the ramp traveled-way width be less than needed for Case I. For example, with condition C and a 400-ft [125-m] radius, the Case II ramp traveled-way width without shoulders is 21 ft [6.4 m]. If a 2-ft [0.6-m] left shoulder and an 8-ft [2.4-m] right shoulder are provided, the minimum ramp traveled-way width should be 15 ft [4.8 m].
- Directional ramps with a ramp design speed over 40 mph [60 km/h] should have a paved right shoulder width of 8 to 10 ft [2.4 to 3.0 m] and a paved left shoulder width of 1 to 6 ft [0.3 to 1.8 m].
- For freeway ramp terminals where the ramp shoulder is narrower than the freeway shoulder, the paved shoulder width of the through lane should be carried into the exit terminal. It should

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also begin within the entrance terminal, with the transition to the narrower ramp shoulder accomplished gradually on the ramp end of the terminal. Abrupt changes should be avoided.

- Ramps should have a lateral offset on the right outside of the edge of the traveled way of at least 6 ft [1.8 m], and preferably 8 to 10 ft [2.4 to 3.0 m], and a lateral offset on the left of at least 4 ft [1.2 m] beyond the edge of traveled way. These lateral offsets may be shifted between the right and left sides of the ramp in some cases to provide the needed horizontal sightline offset.
- Where ramps pass under structures, the total roadway width should be carried through the structure. Desirably, structural supports should be located beyond the clear zone. As a minimum, structural supports should be at least 4 ft [1.2 m] beyond the edge of paved shoulder. The *AASHTO Roadside Design Guide* (*3*) provides guidance on clear zone and the use of roadside barriers.
- Ramps on overpasses should have the full approach roadway width carried over the structure.
- Edge lines or some type of color or texture differentiation between the traveled way and shoulder is desirable.

10.9.6.3.3 Shoulders and Curbs

Shoulders should be provided on ramps and ramp terminals in interchange areas to provide a space that is clear of the traveled way for emergency stopping, to minimize the effect of breakdowns, and to aid drivers who may be confused.

Ramps at interchanges should be designed without curbs. Curbs should be considered only to facilitate particularly difficult drainage situations, such as in urban areas where restrictive ROW favors enclosed drainage. In some cases, curbs are used at the ramp terminals but are omitted along the central ramp portions. Where curbs are not used, full-depth paving should be provided on shoulders because of the frequent use of shoulders for turning movements.

On low-speed facilities, curbs may be placed at the edge of roadway. Vertical curbs are seldom used in conjunction with shoulders, except where pedestrian protection is needed. Where curbs are used on high-speed facilities, sloping curbs should be placed at the outer edge of the shoulder. Because of fewer restrictions and more liberal designs in rural areas, the need for curbs seldom arises. See Section 4.4 for a full discussion of shoulder cross section elements.

10.9.6.4 Ramp Terminals

The terminal of a ramp is that portion adjacent to the through traveled way, including SCLs, tapers, and islands. Ramp terminals may be the at-grade type, as at the crossroad terminal of diamond or partial cloverleaf interchanges, or the free-flow type where ramp traffic merges with or diverges from high-speed through traffic at flat angles. Design elements for the at-grade type are discussed in Chapter 9, and those for the free-flow type are discussed in the following sections.

Terminals are further classified as either single or multilane, according to the number of lanes on the ramp at the terminal, and as either a taper or parallel type, according to the configuration of the speed-change lane.

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10.9.6.4.1 Left-Side Entrances and Exits

Left-side entrances and exits are contrary to driver expectancy when intermixed with right-side entrances and exits and should be avoided, where practical. Left-side entrances and exits for managed lane facilities are also contrary to driver expectancy. See Section 8.4.8 for additional information on managed lanes and transit facilities.

Left-side ramp terminals break up the uniformity of interchange patterns and generally create uncertain operation on through roadways. Left-side entrances and exits are considered satisfactory for CD roads; however, their use on high-speed, free-flow ramp terminals is not recommended. Because left-side entrances and exits are contrary to driver expectancy, special attention should be given to acceleration/deceleration lengths, signing, and the provision for decision sight distance preceding the approach nose of the exit ramp in order to alert the driver that an unusual situation exists. There should be a clear view of the whole of the exit terminal. If it is not practical to provide decision sight distance because of horizontal or vertical curvature or if relocation of decision points is not practical, additional traffic control devices for advance warning of the conditions should be considered.

10.9.6.4.2 Terminal Location and Sight Distance

Where diamond ramps and partial cloverleaf arrangements intersect the crossroad at grade, an at-grade intersection is formed. Desirably, this intersection should be located an adequate distance from the separation structure to provide adequate sight distance for all approaches. Sight distance criteria are detailed in Section 3.2.

Drivers prefer and expect to exit in advance of the separation structure. The use of CD roads and single exits on partial cloverleafs and other types of interchange configurations automatically positions the mainline exit in advance of the separation structure.

Designs that result in an exit concealed behind a crest vertical curve should be avoided, especially on high-speed facilities. Desirably, high-speed entrance ramp terminals should be located on descending grades to aid truck acceleration. Adequate sight distance at entrance terminals should be available so that merging traffic on the ramp can adjust speed to merge into gaps on the main facility.

Loop ramps that are located beyond the structure, as in the conventional cloverleaf or in certain arrangements of partial cloverleafs, usually need a parallel deceleration lane. The actual exit from the auxiliary lane is difficult for drivers to locate even when sight distance is not restricted by a vertical curve. Placing the exit in advance of the structure via a single exit alleviates this concern. See “Two-Exit versus Single-Exit Interchange Design” in Section 10.9.5.

10.9.6.4.3 Ramp Terminal Design

Profiles of ramp terminals should be designed in association with horizontal curves to avoid sight restrictions that will adversely affect operations. At an exit into a ramp on a descending grade, a horizontal curve ahead should not appear suddenly to a driver. Instead, the initial crest vertical curve should be made longer and sight distance over it should be increased so that the location and direction of the horizontal curve are apparent to the driver sufficiently in advance to provide time for the driver to respond appropriately. At an entrance terminal from a ramp on an

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ascending grade, the portion of the ramp intended for acceleration and the ramp terminal should closely parallel the through-lane profile to permit entering drivers to have a clear view of the through road ahead, to the side, and to the rear.

It is desirable that profiles of highway ramp terminals be designed with a platform on the ramp side of the approach nose or merging end. This platform should be at least 200 ft [60 m] in length and should have a profile that does not greatly differ from that of the adjacent through-traffic lane.

A platform area should also be provided at the at-grade terminal of a ramp. The length of this platform should be determined from the type of traffic control and the capacity at the terminal. For further discussion, see Section 9.4.3.

10.9.6.4.4 Traffic Control

On major highways, ramps are arranged to facilitate all turning movements by merging or diverging maneuvers. On minor highways, some of the left-turning movements often are made at grade. The left-turning movements leaving the crossing highway preferably should have median left-turn lanes. For low-volume crossroads, the left-turning movements from ramps normally should be controlled by stop signs. The right-turning movements from ramps into multilane crossroads should be provided with an acceleration lane or generous taper, or should be controlled by stop or yield signs. Ramps approaching stop signs should be nearly perpendicular to the crossroad and be nearly level for storage of several vehicles. Ramp terminals at cross streets can also be controlled by roundabouts.

Traffic signal controls may be needed at ramp terminals on the minor road where there is sufficient volume of through and turning traffic. In such cases, the intersections formed at the terminals should be designed and operated in the same manner as any other traffic-signal-controlled intersection at grade. Signal controls should be avoided on express-type highways and confined to the minor highways on which other intersections are at grade and some of which are signalized. In or near urban areas, signal control is especially appropriate at ramp terminals on streets that cross over or under an expressway. Here the turning movements usually are sizable, and the cost of ROW and improvements is high. As a result, appreciable savings may be realized by the use of diamond ramps with high-type terminals on the expressway and signalized terminals on the cross streets. Warrants for the installation of traffic signals that can be applied to diamond ramp terminals are given in Part 4 of the MUTCD (*9*).

10.9.6.4.5 Distance Between a Free-Flow Terminal and Structure

The terminal of a ramp should not be near the grade-separation structure. If it is not practical to place the exit terminal in advance of the structure, the exiting terminal on the far side of the structure should be well removed to provide drivers leaving the through lanes some distance after passing the structure to see the exit and begin the exit maneuver. Decision sight distance should be provided, where practical. The distance between the structure and the approach nose at the ramp terminal should be sufficient for exiting drivers to leave the through lanes without undue hindrance to through traffic. Such distance also aids drivers who enter from a ramp terminal on the far side of the structure so they have a clear view well back on the through road behind or to the left. Such drivers may be able to see back along the road beyond the limits of the structure,

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but as a general rule, the entering driver's view is obstructed by the crest of the profile at an overpass and by the columns, abutments, and approach walls at an underpass.

The conditions for determining the distance between a structure and the far side approach nose are similar to those discussed for speed-change lanes. A minimum distance between the structure and an exit nose of about the same length as a speed-change taper is suggested. Decision sight distances are desirable but are not rigid controls for ramp design. Topographic or ROW controls may govern the overall shape of the ramp.

While a long separation distance between a structure and an exit ramp terminal is desirable to achieve efficient operations and low crash frequencies, this distance can be too long for certain ramp arrangements such as cloverleaf loop ramps. Unusually large ROW needs as well as increased travel time and length on the loops may result. Where only one loop is needed and it falls on the far side of the structure, a SCL should be developed on the near side of the structure and carried across the structure if sight distance is limited.

The separation distance between a structure and a ramp terminal does not need to be as long for ramp terminals on the near side of a structure as for those beyond the structure. Both the view of the terminal ahead for drivers approaching on the through road and the view back along the road for drivers on an entrance ramp are not affected by the structure. Where an entrance ramp curve on the near side of the structure needs an acceleration lane, the ramp terminal should be located to provide sufficient length for it between the terminal and the structure, or the acceleration lane may be continued through or over the structure. Where ramp terminals on the far side of a structure are located close to it, the horizontal sight line may be limited by the abutment or parapet; available sight distance should, therefore, be checked.

10.9.6.4.6 Distance Between Successive Ramp Terminals

On urban freeways, two or more ramp terminals are often located in close succession. To provide sufficient weaving length and adequate space for signing, a reasonable distance should be provided between successive ramp terminals. Spacing between successive outer ramp terminals is dependent on the classification of the interchanges involved, the function of the ramp pairs (entrance or exit), and weaving potential.

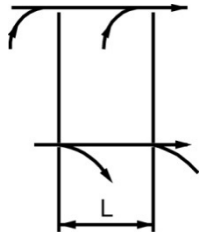
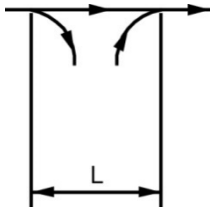
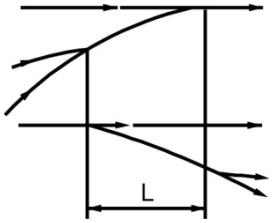
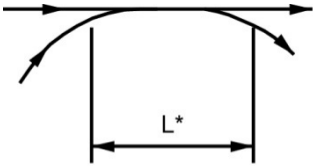
The five possible ramp-pair combinations are: (1) an entrance followed by an entrance (EN-EN), (2) an exit followed by an exit (EX-EX), (3) an exit followed by an entrance (EX-EN), (4) an entrance followed by an exit (EN-EX) (weaving), and (5) turning roadways.

Figure 10-70 presents recommended minimum ramp terminal spacing for the various ramp-pair combinations as they are applicable to interchange classifications. **These recommended minimum ramp terminal spacing values represent a reasonable starting point during planning and early design.** The recommendations presented in Figure 10-70 are based on operational experience and the need for flexibility and adequate signing. They should be checked in accordance with the procedure outlined in the *Highway Capacity Manual* (HCM) (*24*). Also, the procedure for measuring the length of the weaving section is given in the HCM. The distances labeled *L* in the Figure 10-70 are measured between the painted noses (see Figure 10-63). Figure 10-71 shows the definition of the ramp spacing dimension. A minimum distance of

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300 ft [90 m] is recommended between the end of the taper (as shown in Figure 10-72) for the first entrance ramp and the painted nose for the succeeding entrance ramp.

Recent research focused on the EN-EX and EN-EN ramp pair combinations at relatively simple, single-lane service type interchanges. Geometry, traffic operations, safety, and signing were considered in providing a performance based approach to interchange ramp spacing. The ramp terminal spacing needs may be greater for conditions involving multilane ramps such as freeway-to-freeway connections. Ramp spacing distances and the potential geometric feasibility of specific ramp spacing dimensions were provided for simple service type interchanges for each of the ramp pair combinations outlined in Figure 10-70 (*18*).

EN-EN or EX-EX		EX-EN		Turning Roadways		EN-EX (Weaving)			
						 * Not Applicable to Cloverleaf Loop Ramps			
Full Freeway	CDR	Full Freeway	CDR	System Interchange	Service Interchange	System to Service Interchange		Service to Service Interchange	
						Full Freeway	CDR	Full Freeway	CDR
Minimum Lengths Measured between Successive Ramp Terminals									
1000 ft (300 m)	800 ft (240 m)	500 ft (150 m)	400 ft (120 m)	800 ft (240 m)	600 ft (180 m)	2000 ft (600 m)	1600 ft (480 m)	1600 ft (480 m)	1000 ft (300 m)

Notes: CDR—Collector-distributor road
EN—Entrance EX—Exit

Figure 10-70. Recommended Minimum Ramp Terminal Spacing

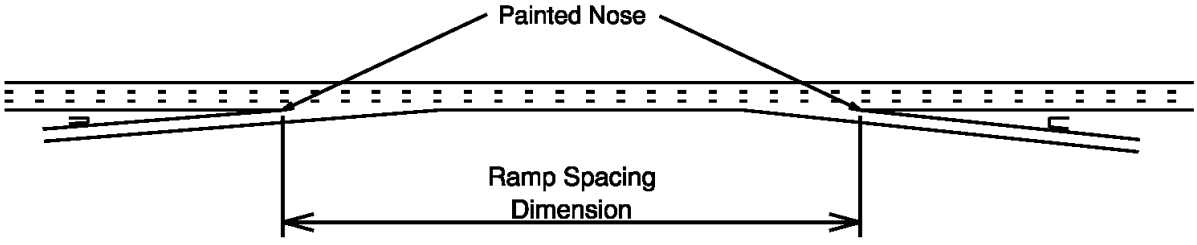


Figure 10-71. Ramp Spacing Dimension

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Where an entrance ramp is followed by an exit ramp, the absolute minimum distance between the successive noses is governed by weaving considerations. The spacing policy for EN-EX ramp combinations is not applicable to cloverleaf loop ramps. For these interchanges, the distance between EN-EX ramp noses is primarily dependent on loop ramp radii and roadway and median widths. A recovery lane beyond the nose of the loop ramp exit is desirable.

When the distance between the successive noses is less than 1,500 ft [450 m], the SCLs should be connected to provide an auxiliary lane. This auxiliary lane improves traffic operation over relatively short sections of the freeway route and is not considered an addition to the basic number of lanes. See "Auxiliary Lanes" of Section 10.9.5 for alternate methods of dropping these lanes.

There are two typical scenarios of an EX-EN combination. The shortest dimension (400 to 500 ft [120 to 150 m]) would be that of an exit followed by the entrance for a "buttonhook" or "gull wing" design where the freeway ramps are serving a local street parallel to the freeway versus a local street crossing the freeway as an over or underpass. The second scenario would be when an exit ramp and subsequent entrance ramp are servicing grade separated ramps (ramp braids). Due to the vertical and horizontal relationships of this configuration, the spacing will generally be greater than the minimum values in Figure 10-70 reflecting a condition where both ramp profiles are changing.

10.9.6.4.7 Speed-Change Lanes

Drivers leaving a highway at an interchange are required to reduce speed as they exit onto a ramp. Drivers entering a highway from a turning roadway accelerate until the desired highway speed is reached. Because the change in speed is usually substantial, provision should be made for acceleration and deceleration to be accomplished on auxiliary lanes to minimize interference with through traffic and to reduce crash potential. Such an auxiliary lane, including tapered areas, may be referred to as a SCL. The terms "SCL," "deceleration lane," or "acceleration lane" as used herein apply broadly to the added lane that joins the traveled way of the highway to the turning roadway and do not necessarily imply a definite lane of uniform width. This additional lane is a part of the elongated ramp terminal area.

A SCL should have sufficient length to enable a driver to make the appropriate change in speed between the highway and the turning roadway. Moreover, in the case of an acceleration lane, there should be additional length to permit adjustments in speeds of both through and entering vehicles so that the entering driver can position the vehicle opposite a gap in the through-traffic stream and then maneuver into the stream before the acceleration lane ends. This latter consideration influences both the configuration and length of an acceleration lane.

Two general forms of SCLs are: (1) the taper type and (2) the parallel type. The taper type provides a direct entry or exit at a flat angle, whereas the parallel type has an added lane for changing speed. Either type, when properly designed, will operate satisfactorily. However, the parallel type is still favored in certain areas, and some agencies use the taper type for exits and the parallel type for entrances. Furthermore, taper-type entrances have been found to encourage merge speeds that are closer to freeway speeds than parallel type entrances (*21*); however, where

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there are mainline volumes that meet or exceed capacity, parallel type entrances allow additional flexibility to drivers in selecting a merge location.

See Section 9.7 for discussion of SCLs applicable to at-grade intersections.

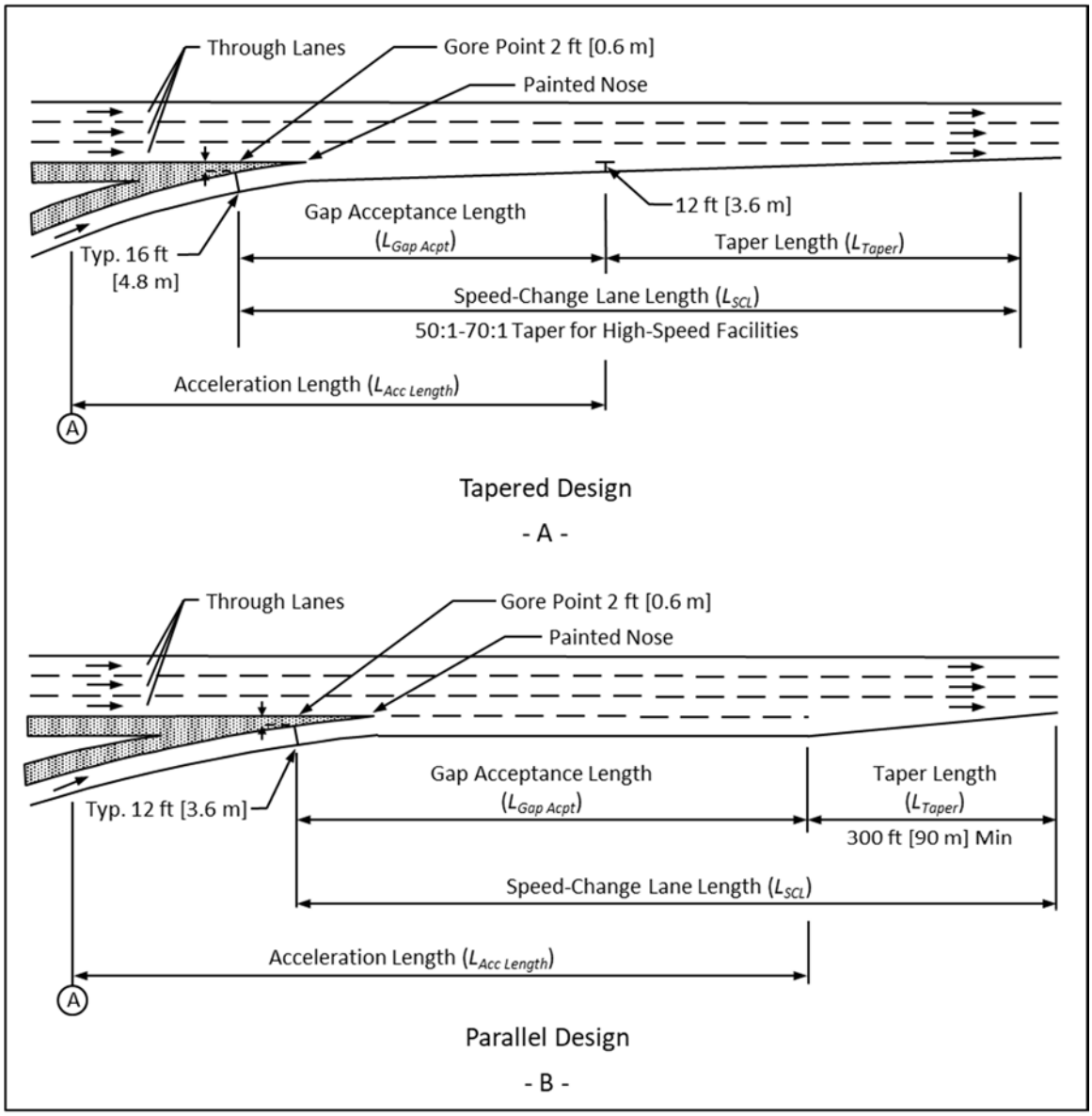
10.9.6.5 Single-Lane Free-Flow Terminals, Entrances

10.9.6.5.1 Taper-Type Entrance

When properly designed, the taper-type entrance usually operates smoothly at all volumes up to and including the design capacity of merging areas. By relatively minor speed adjustment, the entering driver can see and use an available gap in the through-traffic stream. A typical single-lane, taper-type entrance terminal is shown in Figure 10-72A.

The entrance is merged into the freeway with a long, uniform taper. Operational studies show a desirable rate of taper of approximately 50:1 to 70:1 (longitudinal to lateral) between the outer edge of the acceleration lane and the edge of the through-traffic lane. The gap acceptance length, L_g, is also a consideration in the design of taper-type entrances, as illustrated in Figure 10-72A.

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Notes:

1. $L_{Acc\ Length}$ is the recommended acceleration length as shown in Table 10-4 or as adjusted by Table 10-5.
2. Point A controls speed on the ramp. $L_{Acc\ Length}$ should not start back on the curvature of the ramp unless the radius equals 1,000 ft [300 m] or more.
3. $L_{Gap\ Acpt}$ is the recommended gap acceptance length. $L_{Gap\ Acpt}$ should be a minimum of 300 to 500 ft [90 to 150 m].
4. The value of $L_{Acc\ Length}$ or $L_{Gap\ Acpt}$, whichever produces the greater distance downstream from where the gore point equals 2 ft [0.6 m], is suggested for use in the design of the ramp distance.

Figure 10-72. Typical Single-Lane Entrance Ramps

The geometrics of the ramp proper should be such that motorists may attain a merge speed that is within 5 mph [10 km/h] of the operating speed of the freeway by the time they reach the point where the left edge of the ramp joins the traveled way of the freeway. For consistency of

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application, this point of convergence of the left edge of the ramp and the right edge of the through lane may be assumed to occur where the right edge of the ramp traveled way is 12 ft [3.6 m] from the right edge of the through lane of the freeway. While it is desirable for motorists to merge onto the freeway at speeds near the operating speed of the freeway, some motorists may choose to enter the freeway at speeds below the operating speed of the freeway without using the full length of the SCL. Taper-type entrances have been shown to encourage motorists to merge closer to freeway speeds (*21*).

The acceleration length is the distance needed for acceleration upon exiting the controlling feature of the ramp proper or the final section of the ramp proper upstream of the gore point to the point of convergence with the freeway. The acceleration length begins at Point A and ends where the gap acceptance length ends and the taper begins. Point A represents the location where the ramp proper ends and the freeway mainline ramp terminal begins. Conceptually, the acceleration length begins at the end of the controlling curve or feature of the ramp proper. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. For design purposes, it may be assumed that horizontal curves with radii less than 1,000 ft will significantly affect vehicle speeds. If the radius of the final curve of the ramp is greater than 1,000 ft and drivers on the ramp have a clear view of traffic in the right lane of the freeway, the beginning of the acceleration length may be located upstream of the gore point (i.e., along the final curve of the ramp proper).

Table 10-4 shows minimum values of acceleration lengths for entrance terminals. Figure 10-72 illustrates the minimum lengths for gap acceptance. Referring to Figure 10-72, the larger value of the acceleration length ($L_{Acc\ Length}$) or the gap acceptance length ($L_{Gap\ Acpt}$) is suggested for use in the design of the ramp entrance. Where the minimum values for lane width 16 ft [4.8 m]) and taper rate (50:1) are used with high traffic volumes, taper lengths longer than the larger of $L_{Acc\ Length}$ or $L_{Gap\ Acpt}$ may be needed to avoid inferior operation and to reduce abrupt moves when merging into the mainline traffic stream. Where grades are present on ramps, speed-change lengths should be adjusted in accordance with Table 10-5.

The design values in Table 10-4 are conservative estimates based on free-merge conditions (i.e., free-flow conditions) for passenger cars. Additionally, if trucks constitute a substantial percentage of the traffic volume to be selected as the design vehicle, acceleration lane lengths designed to better accommodate heavier design vehicles can be derived using Figures 3-16 and 3-18 in Chapter 3.

10.9.6.5.2 Parallel-Type Entrances

The parallel-type entrance provides an added lane of sufficient length to enable a vehicle to accelerate to near-freeway speed prior to merging. A taper is provided at the end of the added lane. The process of entering the freeway is similar to a lane change to the left. The driver is able to use the side-view and rear-view mirrors to monitor surrounding traffic.

A typical design of a parallel-type entrance is shown in Figure 10-72B. Desirably, a curve with a radius of 1,000 ft [300 m] or more and a length of at least 200 ft [60 m] should be provided in advance of the added lane. If this curve has a short radius, motorists tend to drive directly onto

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the freeway without using much of the gap acceptance length. This behavior results in undesirable merging operations.

The taper at the downstream end of a parallel-type acceleration lane should be a suitable length to guide the vehicle gradually onto the through lane of the freeway. A taper length of approximately 300 ft [90 m] is suitable for design speeds up to 70 mph [110 km/h].

The minimum acceleration lengths for entrance terminals are given in Table 10-4, and the adjustments for grades are given in Table 10-5. The acceleration length is the distance needed for acceleration upon exiting the controlling feature of the ramp proper or the final section of the ramp proper upstream of the gore point to the point of convergence with the freeway. Conceptually, the acceleration length begins at the end of the controlling curve or feature of the ramp proper (Point A) and ends where the gap acceptance length ends and the taper begins. For an entrance ramp, the last curve encountered along the ramp proper that significantly affects vehicle speed is considered the controlling curve. It may be assumed that horizontal curves with radii less than 1,000 ft will significantly affect vehicle speeds. If the radius of the final curve of the ramp is greater than 1,000 ft and drivers on the ramp have an unobstructed view of traffic in the right lane of the freeway, the beginning of the acceleration length may be located upstream of the gore point (i.e., along the final curve of the ramp proper).

The advantages in efficient traffic operations and low crash frequencies of long SCLs provided by parallel type entrances are well recognized. A long SCL provides more time for the merging vehicles to find an opening in the through-traffic stream. An acceleration length of at least 1,200 ft [360 m] plus the taper is desirable wherever it is anticipated that the ramp and freeway will frequently carry traffic volumes approximately equal to the design capacity of the merging area.

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Table 10-4. Minimum Acceleration Lengths for Entrance Terminals with Flat Grades of Less Than Three Percent

U.S Customary													
Highway		Acceleration Length, $L_{Acc\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point) (mph)											
		Stop Condition	15	20	25	30	35	40	45	50	55	60	65
Design Speed ($V_{DS\ Hwy}$) (mph)	Merge Speed ($V_{Acc\ Length(f)}$) (mph)	Average Running Speed (i.e., Initial Speed) Exiting the Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point), $V_{Acc\ Length(i)}$ (mph)											
		0	14	18	22	26	30	36	40	44	48	52	56
30	23	180	140	—	—	—	—	—	—	—	—	—	—
35	27	280	220	160	—	—	—	—	—	—	—	—	—
40	31	360	300	270	210	120	—	—	—	—	—	—	—
45	35	560	490	440	380	280	160	—	—	—	—	—	—
50	39	720	660	610	550	450	350	130	—	—	—	—	—
55	43	960	900	810	780	670	550	320	150	—	—	—	—
60	47	1200	1140	1100	1020	910	800	550	420	180	—	—	—
65	50	1410	1350	1310	1220	1120	1000	770	600	370	135	—	—
70	53	1620	1560	1520	1420	1350	1230	1000	820	580	350	135	—
75	55	1790	1730	1630	1580	1510	1420	1160	1040	780	550	300	—
80	57	2000	1900	1800	1750	1680	1600	1340	1240	980	725	500	200
85	59	2230	2090	1985	1930	1865	1790	1525	1455	1250	940	700	500

NOTE: Uniform 50:1 to 70:1 tapers are recommended where acceleration lengths exceed 1,300 ft.

$V_{DS\ Hwy}$ = design speed of highway (mph)

$V_{Acc\ Length(f)}$ = merge speed (i.e., the speed at which a driver merges with through traffic at the end of the acceleration length (mph)

V_{DS} = design speed of controlling feature on the ramp proper (or the final section of the ramp proper upstream of the gore point) (mph)

$V_{Acc\ Length(i)}$ = average running speed exiting the controlling feature on the ramp proper (or the final section of the ramp proper upstream of the gore point) (mph)

$L_{Acc\ Length}$ = acceleration length (ft)

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Metric											
Highway		Acceleration Length, $L_{Acc\ Length}$ (m), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point) (km/h)									
		Stop Condition	20	30	40	50	60	70	80	90	100
Design Speed ($V_{DS\ Hwy}$) (km/h)	Merge Speed ($V_{Acc\ Length(f)}$) (km/h)	Average Running Speed (i.e., Initial Speed) Exiting the Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point), $V_{Acc\ Length(i)}$ (km/h)									
		0	20	28	35	42	51	63	70	77	85
50	37	60	50	30	—	—	—	—	—		
60	45	95	80	65	45	—	—	—	—		
70	53	150	130	110	90	65	—	—	—		
80	60	200	180	165	145	115	65	—	—		
90	67	260	245	225	205	175	125	35	—		
100	74	345	325	305	285	255	205	110	40		
110	81	430	410	390	370	340	290	200	125	42	
120	88	545	530	515	490	460	410	325	245	170	92
130	92	610	580	550	530	520	500	375	300	220	152
140	96	680	630	605	590	570	550	415	330	286	215

Note: Uniform 50:1 to 70:1 tapers are recommended where lengths of acceleration lanes exceed 400 m.

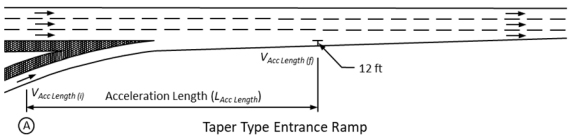
$V_{DS\ Hwy}$ = design speed of highway (km/h)

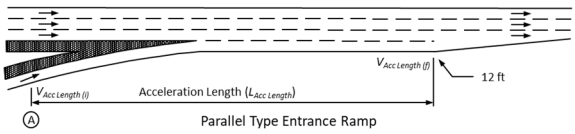
$V_{Acc\ Length(f)}$ = merge speed (i.e., the speed at which a driver merges with through traffic at the end of the acceleration length (km/h)

V_{DS} = design speed of controlling feature on the ramp proper (or the final section of the ramp proper upstream of the gore point) (km/h)

$V_{Acc\ Length(i)}$ = average running speed exiting the controlling feature on the ramp proper (or the final section of the ramp proper upstream of the gore point) (km/h)

$L_{Acc\ Length}$ = acceleration length (m)





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Table 10-5. SCL Adjustment Factors as a Function of Grade

U.S. Customary					
Design Speed of Highway ($V_{DS\ Hwy}$) (mph)	Deceleration Length				
	Ratio of Length on Grade to Length on Level for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point) (mph)[a]				
All Speeds	3 to 4% upgrade 0.9			3 to 4% downgrade 1.2	
All Speeds	5 to 6% upgrade 0.8			5 to 6% downgrade 1.35	
Design Speed of Highway ($V_{DS\ Hwy}$) (mph)	Acceleration Length				
	Ratio of Length on Grade to Length on Level for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Final Section of the Ramp Proper Upstream of the Gore Point) (mph)[a]				
	20	30	40	50	All Speeds
	3 to 4% Upgrade				3 to 4% Downgrade
40	1.3	1.3	—	—	0.7
45	1.3	1.35	—	—	0.675
50	1.3	1.4	1.4	—	0.65
55	1.35	1.45	1.45	—	0.625
60	1.4	1.5	1.5	1.6	0.6
65	1.45	1.55	1.6	1.7	0.6
70	1.5	1.6	1.7	1.8	0.6
75	1.6	1.7	1.8	2.0	0.6
80	1.7	1.8	2.0	2.1	0.6
85	1.8	1.9	2.1	2.2	0.6
	5 to 6% Upgrade				5 to 6% Downgrade
40	1.5	1.5	—	—	0.6
45	1.5	1.6	—	—	0.575
50	1.5	1.7	1.9	—	0.55
55	1.6	1.8	2.05	—	0.525
60	1.7	1.9	2.2	2.5	0.5
65	1.85	2.05	2.4	2.75	0.5
70	2.0	2.2	2.6	3.0	0.5
75	2.15	2.35	2.8	3.25	0.5
80	2.3	2.5	3	3.5	0.5
85	2.45	2.65	3.2	3.75	0.5

[a] Ratio from this table multiplied by the length in Table 10-4 or Table 10-6 gives length of SCL on grade.

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Table 10-5. SCL Adjustment Factors as a Function of Grade (Continued)

Metric						
Design Speed of Highway (km/h)	Deceleration Lanes Ratio of Length on Grade to Length on Level for Design Speed of Turning Curve (km/h)[a]					
All Speeds	3 to 4% upgrade 0.9			3 to 4% downgrade 1.2		
All Speeds	5 to 6% upgrade 0.8			5 to 6% downgrade 1.35		
Design Speed of Highway (km/h)	Acceleration Lanes Ratio of Length on Grade to Length on Level for Design Speed of Turning Curve (km/h)[a]					
	40	50	60	70	80	All Speeds
	3 to 4% Upgrade					3 to 4% Downgrade
60	1.3	1.4	1.4	—	—	0.7
70	1.3	1.4	1.4	1.5	—	0.65
80	1.4	1.5	1.5	1.5	1.6	0.65
90	1.4	1.5	1.5	1.5	1.6	0.6
100	1.5	1.6	1.7	1.7	1.8	0.6
110	1.5	1.6	1.7	1.7	1.8	0.6
120	1.5	1.6	1.7	1.7	1.8	0.6
130	1.6	1.7	1.8	1.8	1.8	0.6
140	1.7	1.7	1.8	1.8	1.8	0.6
	5 to 6% Upgrade					5 to 6% Downgrade
60	1.5	1.5	—	—	—	0.6
70	1.5	1.6	1.7	—	—	0.6
80	1.5	1.7	1.9	1.8	—	0.55
90	1.6	1.8	2.0	2.1	2.2	0.55
100	1.7	1.9	2.2	2.4	2.5	0.5
110	2.0	2.2	2.6	2.8	3.0	0.5
120	2.15	2.35	2.8	3.2	3.5	0.5
130	2.3	2.5	3.0	3.2	3.5	0.5
140	2.45	2.65	3.2	3.2	3.5	0.5

[a] Ratio from this table multiplied by the length in Table 10-4 or Table 10-6 gives length of SCL on grade.

10.9.6.6 Single-Lane Free-Flow Terminals, Exits

The design criteria provided for minimum deceleration lengths at exit terminals assume that there is no deceleration on the mainline prior to exiting. Although it is common that drivers start decelerating prior to leaving the mainline, the designer should not rely on this in the design of the exit ramp.

10.9.6.6.1 Taper-Type Exits

The taper-type exit fits the direct path preferred by most drivers, permitting them to follow an easy path within the diverging area. The taper-type exit terminal beginning with an outer edge alignment break usually provides a clear indication of the point of departure from the through lane and has generally been found to operate smoothly on high-volume freeways. The divergence angle is usually between 2 and 5 degrees.

Studies of this type of terminal show that most vehicles leave the through lane at relatively high speeds, thereby reducing the potential for rear-end collisions as a result of deceleration on the through lane. The speed change can be achieved off the traveled way as the exiting vehicle moves along the taper onto the ramp proper. Figure 10-73A shows a typical design for a taper-type exit.

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Minimum deceleration lengths for various combinations of design speeds for the highway and for the ramp roadway are given in Table 10-6. Grade adjustments are given in Table 10-5. The deceleration length is the distance needed for deceleration after clearing the through-traffic lane of the freeway and before reaching the first location that significantly affects vehicle speed on the ramp proper. The deceleration length is governed by the speed of traffic on the through lane of the freeway and the speed to be attained on the ramp. Deceleration may end in a complete stop, as at a crossroad terminal for a diamond ramp, or may be governed by the operating speed of the curvature of the ramp proper. The deceleration length begins where the width of the auxiliary lane increases to 12 ft [3.6 m] or greater (i.e., the end of taper) and ends at Point D. Point D represents the location where the freeway mainline ramp terminal ends and the ramp proper begins. Point D cannot be located upstream of the gore point but may be located downstream of the gore point.

To determine the boundary between the freeway mainline ramp terminal and the ramp proper, the ramp beyond the gore point should be divided into individual tangent and curve sections. If the horizontal alignment of the ramp is curvilinear, the beginning of the first curve encountered at the gore point or downstream of the gore point that significantly affects vehicle speed is the boundary between the freeway mainline ramp terminal and the ramp proper. It may be assumed that any curve radius less than or equal to 1,000 ft [300 m] will significantly affect vehicle speed. If the first curve that significantly affects vehicle speed begins at the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper is at the gore point. If the beginning of the first curve that significantly affects vehicle speed is downstream of the gore point, then the boundary between the freeway mainline ramp terminal and the ramp proper is downstream of the gore point. If the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, then conceptually the acceleration length extends to the beginning of the functional area of the crossroad ramp terminal.

The divergence zone length is associated with the freeway mainline ramp terminal of an exit ramp (and diverging maneuvers of a system interchange) and allows for deceleration after clearing the through-traffic lane of the freeway up to the gore point. The divergence zone length begins where the width of the auxiliary lane increases to 12 ft [3.6 m] or greater (i.e., the end of taper) and ends at the gore point. The divergence zone length is always less than or equal to the deceleration length. For a taper-type exit ramp, the divergence zone may be relatively short.

Although it is not desirable for vehicles to decelerate on the freeway mainline prior to moving into a deceleration lane, recent research (*21*) has found that this does occur. Because the values in Table 10-6 for minimum deceleration length on exit ramps do not account for deceleration in the through lanes, these design values provide a conservative estimate for design. It is still prudent for the designer to assume that all deceleration takes place in the SCL when determining the minimum deceleration length.

The taper-type exit terminal design can be used advantageously in developing the desired long, narrow, triangular emergency maneuver area just upstream from the exit nose located at a proper offset from both the through lane and separate ramp lane. The taper configuration also works

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well in the length-width superelevation adjustments to obtain a ramp cross slope different from that of the through lane.

The width of the recovery area or the distance between the inner edges of the diverging lanes at the ramp nose is usually 20 to 30 ft [6.0 to 9.0 m]. This entire area should be paved to provide a maneuver and recovery area, but the desired travel path for the ramp roadway should be clearly delineated by pavement markings.

10.9.6.6.2 Parallel-Type Exits

A parallel-type exit terminal usually begins with a taper, followed by an added lane that is parallel to the traveled way. A typical parallel-type exit terminal is shown in Figure 10-73C. This type of terminal provides an inviting exit area, because the foreshortened view of the taper and the added width are very apparent. A parallel-type exit operates best when drivers choose to exit the through lane sufficiently in advance of the exit nose to permit deceleration to occur on the added lane (deceleration lane) and allows them to follow a path similar to that encouraged by a taper design. Drivers who do not exit the through lane sufficiently in advance of the exit nose will likely utilize a more abrupt reverse-curve maneuver, which is somewhat unnatural and can sometimes result in the driver slowing in the through lane. In locations where both the main line and ramp carry high volumes of traffic, the deceleration lane provided by the parallel-type exit provides storage for vehicles that would otherwise undesirably queue up on the through lane or on a shoulder, if available.

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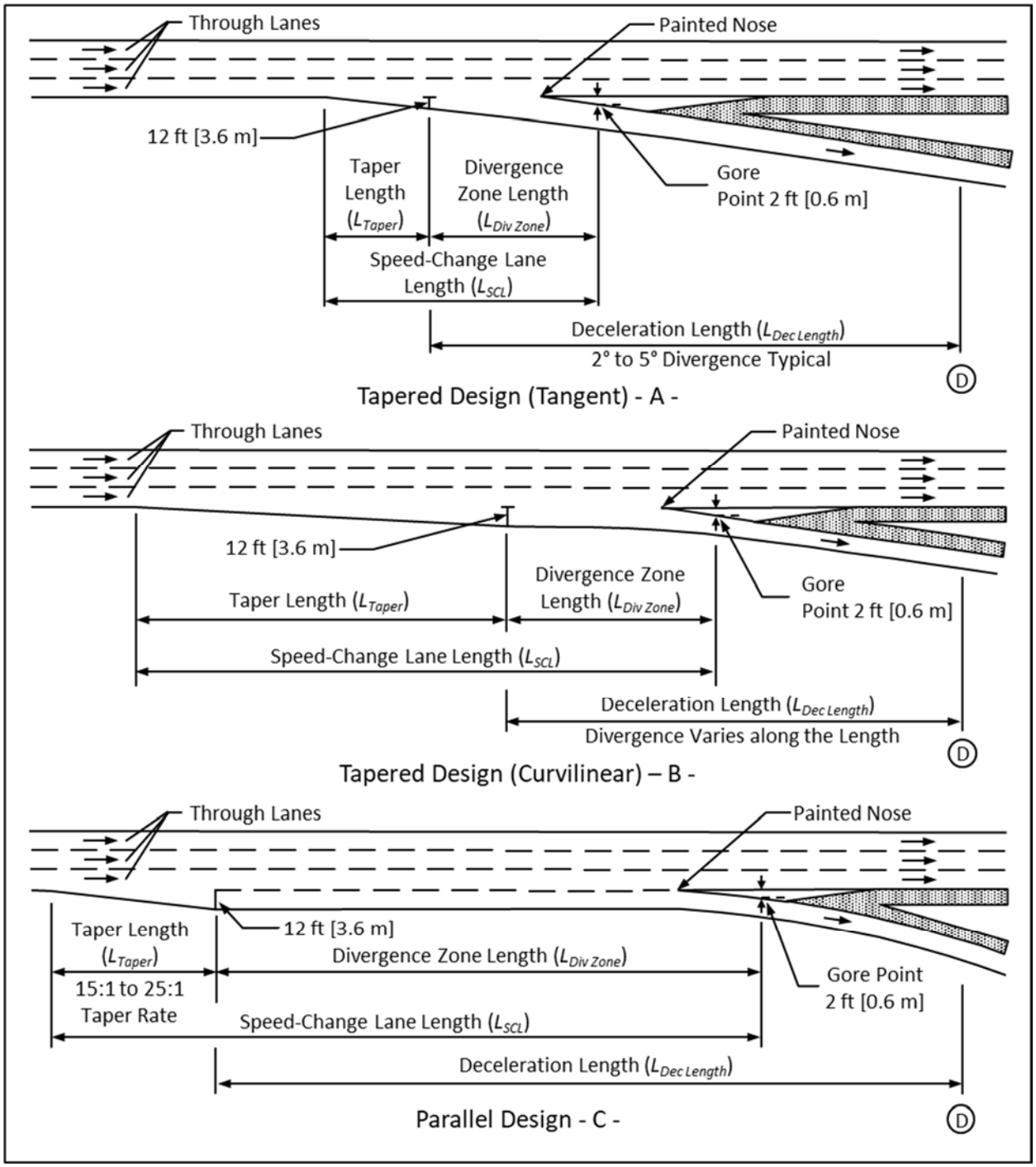


Figure 10-73. Exit Ramps—Single Lane

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Table 10-6. Minimum Deceleration Lengths for Exit Terminals with Flat Grades of Less Than Three Percent

U.S. Customary

Highway Design Speed ($V_{DS\ Hwy}$) (mph)	Diverge Speed ($V_{OS\ Hwy}$) ($V_{Dec\ Lenth\ (i)}$) (mph)	Deceleration Length, $L_{Dec\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point) (mph)								
		Stop Condition	15	20	25	30	35	40	45	50
		Average Running Speed at the End of the Deceleration Length, $V_{Dec\ Length\ (f)}$ (mph), Entering the Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point)								
		0	14	18	22	26	30	36	40	44
30	28	235	200	170	140	—	—	—	—	—
35	32	280	250	210	185	150	—	—	—	—
40	36	320	295	265	235	185	155	—	—	—
45	40	385	350	325	295	250	220	—	—	—
50	44	435	405	385	355	315	285	225	175	—
55	48	480	455	440	410	380	350	285	235	—
60	52	530	500	480	460	430	405	350	300	240
65	55	570	540	520	500	470	440	390	340	280
70	58	615	590	570	550	520	490	440	390	340
75	61	660	635	620	600	575	535	490	440	390
80	64	705	680	665	645	620	580	535	490	440
85	67	750	725	710	690	665	625	580	540	485

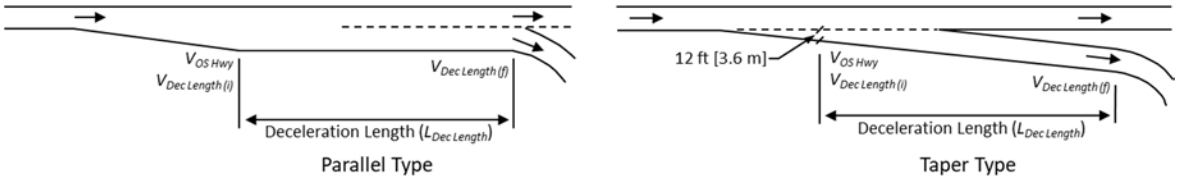
$V_{DS\ Hwy}$ = design speed of highway (mph)
$V_{OS\ Hwy}$ = average operating speed on highway (i.e., diverge speed) ($V_{Dec\ Lenth\ (i)}$) (mph)
V_{DS} = design speed of controlling feature on ramp (mph)
$V_{Dec\ Length\ (f)}$ = average running speed entering the controlling feature on the ramp (or the initial section of the ramp proprer downstream of the gore point) (mph)
$L_{Dec\ Length}$ = deceleration lane length (ft)

Metric

Highway Design Speed ($V_{DS\ Hwy}$) (km/h)	Diverge Speed ($V_{OS\ Hwy}$) ($V_{Dec\ Lenth\ (i)}$) (km/h)	Deceleration Length, $L_{Dec\ Length}$ (ft), for Design Speed (V_{DS}) of Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point) (km/h)							
		Stop Condition	20	30	40	50	60	70	80
		Average Running Speed at the End of the Deceleration Length, $V_{Dec\ Length\ (f)}$ (km/h), Entering the Controlling Feature on the Ramp Proper (or the Initial Section of the Ramp Proper Downstream of the Gore Point)							
		0	20	28	35	42	51	63	70
50	47	75	70	60	45	—	—	—	—
60	55	95	90	80	65	55	—	—	—
70	63	110	105	95	85	70	55	—	—
80	70	130	125	115	100	90	80	55	—
90	77	145	140	135	120	110	100	75	60
100	85	170	165	155	145	135	120	100	85
110	91	180	180	170	160	150	140	120	105
120	98	200	195	185	175	170	155	140	120
130	103	215	210	205	195	185	170	155	135
140	108	230	220	215	210	205	185	165	150

$V_{DS\ Hwy}$ = design speed of highway (km/h)
$V_{OS\ Hwy}$ = average operating speed on highway (i.e., diverge speed) ($V_{Dec\ Lenth\ (i)}$) (km/h)
V_{DS} = design speed of controlling feature on ramp (km/h)
$V_{Dec\ Length\ (f)}$ = average running speed entering the controlling feature on the ramp (or the initial section of the ramp proprer downstream of the gore point) (km/h)
$L_{Dec\ Length}$ = deceleration length (m)

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Minimum deceleration lengths for exit ramps are given in Table 10-6, and adjustments for grades are given in Table 10-5. For a parallel-type exit ramp, the deceleration length begins where the width of the auxiliary lane increases to 12 ft [3.6 m] or greater (i.e., the end of taper) and ends at the beginning of the first curve or feature on the ramp proper that significantly affects vehicle speeds (i.e., Point D). It may be assumed that any curve radius less than or equal to 1,000 ft [300 m] will significantly affect vehicle speed. Where the ramp proper is curved, it is desirable to provide a transition downstream of the gore point. A compound curve may be used with the initial curve desirably having a long radius of about 1,000 ft [300 m] or more. A transition or a long-radius curve is also desirable if the deceleration lane connects with a relatively straight ramp. Where the horizontal alignment of the ramp is relatively straight and has little or no influence on vehicle speeds, conceptually the deceleration length extends to the beginning of the functional area of the crossroad ramp terminal.

Providing deceleration lengths longer than the minimum values listed in Table 10-6 may promote more casual deceleration by exiting drivers, particularly under uncongested or lightly congested conditions. This is not necessarily a negative result, but it may change the operational characteristics of the ramp, as those drivers may maintain higher speeds further into the speed-change lane and possibly into the ramp proper.

The taper portion of a parallel-type deceleration lane should have a taper of approximately 15:1 to 25:1 [longitudinal:transverse]. A long taper indicates the general path to be followed and reduces the unused portion of the deceleration lane. However, a long taper tends to entice the through driver into the deceleration lane. A short taper produces a better “target” to the approaching driver, giving a positive indication of the added lane ahead.

10.9.6.6.3 Free-Flow Terminals on Curves

The previous discussion was based on highways with a tangent alignment. Because the curvature on most freeways is slight, there is usually no need to make any appreciable adjustments at ramp terminals on curves. However, where the curves on a freeway are relatively sharp and there are exits and entrances located on these curves, some adjustments in design may be desirable to avoid operational difficulties.

On freeways having design speeds of 60 mph [100 km/h] or more, the curves are sufficiently gentle so that either the parallel type or the taper type of speed-change lane is suitable. With the parallel type, the design is about the same as that on tangent and the added lane is usually on the same curvature as the main line. With the taper type, the dimensions applicable to terminals located on tangent alignment are also suitable for use on curves. A method for developing the alignment of tapered speed-change lanes on curves is illustrated in Figure 10-74. On curved sections, the ramp is tapered at the same rate relative to the through-traffic lanes as on tangent sections.

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Wherever a part of a tapered speed-change lane falls on curved alignment, it is desirable that the entire length be within the limits of the curve. Where the taper is introduced on tangent alignment just upstream from the beginning of the curve, the outer edge of the taper will appear as a kink at the point of curvature.

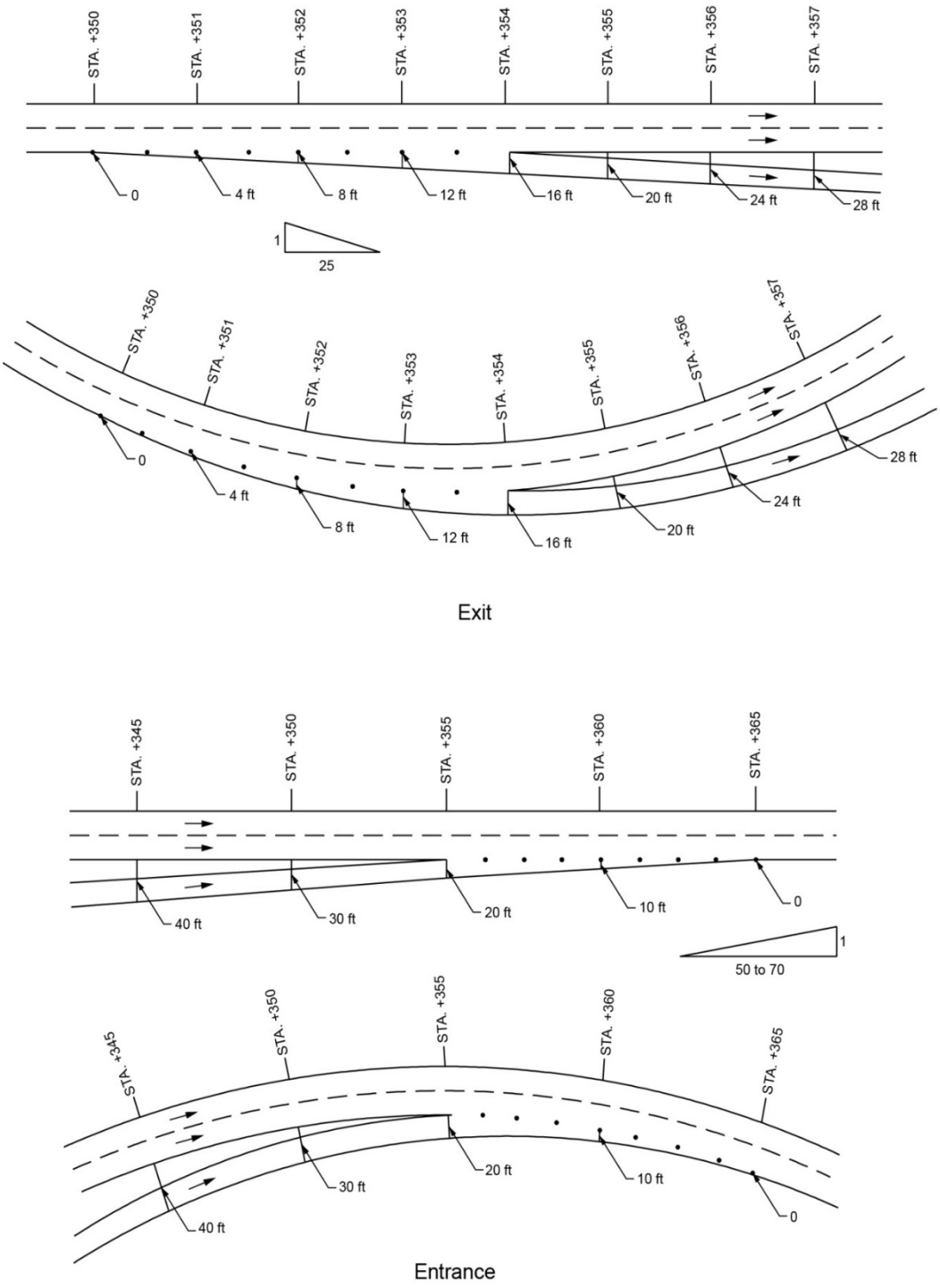
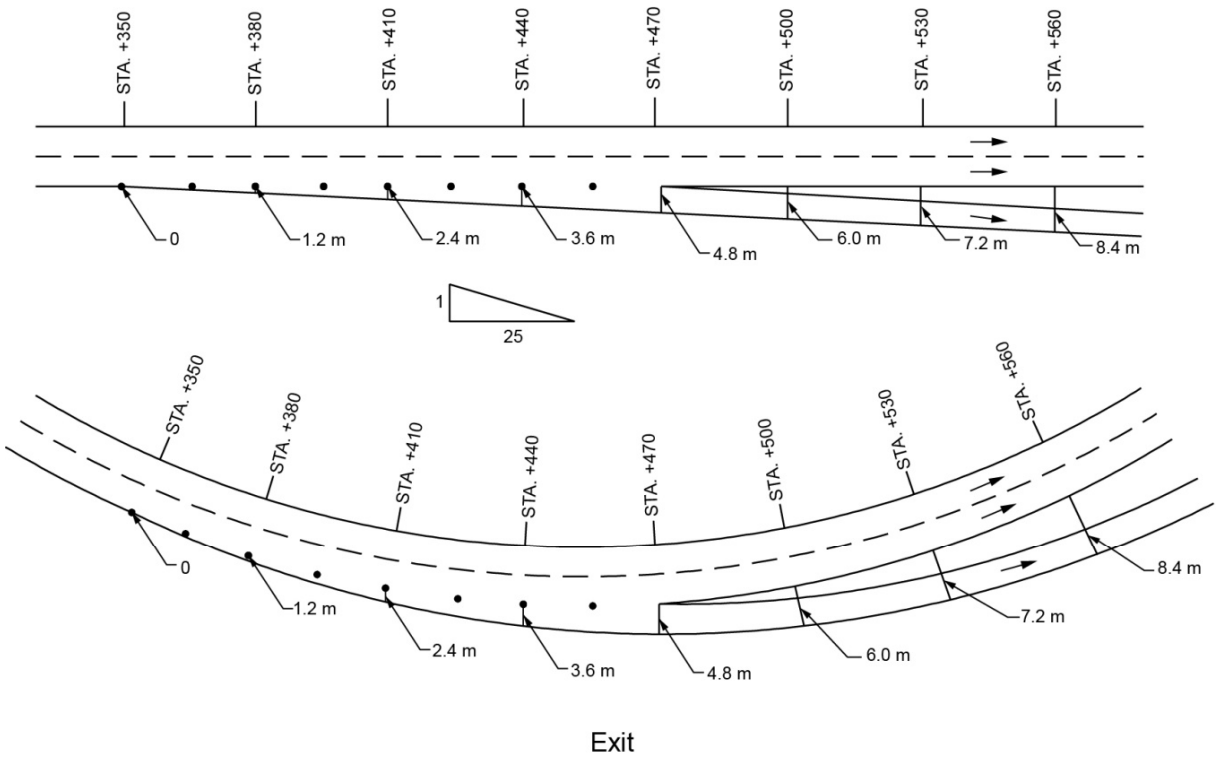


Figure 10-74a. Layout of Taper-Type Terminals on Curves (U.S. Customary)

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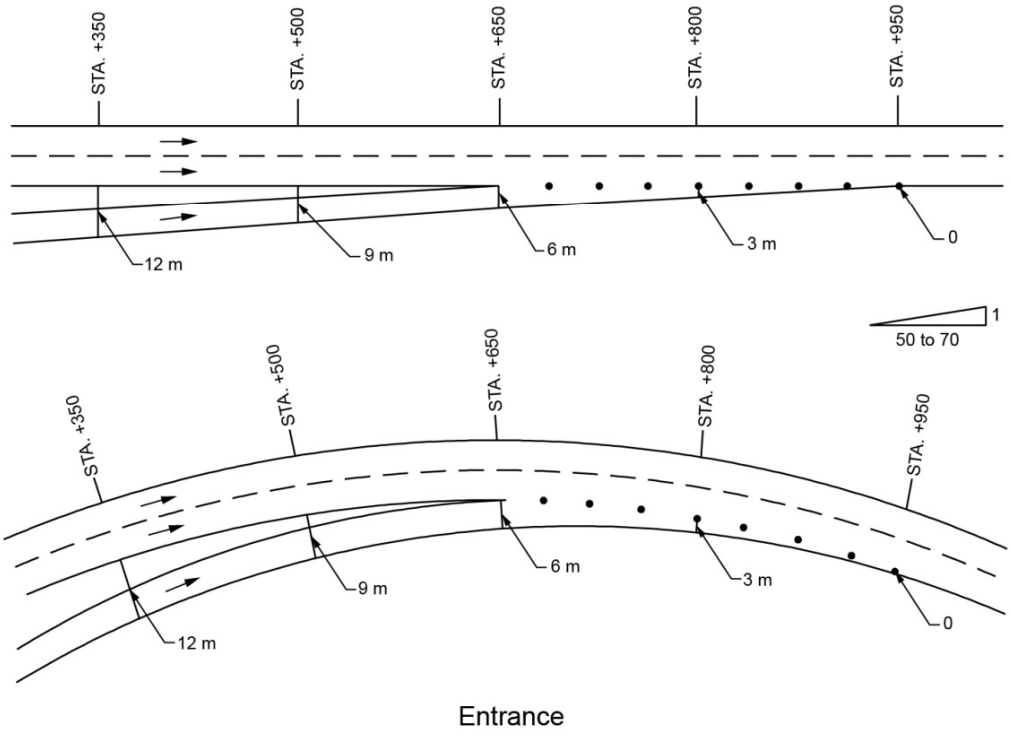


Figure 10-74b. Layout of Taper-Type Terminals on Curves (Metric)

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At ramp terminals on relatively sharp curves, such as those that may occur on freeways having a design speed of 50 mph [80 km/h], the parallel type of speed-change lane has an advantage over the taper type. At exits the parallel type is less likely to confuse through traffic, and at entrances this type will usually result in smoother merging operations. Parallel-type speed-change lanes at ramp terminals on curves are illustrated in Figure 10-75.

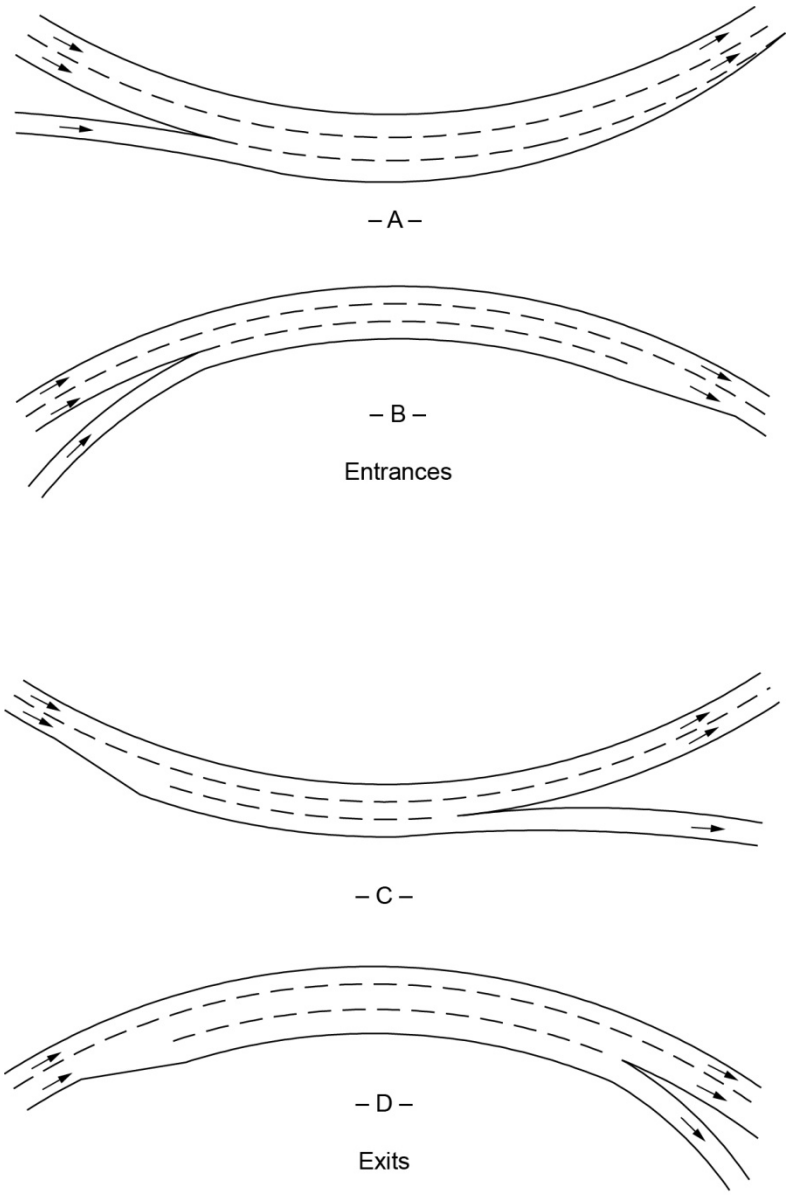


Figure 10-75. Parallel-Type Ramp Terminals on Curves

Entrances on curved sections of highway generally operate better than exits. Figures 10-75A and 10-75B show entrances with the highway curving to the left and right, respectively. It is important that the approach curve on the ramp has a very long radius as it joins the acceleration lane. This aligns the entering vehicle with the acceleration lane and lessens the chances of

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motorists entering directly onto the through lanes. The taper at the end of the acceleration lane should be long, preferably about 300 ft [90 m] in length. When reverse-curve alignment occurs between the ramp and speed-change lane, an intervening tangent should be used to aid in superelevation transition.

An exit may be particularly troublesome where the highway curves to the left (Figure 10-75C) because traffic on the outside lane tends to follow the ramp. Exits on left-turning curves should be avoided, if practical. Caution should be used in positioning a taper-type deceleration lane on the outside of a left-turning mainline curve. The design should provide a definite break in the right edge of the traveled way to provide a visual cue to the through driver to avoid being inadvertently led off the through roadway. To make the deceleration lane more apparent to approaching motorists, the taper should be shorter, preferably no more than 100 ft [30 m] in length. The deceleration lane should begin either upstream or downstream from the PC. It should not begin right at the PC, as the deceleration lane appears to be an extension of the tangent, and motorists are more likely to be confused. The ramp proper should begin with a section of tangent or a long-radius curve to permit a long and gradual reversing of the superelevation.

An alternate design, which will usually avoid operational concerns, is to locate the exit terminal a considerable distance upstream from the PC. In this design, a separate and parallel ramp roadway is provided to connect with the ramp proper.

With the highway curving to the right and the exit located on the right (Figure 10-75D), there is a tendency for vehicles to exit inadvertently. Again, the taper should be short to provide additional “target” value for the deceleration lane. With this configuration, the superelevation of the deceleration lane is readily achieved by continuing the rate from the traveled way and generally increasing it to the appropriate rate for the ramp curve.

10.9.6.6.4 Multilane Free-Flow Terminals

Multilane terminals are appropriate where traffic is too great for single-lane operation. Other considerations that may call for multilane terminals are through-route continuity, queuing on long ramps, lane-balance, and design flexibility. The most common multilane terminals consist of two-lane entrances and exits at freeways. Other multilane terminals are sometimes termed “major forks” and “branch connections.” The latter terms denote a separating and joining of two major routes.

10.9.6.6.5 Two-Lane Entrances

Two-lane entrances are warranted for two situations: either as branch connections or because of capacity needs for the on-ramp. To satisfy lane-balance needs, at least one additional lane should be provided downstream. This addition may be a basic lane, if needed for capacity, or an auxiliary lane that may be reduced 2,500 to 3,000 ft [750 to 900 m] downstream from the entrance or at the next interchange. In some instances, two additional lanes may be needed because of capacity considerations.

If the two-lane entrance is preceded by a two-lane exit, there is probably no need to increase the basic number of lanes on the freeway from a capacity standpoint. In this case, the added lane that results from the two-lane entrance is considered an auxiliary lane, and it may be reduced

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approximately 2,500 ft [750 m] or more downstream from the entrance. Details of lane reductions are presented in Section 10.9.5.11.

Figure 10-76 illustrates simple two-lane entrance terminals where a lane has been added to the freeway. The number of lanes on the freeway has little or no effect on terminal design. Figure 10-76A presents a taper-type entrance and Figure 10-76B shows a parallel-type entrance. Intermixing of the two designs is not recommended within a system route or an urban-area system.

The basic form or layout of a two-lane taper-type entrance, as shown in Figure 10-76A, is the same as a single-lane taper, as described earlier in this section, with a second lane added to the right or outer side and continued as an added or auxiliary lane on the freeway. Table 10-4 shows minimum acceleration lengths for entrance ramps. The gap acceptance length is also a consideration as illustrated in Figure 10-76A. Where ramp grades are involved, the lengths should be adjusted as shown in Table 10-5. As in the case of a single-lane entrance, it is most desirable that the geometrics of the ramp proper permit motorists to attain the approximate running speed of the freeway before reaching the tapered section.

With the parallel type of two-lane entrance, as shown in Figure 10-76B, the left lane of the ramp is continued onto the freeway as an added lane. The right lane of the ramp is carried as a parallel lane for at least 300 to 500 ft [90 to 150 m] and terminated by a tapered section at least 300 ft [90 m] in length. The length of the right lane should, as a minimum, be determined from the acceleration length or gap acceptance length, as shown in Figure 10-76B. Major factors in determining the appropriate length are the traffic volume on the ramp and the traffic volume on the freeway.

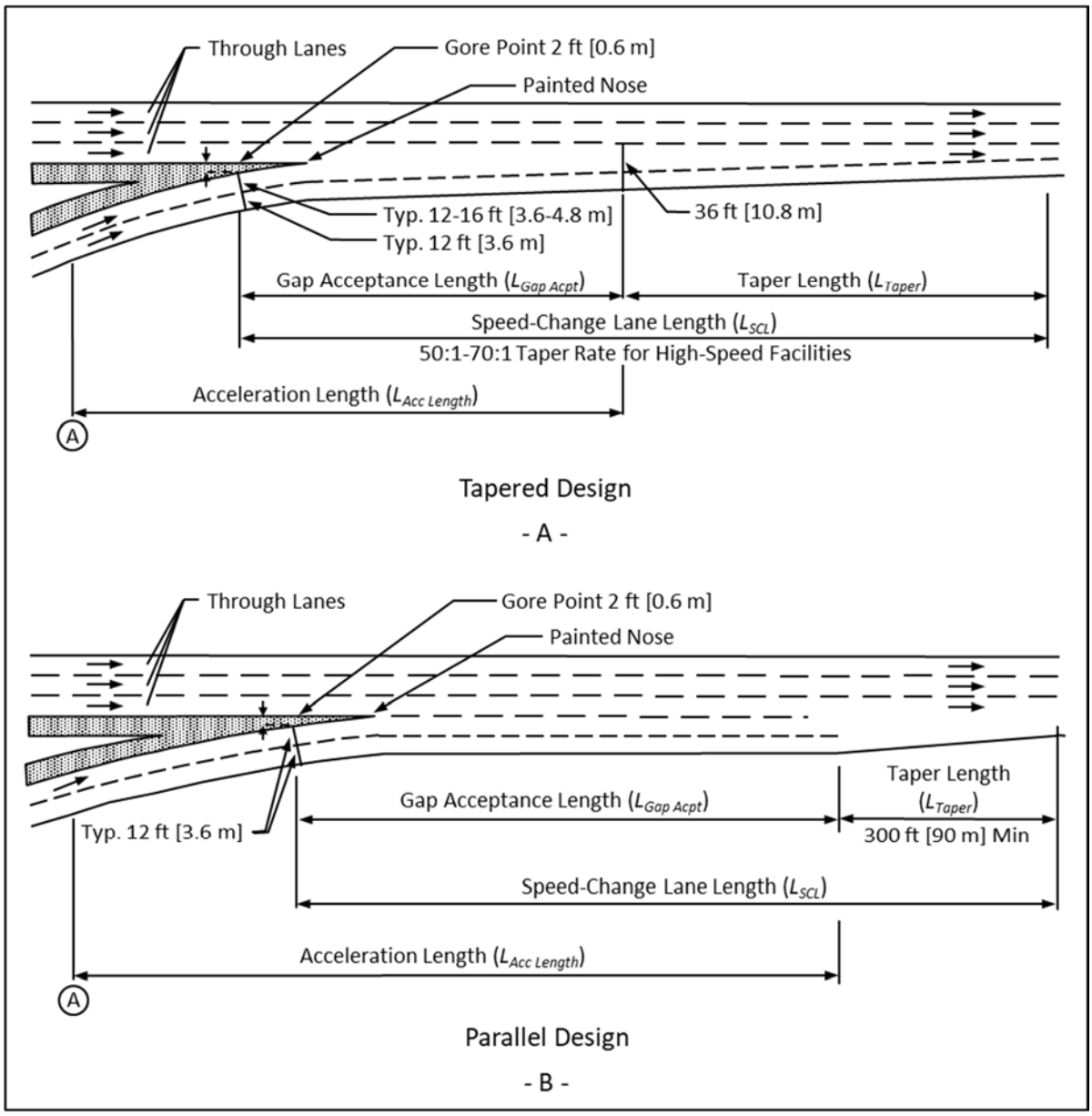
When the volume of the two-lane ramp, either the taper type or parallel type, exceeds the capacity of a through lane as specified in the HCM (*24*), it is suggested that the value for L_g (Figure 10-76) be in the range of 900 to 2,000 ft [300 to 665 m] to allow sufficient time and distance for vehicles in the left ramp lane to move into the mainline lanes. This opens space and provides time for vehicles in the right ramp lane to move into the left ramp lane. Following the termination of the left ramp lane, an additional distance in the range of 900 to 2,000 ft [300 to 665 m] should be provided, plus a taper before terminating the right ramp lane.

Although both the taper type and the parallel type of two-lane entrances will operate efficiently when properly designed, some designers prefer the parallel type. This is based on the premise that the taper type involves an “inside merge” with traffic traveling on both sides of the merging lanes. If either vehicle involved with the merging movement abandons the merge, traffic in the adjacent lanes could prevent the merging vehicles from escaping to the adjacent lanes. By contrast, the parallel type allows the merging vehicle to escape to the right shoulder without any interference.

Where the predominant two-lane entrances in a particular state or locality are of the parallel type and, therefore, drivers are accustomed to that type of entrance, a taper-type entrance would violate driver expectancy, and vice versa. Thus, a particular type of entrance terminal is sometimes criticized as being unsatisfactory when in fact the difficulty may be lack of

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uniformity. Either form of two-lane entrance is satisfactory if used exclusively within an area or a region, but they should not be intermixed along a given route.



Notes:

1. $L_{Acc\ Length}$ is the acceleration length as shown in Table 10-4 or as adjusted by Table 10-5.
2. Point A controls speed on the ramp. $L_{Acc\ Length}$ should not start back on the curvature of the ramp unless the radius equals 1000 ft [300 m] or more.
3. $L_{Gap\ Acpt}$ is the gap acceptance length. $L_{Gap\ Acpt}$ should be a minimum of 300 to 500 ft [90 to 150 m], with suggested $L_{Gap\ Acpt}$ values up to 2,000 ft [600 m] for high-volume conditions.
4. The value of $L_{Acc\ Length}$ or $L_{Gap\ Acpt}$, whichever produces the greater distance downstream from the gore point, is suggested for use in the design of the ramp entrance.

Figure 10-76. Typical Two-Lane Entrance Ramps

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10.9.6.6.6 Two-Lane Exits

Where the traffic volume leaving the freeway at an exit terminal exceeds the design capacity of a single lane, a two-lane exit terminal should be provided. To satisfy lane-balance needs and not reduce the basic number of through lanes, it is usually appropriate to add an auxiliary lane upstream from the exit. A distance of approximately 1,500 ft [450 m] is recommended to develop the full capacity of a two-lane exit. As with single-lane exits, attention should be given to obtaining the appropriate deceleration distance between the exit and first horizontal curve on the ramp. Typical designs for two-lane exit terminals are shown in Figure 10-77; the taper is illustrated in Figure 10-77A and the parallel type in Figure 10-77B.

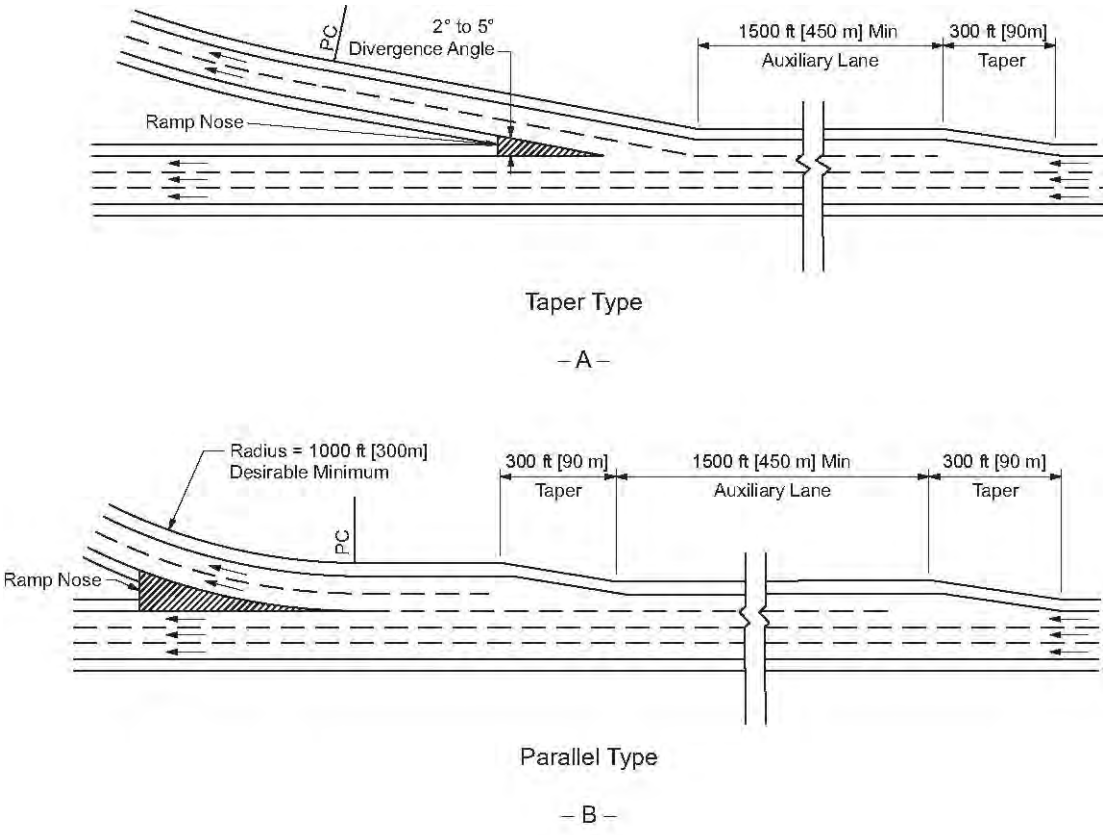


Figure 10-77. Two-Lane Exit Terminals

In cases where the basic number of lanes is to be reduced beyond a two-lane exit, the basic number of lanes should be carried beyond the exit before the lane reduction of the outer lane. This design provides a recovery area for any through vehicles that remain in that lane. This was discussed in Section 10.9.5.11.

With the parallel type of two-lane exit, as shown in Figure 10-77B, the operation is different from the taper type in that traffic in the outer through lane of the freeway must change lanes to exit. In fact, an exiting motorist is required to move two lanes to the right to use the right lane of the ramp. Thus, considerable lane changing is needed in order for the exit to operate efficiently. This entire operation takes place over a substantial length of highway, which is dependent in part on the total traffic volume on the freeway and especially on the volume using the exit ramp. The

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total length from the beginning of the first taper to the point where the ramp traveled way departs from the right-hand through lane of the freeway should range from 2,500 ft [750 m] for turning volumes of 1,500 veh/h or less upward to 3,500 ft [1,000 m] for turning volumes of 3,000 veh/h.

10.9.6.6.7 Two-Lane Terminals on Curved Alignment

The design of ramp terminals where the freeway is on curved alignment is discussed under single-lane terminals. The same principles of design, in which offsets from the edge of roadway are used, may be used in the layout of two-lane terminals.

10.9.6.6.8 Major Forks and Branch Connections

A major fork is defined as the bifurcation of a directional roadway of a terminating freeway route into two directional multilane ramps that connect to another freeway, or of a freeway route into two separate freeway routes of about equal importance.

When designing major forks, the nose should be placed in direct alignment with the centerline of one of the interior lanes, as illustrated in Figures 10-78A, 10-78B, or 10-78C, where the horizontal alignments of the two departing roadways are in curves. This interior lane is continued as a full-width lane, both left and right of the gore. Thus, the width of this interior lane will be at least 24 ft [7.2 m] at the painted nose (prolongation of pavement edge stripes) and preferably not over 28 ft [8.4 m]. The length over which the widening from 12 to 24 ft [3.6 to 7.2 m] takes place should be within the range of 1,000 or 1,800 ft [300 or 540 m]. However, in the case where at least one of the approaches is on a tangent alignment and continues on a tangent, a true optional interior lane cannot be physically developed. As such, the principles of the two-lane exit facility should be used as shown in Figure 10-78D, depending on the nose width, with suggested lengths up to 2,000 ft [600 m] in high-volume conditions. When the approach to a major fork has four or more lanes, the development of an interior option lane may present signing complications. Adding an auxiliary lane and having dedicated lanes in advance of the split (no option lane) may allow for simpler signing to improve driver guidance into their intended lanes.

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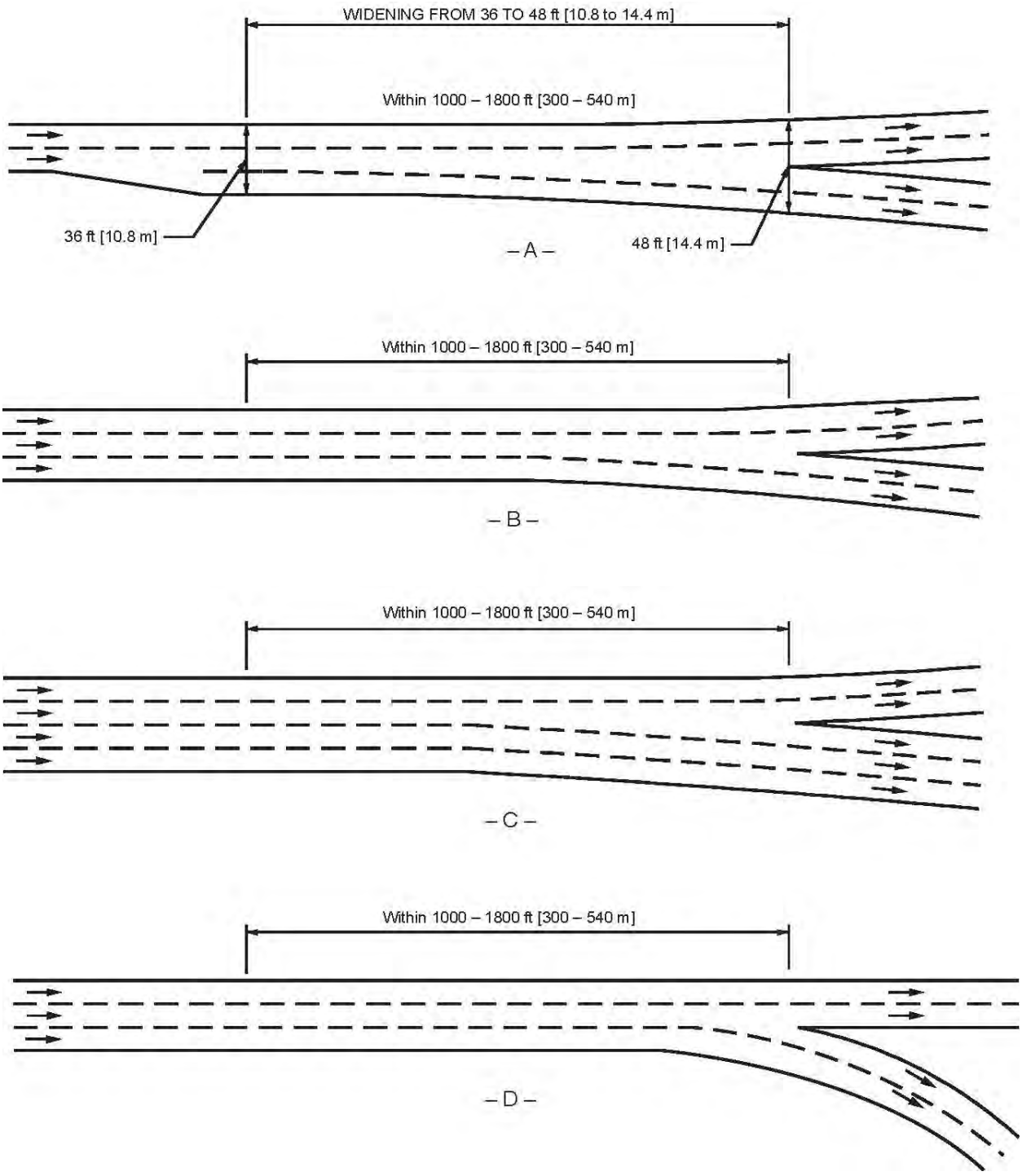


Figure 10-78. Major Forks

In the case of a two-lane roadway separating into two, two-lane routes, there is no interior lane. In such cases, it is advisable to widen the approach roadway to three lanes, thus creating an interior lane. The lane is added on the side of the fork that serves the lesser traffic volume. In Figure 10-78A, the right (lower) fork would be the more lightly traveled of the two. The widening from 36 ft [10.8 m] for the approach roadway to about 48 or 50 ft [14.4 or 15.0 m] at

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the painted nose should be accomplished in a continuous sweeping curve with no reverse curvature in the alignment of the roadway edges.

A branch connection is defined by (1) the beginning of a directional roadway of a freeway formed by the convergence of two directional multilane ramps from another freeway or by (2) the converging of two freeway routes to form a single freeway route.

The number of lanes downstream from the point of convergence may be one lane less than the combined total on the two approach roadways. In some cases, the traffic demand may indicate that the number of lanes going away from the merging area be equal to the sum of the number of lanes on the two roadways approaching it, and a design of this type will operate efficiently. Such a design is illustrated in Figure 10-79A.

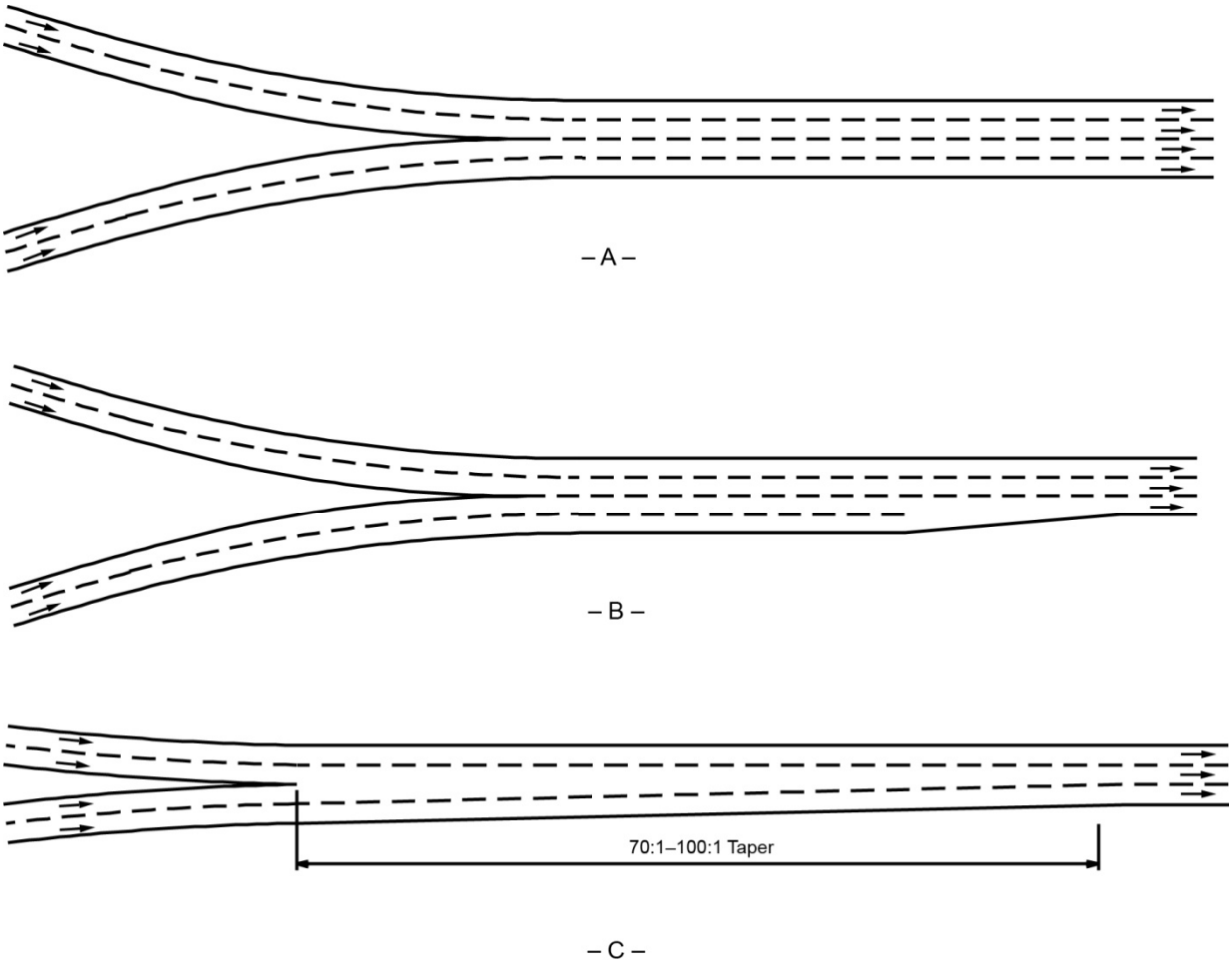


Figure 10-79. Branch Connections

Where a lane is to be eliminated, the most common method for accomplishing this is a lane reduction, as discussed in Section 10.9.5.11. The lane that is terminated will ordinarily be the exterior lane from the roadway serving the lowest volume per lane. However, some

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considerations should also be given to the fact that the outer lane from the roadway entering from the right is the slow-speed lane for that roadway, whereas the opposite is true for the roadway entering from the left. If the traffic volumes per lane are about equal, it would be proper to terminate the lane on the right, as shown in Figure 10-79B. In any case, consistency within an area or region is often more important than volume per lane since the latter may change with the specific design or with traffic demand changing over time. The lane being terminated should be carried at full lane width for a distance of approximately 1,000 ft [300 m] before being tapered out.

Another consideration is the possibility of a high-speed inside merge, as in Figure 10-79C. This merge should be treated as any other high-speed merging situation; see the discussion of the advantages of parallel-type entrance in Section 10.9.6.6.5.

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